In [1]:
```
#initiate spark
import pyspark
from pyspark import SparkContext, SparkConf
from pyspark.sql import SparkSession
c = pyspark.SparkConf().setAppName("test_app").setMaster("local")
sc = pyspark.SparkContext(conf = c)
spark = SparkSession(sc)
```

In [2]:
```
from pyspark.sql.functions import *
```

In [3]:
```
import pandas as pd
import numpy as np
import datetime
import warnings
warnings.filterwarnings('ignore')
```

In [4]:
```
from pyspark.sql.window import Window
```

In [5]:
```
# Loading Data Sets with PySpark

s_boxf = spark.read.csv("D:\Data Engineer books\CORE TOPICS\PYSPARK\PySpark Assignment\Assignment 3\Boxoffice_Fa
                        header = True, inferSchema = True, escape = '"')
s_dir = spark.read.csv("D:\Data Engineer books\CORE TOPICS\PYSPARK\PySpark Assignment\Assignment 3\Director_dim
                        header = True, inferSchema = True, escape = '"')
s_gen = spark.read.csv("D:\Data Engineer books\CORE TOPICS\PYSPARK\PySpark Assignment\Assignment 3\Genere_dim.cs
                        header = True, inferSchema = True, escape = '"')
s_lan = spark.read.csv("D:\Data Engineer books\CORE TOPICS\PYSPARK\PySpark Assignment\Assignment 3\Language_dim
                        header = True, inferSchema = True, escape = '"')
```

In [6]:
```
# Loading Data Sets with Pandas

p_boxf = pd.read_csv("D:\Data Engineer books\CORE TOPICS\PYSPARK\PySpark Assignment\Assignment 3\Boxoffice_Fact
p_dir = pd.read_csv("D:\Data Engineer books\CORE TOPICS\PYSPARK\PySpark Assignment\Assignment 3\Director_dim.csv
p_gen = pd.read_csv("D:\Data Engineer books\CORE TOPICS\PYSPARK\PySpark Assignment\Assignment 3\Genere_dim.csv"
p_lan = pd.read_csv("D:\Data Engineer books\CORE TOPICS\PYSPARK\PySpark Assignment\Assignment 3\Language_dim.csv
```

In [7]:
```
# Checking Sample Data With Pandas
p_boxf.head(5)
```

Out[7]:

	FilmID	Title	Release Date	DirectorID	Lead Actor/Actress	LanguageID	Industry	GenreID	Budget in Crores	First Day Collection Worldwide in Crores	Worldwide Collection in Crores	Overs Collect in Cro
0	34	Sanju	29-Jun-18	101	Ranbir Kapoor	501	Bollywood	623	100	54.4	588.5	
1	36	Simmba	28-Dec-18	105	Ranveer Singh	501	Bollywood	605	125	20.2	390	
2	225	Janatha Garage	01-Sep-16	140	N.T. Rama Rao Jr.	503	Kollywood	605	50	45	140	
3	13	Kaththi	22-Oct-14	107	Joseph Vijay	503	Kollywood	606	70	25	117.5	
4	31	Maanikya	01-May-14	130	V. Ravichandran	507	Sandalwood	606	15	3.5	43	

In [8]:
```
# Checking Sample Data With PySpark
s_boxf.show(5,False)
```

```
+------+--------------+------------+----------+------------------+----------+----------+-------+----------------
+-----------------------------------------+-----------------------------+-----------------------------+-------
-------------------------+-----------+----------+--------------+------------------+
|FilmID|Title         |Release Date|DirectorID|Lead Actor/Actress|LanguageID|Industry  |GenreID|Budget in Crores
|First Day Collection Worldwide  in Crores|Worldwide Collection  in Crores|Overseas Collection in Crores|India G
ross Collection in Crores|Verdict    |IMDb Rating|Runtime (mins)|OTT Platform      |
+------+--------------+------------+----------+------------------+----------+----------+-------+----------------
+-----------------------------------------+-----------------------------+-----------------------------+-------
-------------------------+-----------+----------+--------------+------------------+
|34    |Sanju         |29-Jun-18   |101       |Ranbir Kapoor     |501       |Bollywood |623    |100
|54.4                                     |588.5                        |150                          |438.5
|Blockbuster|7.6        |155           |Netflix           |
|36    |Simmba        |28-Dec-18   |105       |Ranveer Singh     |501       |Bollywood |605    |125
|20.2                                     |390                          |95                           |295
|Blockbuster|5.5        |158           |ZEE5              |
|225   |Janatha Garage|01-Sep-16   |140       |N.T. Rama Rao Jr. |503       |Kollywood |605    |50
|45                                       |140                          |15                           |125
|Blockbuster|7.2        |162           |Disney+ Hotstar   |
|13    |Kaththi       |22-Oct-14   |107       |Joseph Vijay      |503       |Kollywood |606    |70
|25                                       |117.5                        |29                           |88.5
|Blockbuster|8.1        |166           |Amazon Prime Video|
|31    |Maanikya      |01-May-14   |130       |V. Ravichandran   |507       |Sandalwood|606    |15
|3.5                                      |43                           |0                            |43
|Blockbuster|6.5        |165           |ZEE5              |
+------+--------------+------------+----------+------------------+----------+----------+-------+----------------
+-----------------------------------------+-----------------------------+-----------------------------+-------
-------------------------+-----------+----------+--------------+------------------+
only showing top 5 rows
```

In [9]:
```
# Checking Volume of Data With Pandas
p_boxf.shape
```

Out[9]:
```
(604, 17)
```

In [10]:
```
# Checking Volume of Data With PySprak
s_boxf.count(),len(s_boxf.columns)
```

Out[10]:
```
(604, 17)
```

In [11]:
```
# checking null Values with Pandas
p_boxf.isnull().sum()
```

Out[11]:
```
FilmID                                        0
Title                                         0
Release Date                                  0
DirectorID                                    0
Lead Actor/Actress                            0
LanguageID                                    0
Industry                                      0
GenreID                                       0
Budget in Crores                              0
First Day Collection Worldwide  in Crores     0
Worldwide Collection  in Crores               0
Overseas Collection in Crores                 9
India Gross Collection in Crores              0
Verdict                                       0
IMDb Rating                                   0
Runtime (mins)                                0
OTT Platform                                  0
dtype: int64
```

In [12]:
```
# Writing Function for checking missing values
def checking_missing_values(df,list_col):
    missing_values = {}
    for i in list_col:
        a = df.filter(col(i).isNull()).count()
        missing_values[i] = a
    return missing_values
```

In [13]:
```
# Checking Missing Values With PySpark
checking_missing_values(s_boxf.cache(),s_boxf.columns)
```

Out[13]:
```
{'FilmID': 0,
 'Title': 0,
 'Release Date': 0,
 'DirectorID': 0,
 'Lead Actor/Actress': 0,
 'LanguageID': 0,
 'Industry': 0,
 'GenreID': 0,
 'Budget in Crores': 0,
 'First Day Collection Worldwide  in Crores': 0,
 'Worldwide Collection  in Crores': 0,
 'Overseas Collection in Crores': 9,
 'India Gross Collection in Crores': 0,
 'Verdict': 0,
 'IMDb Rating': 0,
 'Runtime (mins)': 0,
 'OTT Platform': 0}
```

In [14]:
```
# Checking data types with Pandas
p_boxf.dtypes
```

Out[14]:
```
FilmID                                          int64
Title                                          object
Release Date                                   object
DirectorID                                      int64
Lead Actor/Actress                             object
LanguageID                                      int64
Industry                                       object
GenreID                                         int64
Budget in Crores                               object
First Day Collection Worldwide  in Crores      object
Worldwide Collection  in Crores                object
Overseas Collection in Crores                  object
India Gross Collection in Crores               object
Verdict                                        object
IMDb Rating                                   float64
Runtime (mins)                                  int64
OTT Platform                                   object
dtype: object
```

In [15]:
```
# Checking data Type with PySpak
s_boxf.printSchema()
```

```
root
 |-- FilmID: integer (nullable = true)
 |-- Title: string (nullable = true)
 |-- Release Date: string (nullable = true)
 |-- DirectorID: integer (nullable = true)
 |-- Lead Actor/Actress: string (nullable = true)
 |-- LanguageID: integer (nullable = true)
 |-- Industry: string (nullable = true)
 |-- GenreID: integer (nullable = true)
 |-- Budget in Crores: string (nullable = true)
 |-- First Day Collection Worldwide  in Crores: string (nullable = true)
 |-- Worldwide Collection  in Crores: string (nullable = true)
 |-- Overseas Collection in Crores: string (nullable = true)
 |-- India Gross Collection in Crores: string (nullable = true)
 |-- Verdict: string (nullable = true)
 |-- IMDb Rating: double (nullable = true)
 |-- Runtime (mins): integer (nullable = true)
 |-- OTT Platform: string (nullable = true)
```

Data Cleaning With Pandas

Changing the Data Type with Pandas

In [16]:
```
# checking how many null values are present in the data frame with Pandas
p_boxf["Budget in Crores"].str.isnumeric().sum()
```

Out[16]:
```
np.int64(587)
```

In [17]:
```
# Replacing the values in the data frame
p_boxf["Budget in Crores"] = p_boxf["Budget in Crores"].str.replace(":","",regex = True).astype(float)
```

In [18]:
```
p_boxf.dtypes
```

Out[18]:
```
FilmID                                          int64
Title                                          object
Release Date                                   object
DirectorID                                      int64
Lead Actor/Actress                             object
LanguageID                                      int64
Industry                                       object
GenreID                                         int64
Budget in Crores                             float64
First Day Collection Worldwide  in Crores      object
Worldwide Collection  in Crores                object
Overseas Collection in Crores                  object
India Gross Collection in Crores               object
Verdict                                        object
IMDb Rating                                  float64
Runtime (mins)                                  int64
OTT Platform                                   object
dtype: object
```

In [19]:
```
p_boxf["First Day Collection Worldwide  in Crores"] = p_boxf["First Day Collection Worldwide  in Crores"]\
    .str.replace("[: ]","",regex = True).astype(float)
```

In [20]:
```
p_boxf.dtypes
```

Out[20]:
```
FilmID                                          int64
Title                                          object
Release Date                                   object
DirectorID                                      int64
Lead Actor/Actress                             object
LanguageID                                      int64
Industry                                       object
GenreID                                         int64
Budget in Crores                             float64
First Day Collection Worldwide  in Crores    float64
Worldwide Collection  in Crores                object
Overseas Collection in Crores                  object
India Gross Collection in Crores               object
Verdict                                        object
IMDb Rating                                  float64
Runtime (mins)                                  int64
OTT Platform                                   object
dtype: object
```

In [21]:
```
p_boxf["Worldwide Collection  in Crores"] = p_boxf["Worldwide Collection  in Crores"]\
    .str.replace("[: ]","",regex = True).astype(float)
p_boxf["Overseas Collection in Crores"]=p_boxf["Overseas Collection in Crores"]\
    .str.replace("[: ]","",regex = True).astype(float)
p_boxf["India Gross Collection in Crores"]=p_boxf["India Gross Collection in Crores"]\
    .str.replace("[: ]","",regex=True).astype(float)
```

In [22]:
```
p_boxf.dtypes
```

Out[22]:
```
FilmID                                          int64
Title                                          object
Release Date                                   object
DirectorID                                      int64
Lead Actor/Actress                             object
LanguageID                                      int64
Industry                                       object
GenreID                                         int64
Budget in Crores                             float64
First Day Collection Worldwide  in Crores    float64
Worldwide Collection  in Crores              float64
Overseas Collection in Crores                float64
India Gross Collection in Crores             float64
Verdict                                        object
IMDb Rating                                  float64
Runtime (mins)                                  int64
OTT Platform                                   object
dtype: object
```

In [23]:
```
list(p_boxf["Verdict"].unique())
```

Out[23]:
```
['Blockbuster',
 'All Time Blockbuster',
 'Below Average',
 ': Disaster',
 ': Blockbuster',
 ': All Time Blockbuster',
 'Super Hit',
 ': Flop',
 'Above Average',
 'Flop',
 ': Hit',
 ':  Super Hit',
 ': Super Hit',
 ': Average',
 'Disaster',
 'Average',
 'Hit',
 ': Below Average',
 ': Above Average',
 ' Below Average',
 'Super Hitt']
```

In [24]:
```
p_boxf["Verdict"]= p_boxf["Verdict"].str.replace("[: ]","",regex = True)
```

In [25]:
```
list(p_boxf["Verdict"].unique())
```

Out[25]:
```
['Blockbuster',
 'AllTimeBlockbuster',
 'BelowAverage',
 'Disaster',
 'SuperHit',
 'Flop',
 'AboveAverage',
 'Hit',
 'Average',
 'SuperHitt']
```

In [26]:
```
p_boxf["Verdict"]= p_boxf["Verdict"].replace("SuperHitt","SuperHit")
```

In [27]:
```
list(p_boxf["Verdict"].unique())
```

Out[27]:
```
['Blockbuster',
 'AllTimeBlockbuster',
 'BelowAverage',
 'Disaster',
 'SuperHit',
 'Flop',
 'AboveAverage',
 'Hit',
 'Average']
```

In [28]:
```
list(p_boxf["OTT Platform"].unique())
```

Out[28]:
```
['Netflix',
 'ZEE5',
 'Disney+ Hotstar',
 'Amazon Prime Video',
 'Aha  ',
 'Google Play Movies',
 'Amazon Prime Video ',
 'Disney+ Hotstar ',
 'Netflix ',
 'Sun NXT',
 'ZEE5  ',
 'Sun NXT  ',
 'Amazon Prime Video  ',
 'SonyLIV  ',
 'Netflix  ',
 'Voot  ',
 'Disney+ Hotstar  ',
 'aha  ',
 'ZEE5 ',
 'SonyLIV',
 'ZEE5      ',
 'JioCinema',
 'Aha']
```

In [29]:
```
p_boxf["OTT Platform"] = p_boxf["OTT Platform"].str.strip()
```

In [30]:
```
list(p_boxf["OTT Platform"].unique())
```

Out[30]:
```
['Netflix',
 'ZEE5',
 'Disney+ Hotstar',
 'Amazon Prime Video',
 'Aha',
 'Google Play Movies',
 'Sun NXT',
 'SonyLIV',
 'Voot',
 'aha',
 'JioCinema']
```

In [31]:
```
p_boxf["OTT Platform"] = p_boxf["OTT Platform"].replace("aha","Aha")
```

In [32]:
```
list(p_boxf["OTT Platform"].unique())
```

Out[32]:
```
['Netflix',
 'ZEE5',
 'Disney+ Hotstar',
 'Amazon Prime Video',
 'Aha',
 'Google Play Movies',
 'Sun NXT',
 'SonyLIV',
 'Voot',
 'JioCinema']
```

In [33]:
```
# checking the LanguageID columns
p_boxf["LanguageID"].unique()
```

Out[33]:
```
array([501, 503, 507, 505, 509, 511, 515, 517, 513, 519])
```

In [34]:
```
# checking the language table how many languge in avlilbe in data frame
p_lan.head(10)
```

Out[34]:

	Language	LanguageID
0	Hindi	501
1	Tamil	503
2	Telugu	505
3	Kannada	507
4	Malayalam	509

In [35]:
```
# grouping languageID with the Industry to get the wrong matching languageID
p_boxf[["LanguageID","Industry"]].groupby("LanguageID").head(20)
```

Out[35]:

	LanguageID	Industry
0	501	Bollywood
1	501	Bollywood
2	503	Kollywood
3	503	Kollywood
4	507	Sandalwood
...	...	...
241	511	Tollywood
242	513	Mollywood
253	517	Kollywood
255	519	Sandalwood
256	513	Mollywood

151 rows × 2 columns

In [36]:
```
# 511 --> 505
# 513 --> 509
# 515 --> 501
# 517 --> 503
# 519 --> 507
```

In [37]:
```
# replacing the languageID with the representative industry
p_boxf["LanguageID"] = p_boxf["LanguageID"].replace({511:505,513:509,515:501,517:503,519:507})
```

In [38]:
```
# checking the unique values in languageID
p_boxf["LanguageID"].unique()
```

Out[38]:
```
array([501, 503, 507, 505, 509])
```

In [39]:
```
p_boxf.dtypes
```

Out[39]:
```
FilmID                                          int64
Title                                          object
Release Date                                   object
DirectorID                                      int64
Lead Actor/Actress                             object
LanguageID                                      int64
Industry                                       object
GenreID                                         int64
Budget in Crores                              float64
First Day Collection Worldwide  in Crores     float64
Worldwide Collection  in Crores               float64
Overseas Collection in Crores                 float64
India Gross Collection in Crores              float64
Verdict                                        object
IMDb Rating                                   float64
Runtime (mins)                                  int64
OTT Platform                                   object
dtype: object
```

In [40]:
```
# changing the realse Date  data type
p_boxf["Release Date"] = pd.to_datetime(p_boxf["Release Date"])
```

In [41]:
```
p_boxf.dtypes
```

Out[41]:
```
FilmID                                                   int64
Title                                                   object
Release Date                                  datetime64[ns]
DirectorID                                               int64
Lead Actor/Actress                                      object
LanguageID                                               int64
Industry                                                object
GenreID                                                  int64
Budget in Crores                                       float64
First Day Collection Worldwide  in Crores              float64
Worldwide Collection  in Crores                        float64
Overseas Collection in Crores                          float64
India Gross Collection in Crores                       float64
Verdict                                                 object
IMDb Rating                                            float64
Runtime (mins)                                           int64
OTT Platform                                            object
dtype: object
```

Data Cleaning With PySpark

Changing Data Types with PySpark

In [42]:
```
s_boxf.printSchema()
```

```
root
 |-- FilmID: integer (nullable = true)
 |-- Title: string (nullable = true)
 |-- Release Date: string (nullable = true)
 |-- DirectorID: integer (nullable = true)
 |-- Lead Actor/Actress: string (nullable = true)
 |-- LanguageID: integer (nullable = true)
 |-- Industry: string (nullable = true)
 |-- GenreID: integer (nullable = true)
 |-- Budget in Crores: string (nullable = true)
 |-- First Day Collection Worldwide  in Crores: string (nullable = true)
 |-- Worldwide Collection  in Crores: string (nullable = true)
 |-- Overseas Collection in Crores: string (nullable = true)
 |-- India Gross Collection in Crores: string (nullable = true)
 |-- Verdict: string (nullable = true)
 |-- IMDb Rating: double (nullable = true)
 |-- Runtime (mins): integer (nullable = true)
 |-- OTT Platform: string (nullable = true)
```

In [43]:
```
cols = ["Budget in Crores",\
     "First Day Collection Worldwide  in Crores",\
     "Worldwide Collection  in Crores",\
     "Overseas Collection in Crores",\
     "India Gross Collection in Crores"]
for c in cols:
```

```
    s_boxf = s_boxf.withColumn(c,regexp_replace(col(c),r"[: ]","").cast("double"))
```

In [44]:
```
s_boxf.printSchema()
```

```
root
 |-- FilmID: integer (nullable = true)
 |-- Title: string (nullable = true)
 |-- Release Date: string (nullable = true)
 |-- DirectorID: integer (nullable = true)
 |-- Lead Actor/Actress: string (nullable = true)
 |-- LanguageID: integer (nullable = true)
 |-- Industry: string (nullable = true)
 |-- GenreID: integer (nullable = true)
 |-- Budget in Crores: double (nullable = true)
 |-- First Day Collection Worldwide  in Crores: double (nullable = true)
 |-- Worldwide Collection  in Crores: double (nullable = true)
 |-- Overseas Collection in Crores: double (nullable = true)
 |-- India Gross Collection in Crores: double (nullable = true)
 |-- Verdict: string (nullable = true)
 |-- IMDb Rating: double (nullable = true)
 |-- Runtime (mins): integer (nullable = true)
 |-- OTT Platform: string (nullable = true)
```

In [45]:
```
s_boxf.select("Verdict").distinct().show()
```

```
+--------------------+
|             Verdict|
+--------------------+
|          Super Hitt|
|        :  Super Hit|
|          : Disaster|
|              : Flop|
|       Above Average|
|             Average|
|           : Average|
|       Below Average|
|     : Above Average|
|     : Below Average|
|       Below Average|
|             Disaster|
|                : Hit|
|All Time Blockbuster|
|       : Blockbuster|
|           Super Hit|
|                Flop|
|: All Time Blockb...|
|         Blockbuster|
|         : Super Hit|
+--------------------+
only showing top 20 rows
```

In [46]:
```
cols = ["Verdict","OTT Platform"]
for c in cols:
    s_boxf = s_boxf.withColumn(c,initcap(trim(regexp_replace(col(c),":",""))))
```

In [47]:
```
s_boxf.select("Verdict").distinct().show()
```

```
+--------------------+
|             Verdict|
+--------------------+
|          Super Hitt|
|       Above Average|
|             Average|
|       Below Average|
|             Disaster|
|All Time Blockbuster|
|           Super Hit|
|                Flop|
|         Blockbuster|
|                 Hit|
+--------------------+
```

In [48]:
```
s_boxf = s_boxf.withColumn("Verdict",when(col("Verdict")=="Super Hitt","Super Hit").otherwise(col("Verdict")))
```

In [49]:
```
s_boxf.select("Verdict").distinct().show()
```

```
+--------------------+
|             Verdict|
+--------------------+
|       Above Average|
|             Average|
|       Below Average|
|            Disaster|
|All Time Blockbuster|
|           Super Hit|
|                Flop|
|         Blockbuster|
|                 Hit|
+--------------------+
```

In [50]:
```
s_boxf.select("OTT Platform").distinct().show()
```

```
+------------------+
|      OTT Platform|
+------------------+
|           Sun Nxt|
|           Sonyliv|
|              Voot|
|Google Play Movies|
|         Jiocinema|
|              Zee5|
|Amazon Prime Video|
|   Disney+ Hotstar|
|               Aha|
|           Netflix|
+------------------+
```

In [51]:
```
s_boxf.select("LanguageID").distinct().show()
```

```
+----------+
|LanguageID|
+----------+
|       513|
|       501|
|       519|
|       507|
|       511|
|       505|
|       503|
|       509|
|       515|
|       517|
+----------+
```

In [52]:
```
s_boxf = s_boxf.replace({511:505,513:509,515:501,517:503,519:507},subset=["LanguageID"])
```

In [53]:
```
s_boxf.select("LanguageID").distinct().show()
```

```
+----------+
|LanguageID|
+----------+
|       501|
|       507|
|       505|
|       503|
|       509|
+----------+
```

In [54]:
```
s_boxf.printSchema()
```

```
root
 |-- FilmID: integer (nullable = true)
 |-- Title: string (nullable = true)
 |-- Release Date: string (nullable = true)
 |-- DirectorID: integer (nullable = true)
 |-- Lead Actor/Actress: string (nullable = true)
 |-- LanguageID: integer (nullable = true)
 |-- Industry: string (nullable = true)
 |-- GenreID: integer (nullable = true)
 |-- Budget in Crores: double (nullable = true)
 |-- First Day Collection Worldwide  in Crores: double (nullable = true)
 |-- Worldwide Collection  in Crores: double (nullable = true)
 |-- Overseas Collection in Crores: double (nullable = true)
 |-- India Gross Collection in Crores: double (nullable = true)
 |-- Verdict: string (nullable = true)
 |-- IMDb Rating: double (nullable = true)
 |-- Runtime (mins): integer (nullable = true)
 |-- OTT Platform: string (nullable = true)
```

In [55]:
```
# Converting PySpark datasets to Spark SQL datasets
s_boxf.createOrReplaceTempView("boxf_tb")
s_dir.createOrReplaceTempView("dir_tb")
s_gen.createOrReplaceTempView("gen_tb")
s_lan.createOrReplaceTempView("lan_tb")
```

1 Write a query to get Total films released ?

In [56]:
```
# With Pandas
p_boxf["Title"].nunique()
```

Out[56]:
```
600
```

In [57]:
```
# With PySpark
s_boxf.select("Title").distinct().count()
```

Out[57]:
```
600
```

In [58]:
```
# With Spark SQL
spark.sql("select count(Title) as Total_Films from boxf_tb").show()
```

```
+-----------+
|Total_Films|
+-----------+
|        604|
+-----------+
```

2 Write a query to get Total budget ?

In [59]:
```
# With Pandas
Total_Budget = p_boxf["Budget in Crores"].sum()
print("Total_Budget: ",Total_Budget)
```

```
Total_Budget:  34653.0
```

In [60]:
```
# With PySpark
Total_Budget = s_boxf.select("Budget in Crores").agg(sum("Budget in Crores").alias("Total_Budget")).show()
print("Total_Budget: ",Total_Budget)
```

```
+------------+
|Total_Budget|
+------------+
|     34653.0|
+------------+

Total_Budget:  None
```

In [61]:
```
# With Spark SQL
spark.sql("""select sum(`Budget in Crores`) as Total_Budget from boxf_tb""").show()
```

```
+------------+
|Total_Budget|
+------------+
|     34653.0|
+------------+
```

3 Write a query to get Total worldwide collection ?

In [62]:
```
# With Pandas
Total_WWC = p_boxf["Worldwide Collection  in Crores"].sum()
```

```
print("Total worldwide collection: ", Total_WWC)
```

```
Total worldwide collection:  79878.79
```

In [63]:
```
# WIith PySpark
s_boxf.select("Worldwide Collection  in Crores")\
    .agg(sum("Worldwide Collection  in Crores")\
        .alias("Total worldwide collection")).show()
```

```
+--------------------------+
|Total worldwide collection|
+--------------------------+
|         79878.78999999998|
+--------------------------+
```

In [64]:
```
# With Spark SQL
spark.sql("select sum(`Worldwide Collection  in Crores`)\
as Total_worldwide_collection from boxf_tb").show()
```

```
+--------------------------+
|Total_worldwide_collection|
+--------------------------+
|         79878.78999999998|
+--------------------------+
```

4 Write a query to get Total First day collection worldwide ?

In [65]:
```
# With Pandas
Total_FDCW = p_boxf["First Day Collection Worldwide  in Crores"].sum()
print("Total First day collection worldwide: ",Total_FDCW)
```

```
Total First day collection worldwide:  11070.27
```

In [66]:
```
# With PySpark
s_boxf.select("First Day Collection Worldwide  in Crores")\
    .agg(sum("First Day Collection Worldwide  in Crores")\
        .alias("Total First day collection worldwide")).show()
```

```
+-----------------------------------+
|Total First day collection worldwide|
+-----------------------------------+
|                  11070.270000000006|
+-----------------------------------+
```

In [67]:
```
# With Spark SQL
spark.sql("select sum(`First Day Collection Worldwide  in Crores`)\
as Total_First_day_collection_worldwide from boxf_tb").show()
```

```
+-----------------------------------+
|Total_First_day_collection_worldwide|
+-----------------------------------+
|                  11070.270000000006|
+-----------------------------------+
```

5 Write a query to get Total Overseas collection ?

In [68]:
```
# With Pandas
Total_OC = p_boxf["Overseas Collection in Crores"].sum()
print("Total Overseas collection:",Total_OC)
```

```
Total Overseas collection: 20953.07
```

In [69]:
```
# With PySpark
s_boxf.select("Overseas Collection in Crores")\
    .agg(sum("Overseas Collection in Crores")
        .alias("Total_Overseas_collection")).show()
```

```
+-------------------------+
|Total_Overseas_collection|
+-------------------------+
|        20953.07000000001|
+-------------------------+
```

In [70]:
```
# With Spark SQL
spark.sql("select sum(`Overseas Collection in Crores`)\
as Total_OVerseas_Collection from boxf_tb").show()
```

```
+-------------------------+
|Total_OVerseas_Collection|
+-------------------------+
|        20953.07000000001|
+-------------------------+
```

6 Write a query to get Total India Gross collection ?

In [71]:
```
# With Pandas
Total_IGC = p_boxf["India Gross Collection in Crores"].sum()
print("Total India Gross collection: ",Total_IGC)
```

```
Total India Gross collection:  60064.1
```

In [72]:
```
# With PySpark
s_boxf.select("India Gross Collection in Crores")\
    .agg(sum("India Gross Collection in Crores")\
        .alias("Total India Gross collection")).show()
```

```
+----------------------------+
|Total India Gross collection|
+----------------------------+
|           60064.10000000004|
+----------------------------+
```

In [73]:
```
# With Spark SQL
spark.sql("select sum(`India Gross Collection in Crores`)as \
Total_India_Gross_collection from boxf_tb").show()
```

```
+----------------------------+
|Total_India_Gross_collection|
+----------------------------+
|           60064.10000000004|
+----------------------------+
```

7. Top 10 filmsbased on world wide collections. Display films,collections

In [74]:
```
# With Pandas
p_boxf[['Title','First Day Collection Worldwide  in Crores',\
        'Worldwide Collection  in Crores',\
        'Overseas Collection in Crores',\
        'India Gross Collection in Crores']]\
    .sort_values("Worldwide Collection  in Crores",ascending = False)\
    .reset_index().head(10)
```

Out[74]:

	index	Title	First Day Collection Worldwide in Crores	Worldwide Collection in Crores	Overseas Collection in Crores	India Gross Collection in Crores
0	524	Dangal	58.0	2122.30	1535.30	587.00
1	396	Bāhubali 2: The Conclusion	217.0	1788.00	371.00	1417.00
2	45	RRR (Rise Roar Revolt)	223.0	1230.00	314.15	915.85
3	149	K.G.F: Chapter 2	159.0	1215.00	214.15	1000.85
4	572	Jawan	129.1	1160.00	400.00	760.00
5	573	Pathaan	104.8	1055.00	397.50	657.50
6	126	Kalki 2898-AD	177.7	1042.25	275.00	767.25
7	503	Bajrangi Bhaijaan	27.2	922.10	489.70	432.40
8	114	Animal	116.0	915.00	255.00	660.00
9	46	Secret Superstar	5.0	912.60	831.40	81.20

In [75]:
```
# With PySpark
s_boxf.select('Title','First Day Collection Worldwide  in Crores',\
        'Worldwide Collection  in Crores',\
        'Overseas Collection in Crores',\
        'India Gross Collection in Crores')\
    .sort(desc('Worldwide Collection  in Crores')).show(10,False)
```

```
+-------------------------+----------------------------------------+-------------------------------+----------
-------------------+-------------------------------+
|Title                    |First Day Collection Worldwide  in Crores|Worldwide Collection  in Crores|Overseas C
ollection in Crores|India Gross Collection in Crores|
+-------------------------+----------------------------------------+-------------------------------+----------
-------------------+-------------------------------+
|Dangal                   |58.0                                    |2122.3                         |1535.3
|587.0                          |
|Bãhubali 2: The Conclusion|217.0                                   |1788.0                         |371.0
|1417.0                         |
|RRR (Rise Roar Revolt)   |223.0                                   |1230.0                         |314.15
|915.85                         |
|K.G.F: Chapter 2         |159.0                                   |1215.0                         |214.15
|1000.85                        |
|Jawan                    |129.1                                   |1160.0                         |400.0
|760.0                          |
|Pathaan                  |104.8                                   |1055.0                         |397.5
|657.5                          |
|Kalki 2898-AD            |177.7                                   |1042.25                        |275.0
|767.25                         |
|Bajrangi Bhaijaan        |27.2                                    |922.1                          |489.7
|432.4                          |
|Animal                   |116.0                                   |915.0                          |255.0
|660.0                          |
|Secret Superstar         |5.0                                     |912.6                          |831.4
|81.2                           |
+-------------------------+----------------------------------------+-------------------------------+----------
-------------------+-------------------------------+
only showing top 10 rows
```

In [76]:
```
# With Spark SQL
spark.sql("select title,`First Day Collection Worldwide  in Crores`,\
`Worldwide Collection  in Crores`,`Overseas Collection in Crores`,\
`India Gross Collection in Crores` from boxf_tb \
order by `Worldwide Collection  in Crores` desc limit 10").show()
```

```
+--------------------+----------------------------------------+-------------------------------+----------------
-------------+-------------------------------+
|               title|First Day Collection Worldwide  in Crores|Worldwide Collection  in Crores|Overseas Collect
ion in Crores|India Gross Collection in Crores|
+--------------------+----------------------------------------+-------------------------------+----------------
-------------+-------------------------------+
|              Dangal|                                    58.0|                         2122.3|
1535.3|                          587.0|
|Bãhubali 2: The C...|                                   217.0|                         1788.0|
371.0|                         1417.0|
|RRR (Rise Roar Re...|                                   223.0|                         1230.0|
314.15|                         915.85|
|    K.G.F: Chapter 2|                                   159.0|                         1215.0|
214.15|                         1000.85|
|               Jawan|                                   129.1|                         1160.0|
400.0|                          760.0|
|             Pathaan|                                   104.8|                         1055.0|
397.5|                          657.5|
|       Kalki 2898-AD|                                   177.7|                        1042.25|
275.0|                         767.25|
|   Bajrangi Bhaijaan|                                    27.2|                          922.1|
489.7|                          432.4|
|              Animal|                                   116.0|                          915.0|
255.0|                          660.0|
|    Secret Superstar|                                     5.0|                          912.6|
831.4|                           81.2|
+--------------------+----------------------------------------+-------------------------------+----------------
-------------+-------------------------------+
```

8 Total Number of films released by year

In [77]:
```
p_boxf["Year"] = p_boxf["Release Date"].dt.year
```

In [78]:
```
p_boxf.groupby("Year").agg(Total_Films = ("FilmID","count")).reset_index()
```

Out[78]:

	Year	Total_Films
0	2014	32
1	2015	32
2	2016	41
3	2017	72
4	2018	71
5	2019	74
6	2020	27
7	2021	31
8	2022	80
9	2023	79
10	2024	65

In [79]:

```
s_boxf = s_boxf.withColumn(
    "Release Date",
    expr("try_to_timestamp(`Release Date`, 'dd-MMM-yy')")
)

s_boxf.withColumn("Year", year(col("Release Date"))) \
      .groupBy("Year") \
      .agg(count("FilmID").alias("Total_Films")) \
      .show()
```

```
+----+-----------+
|Year|Total_Films|
+----+-----------+
|2018|         71|
|2015|         32|
|2023|         79|
|2022|         80|
|2014|         32|
|2019|         74|
|2020|         27|
|2016|         41|
|2024|         65|
|2017|         72|
|2021|         31|
+----+-----------+
```

In [80]:

```
# With Spark SQL
spark.sql("SELECT YEAR(try_to_timestamp\
(`Release Date`, 'dd-MMM-yy')) AS Yr,\
COUNT(FilmID) AS Film_count \
FROM boxf_tb \
GROUP BY Yr \
ORDER BY Yr").show()
```

```
+----+----------+
|  Yr|Film_count|
+----+----------+
|2014|        32|
|2015|        32|
|2016|        41|
|2017|        72|
|2018|        71|
|2019|        74|
|2020|        27|
|2021|        31|
|2022|        80|
|2023|        79|
|2024|        65|
+----+----------+
```

9 Top 10 filmsbased on india collections.Display films,collections

In [81]:

```
# With Pandas
p_boxf[['Title','First Day Collection Worldwide  in Crores',\
        'Worldwide Collection  in Crores',\
        'Overseas Collection in Crores',\
        'India Gross Collection in Crores']]\
    .sort_values('India Gross Collection in Crores',ascending = False)\
    .reset_index().head(10)
```

Out[81]:

	index	Title	First Day Collection Worldwide in Crores	Worldwide Collection in Crores	Overseas Collection in Crores	India Gross Collection in Crores
0	396	Bãhubali 2: The Conclusion	217.0	1788.00	371.00	1417.00
1	149	K.G.F: Chapter 2	159.0	1215.00	214.15	1000.85
2	45	RRR (Rise Roar Revolt)	223.0	1230.00	314.15	915.85
3	126	Kalki 2898-AD	177.7	1042.25	275.00	767.25
4	572	Jawan	129.1	1160.00	400.00	760.00
5	127	Stree 2: Sarkate Ka Aatank	80.0	857.07	144.00	713.07
6	114	Animal	116.0	915.00	255.00	660.00
7	573	Pathaan	104.8	1055.00	397.50	657.50
8	115	Gadar 2	53.3	686.00	65.50	620.50
9	524	Dangal	58.0	2122.30	1535.30	587.00

In [82]:

```
# With PySpark
s_boxf.select('Title','First Day Collection Worldwide  in Crores',\
              'Worldwide Collection  in Crores',\
              'Overseas Collection in Crores',\
              'India Gross Collection in Crores')\
    .sort(desc('India Gross Collection in Crores')).show(10)
```

```
+--------------------+----------------------------------------+-----------------------------+-----------------
-------------+-------------------------------+
|               Title|First Day Collection Worldwide  in Crores|Worldwide Collection  in Crores|Overseas Collect
ion in Crores|India Gross Collection in Crores|
+--------------------+----------------------------------------+-----------------------------+-----------------
-------------+-------------------------------+
|Bãhubali 2: The C...|                                   217.0|                        1788.0|
371.0|                         1417.0|
|    K.G.F: Chapter 2|                                   159.0|                        1215.0|
214.15|                         1000.85|
|RRR (Rise Roar Re...|                                   223.0|                        1230.0|
314.15|                          915.85|
|       Kalki 2898-AD|                                   177.7|                       1042.25|
275.0|                          767.25|
|               Jawan|                                   129.1|                        1160.0|
400.0|                           760.0|
|Stree 2: Sarkate ...|                                    80.0|                        857.07|
144.0|                          713.07|
|              Animal|                                   116.0|                         915.0|
255.0|                           660.0|
|            Pathaan|                                   104.8|                        1055.0|
397.5|                           657.5|
|            Gadar 2|                                    53.3|                         686.0|
65.5|                          620.5|
|             Dangal|                                    58.0|                        2122.3|
1535.3|                           587.0|
+--------------------+----------------------------------------+-----------------------------+-----------------
-------------+-------------------------------+
only showing top 10 rows
```

In [83]:

```
# With Spark SQL
spark.sql("select FilmID,`First Day Collection Worldwide  in Crores`,\
`Worldwide Collection  in Crores`,`Overseas Collection in Crores`,\
`India Gross Collection in Crores` from boxf_tb \
order by `India Gross Collection in Crores` desc limit 10").show()
```

```
+------+----------------------------------------+-----------------------------+----------------------------+
--------------------------------+
|FilmID|First Day Collection Worldwide  in Crores|Worldwide Collection  in Crores|Overseas Collection in Crores|
India Gross Collection in Crores|
+------+----------------------------------------+-----------------------------+----------------------------+
--------------------------------+
|   104|                                   217.0|                       1788.0|                       371.0|
1417.0|
|   287|                                   159.0|                       1215.0|                      214.15|
1000.85|
|   286|                                   223.0|                       1230.0|                      314.15|
915.85|
|   540|                                   177.7|                      1042.25|                       275.0|
767.25|
|   463|                                   129.1|                       1160.0|                       400.0|
760.0|
|   541|                                    80.0|                       857.07|                       144.0|
713.07|
|   465|                                   116.0|                        915.0|                       255.0|
660.0|
|   464|                                   104.8|                       1055.0|                       397.5|
657.5|
|   466|                                    53.3|                        686.0|                        65.5|
620.5|
|   213|                                    58.0|                       2122.3|                      1535.3|
587.0|
+------+----------------------------------------+-----------------------------+----------------------------+
--------------------------------+
```

10 Top 10 filmsbased on overses collections.Display films,collections

In [84]:
```
# With Pandas
p_boxf[['Title','First Day Collection Worldwide  in Crores',\
        'Worldwide Collection  in Crores',\
        'Overseas Collection in Crores',\
        'India Gross Collection in Crores']]\
    .sort_values('Overseas Collection in Crores',ascending = False)\
    .reset_index().head(10)
```

Out[84]:

	index	Title	First Day Collection Worldwide in Crores	Worldwide Collection in Crores	Overseas Collection in Crores	India Gross Collection in Crores
0	524	Dangal	58.0	2122.30	1535.30	587.00
1	46	Secret Superstar	5.0	912.60	831.40	81.20
2	503	Bajrangi Bhaijaan	27.2	922.10	489.70	432.40
3	572	Jawan	129.1	1160.00	400.00	760.00
4	573	Pathaan	104.8	1055.00	397.50	657.50
5	396	Bāhubali 2: The Conclusion	217.0	1788.00	371.00	1417.00
6	45	RRR (Rise Roar Revolt)	223.0	1230.00	314.15	915.85
7	460	PK	26.6	792.00	303.00	489.00
8	126	Kalki 2898-AD	177.7	1042.25	275.00	767.25
9	114	Animal	116.0	915.00	255.00	660.00

In [85]:
```
# With PySpark
s_boxf.select('Title','First Day Collection Worldwide  in Crores',\
              'Worldwide Collection  in Crores',\
              'Overseas Collection in Crores',\
              'India Gross Collection in Crores')\
    .sort(desc('Overseas Collection in Crores')).show(10)
```

```
+--------------------+----------------------------------------+-----------------------------+-----------------
-------------+-------------------------------+
|               Title|First Day Collection Worldwide  in Crores|Worldwide Collection  in Crores|Overseas Collect
ion in Crores|India Gross Collection in Crores|
+--------------------+----------------------------------------+-----------------------------+-----------------
-------------+-------------------------------+
|              Dangal|                                    58.0|                        2122.3|
1535.3|                           587.0|
|    Secret Superstar|                                     5.0|                         912.6|
831.4|                            81.2|
|   Bajrangi Bhaijaan|                                    27.2|                         922.1|
489.7|                           432.4|
|               Jawan|                                   129.1|                        1160.0|
400.0|                           760.0|
|             Pathaan|                                   104.8|                        1055.0|
397.5|                           657.5|
|Bähubali 2: The C...|                                   217.0|                        1788.0|
371.0|                          1417.0|
|RRR (Rise Roar Re...|                                   223.0|                        1230.0|
314.15|                          915.85|
|                  PK|                                    26.6|                         792.0|
303.0|                           489.0|
|       Kalki 2898-AD|                                   177.7|                       1042.25|
275.0|                          767.25|
|              Animal|                                   116.0|                         915.0|
255.0|                           660.0|
+--------------------+----------------------------------------+-----------------------------+-----------------
-------------+-------------------------------+
only showing top 10 rows
```

In [86]:
```python
# With Spark SQL
spark.sql("select FilmID,`First Day Collection Worldwide  in Crores`,\
`Worldwide Collection  in Crores`,`Overseas Collection in Crores`,\
`India Gross Collection in Crores` from boxf_tb \
order by `Overseas Collection in Crores` desc limit 10").show()
```

```
+------+----------------------------------------+-----------------------------+----------------------------+
--------------------------------+
|FilmID|First Day Collection Worldwide  in Crores|Worldwide Collection  in Crores|Overseas Collection in Crores|
India Gross Collection in Crores|
+------+----------------------------------------+-----------------------------+----------------------------+
--------------------------------+
|   213|                                    58.0|                        2122.3|                      1535.3|
587.0|
|   105|                                     5.0|                         912.6|                       831.4|
81.2|
|   181|                                    27.2|                         922.1|                       489.7|
432.4|
|   463|                                   129.1|                        1160.0|                       400.0|
760.0|
|   464|                                   104.8|                        1055.0|                       397.5|
657.5|
|   104|                                   217.0|                        1788.0|                       371.0|
1417.0|
|   286|                                   223.0|                        1230.0|                      314.15|
915.85|
|     1|                                    26.6|                         792.0|                       303.0|
489.0|
|   540|                                   177.7|                       1042.25|                       275.0|
767.25|
|   465|                                   116.0|                         915.0|                       255.0|
660.0|
+------+----------------------------------------+-----------------------------+----------------------------+
--------------------------------+
```

11 Top 10 filmsbased on firstday collections.Display films,collections

In [87]:
```python
# With Pandas
p_boxf[['Title','First Day Collection Worldwide  in Crores',\
        'Worldwide Collection  in Crores',\
        'Overseas Collection in Crores',\
        'India Gross Collection in Crores']]\
    .sort_values('First Day Collection Worldwide  in Crores',ascending = False)\
    .reset_index().head(10)
```

Out[87]:

	index	Title	First Day Collection Worldwide in Crores	Worldwide Collection in Crores	Overseas Collection in Crores	India Gross Collection in Crores
0	45	RRR (Rise Roar Revolt)	223.0	1230.00	314.15	915.85
1	396	Bãhubali 2: The Conclusion	217.0	1788.00	371.00	1417.00
2	126	Kalki 2898-AD	177.7	1042.25	275.00	767.25
3	149	K.G.F: Chapter 2	159.0	1215.00	214.15	1000.85
4	212	Salaar: Part 1 - Ceasefire	158.1	617.75	130.00	487.75
5	116	Leo	142.7	605.90	204.00	401.90
6	455	Devara: Part 1	142.0	421.60	78.85	342.75
7	572	Jawan	129.1	1160.00	400.00	760.00
8	213	Adipurush	127.5	393.00	50.00	343.00
9	114	Animal	116.0	915.00	255.00	660.00

In [88]:

```
# With PySpark
s_boxf.select('Title','First Day Collection Worldwide  in Crores',\
              'Worldwide Collection  in Crores',\
              'Overseas Collection in Crores',\
              'India Gross Collection in Crores')\
    .sort(desc('First Day Collection Worldwide  in Crores')).show(10)
```

```
+--------------------+----------------------------------------+------------------------------+-----------------
-------------+------------------------------+
|               Title|First Day Collection Worldwide  in Crores|Worldwide Collection  in Crores|Overseas Collect
ion in Crores|India Gross Collection in Crores|
+--------------------+----------------------------------------+------------------------------+-----------------
-------------+------------------------------+
|RRR (Rise Roar Re...|                                   223.0|                        1230.0|
314.15|                         915.85|
|Bãhubali 2: The C...|                                   217.0|                        1788.0|
371.0|                        1417.0|
|       Kalki 2898-AD|                                   177.7|                       1042.25|
275.0|                        767.25|
|    K.G.F: Chapter 2|                                   159.0|                        1215.0|
214.15|                       1000.85|
|Salaar: Part 1 - ...|                                   158.1|                        617.75|
130.0|                        487.75|
|                 Leo|                                   142.7|                         605.9|
204.0|                         401.9|
|      Devara: Part 1|                                   142.0|                         421.6|
78.85|                        342.75|
|               Jawan|                                   129.1|                        1160.0|
400.0|                         760.0|
|           Adipurush|                                   127.5|                         393.0|
50.0|                        343.0|
|              Animal|                                   116.0|                         915.0|
255.0|                         660.0|
+--------------------+----------------------------------------+------------------------------+-----------------
-------------+------------------------------+
only showing top 10 rows
```

In [89]:

```
# With Spark SQL
spark.sql("select FilmID,`First Day Collection Worldwide  in Crores`,\
`Worldwide Collection  in Crores`,`Overseas Collection in Crores`,\
`India Gross Collection in Crores` from boxf_tb \
order by `First Day Collection Worldwide  in Crores` desc limit 10").show()
```

```
+------+----------------------------------------+-----------------------------+----------------------------+
--------------------------------+
|FilmID|First Day Collection Worldwide  in Crores|Worldwide Collection  in Crores|Overseas Collection in Crores|
India Gross Collection in Crores|
+------+----------------------------------------+-----------------------------+----------------------------+
--------------------------------+
|   286|                                   223.0|                       1230.0|                      314.15|
915.85|
|   104|                                   217.0|                       1788.0|                       371.0|
1417.0|
|   540|                                   177.7|                      1042.25|                       275.0|
767.25|
|   287|                                   159.0|                       1215.0|                      214.15|
1000.85|
|   467|                                   158.1|                       617.75|                       130.0|
487.75|
|   468|                                   142.7|                        605.9|                       204.0|
401.9|
|   543|                                   142.0|                        421.6|                       78.85|
342.75|
|   463|                                   129.1|                       1160.0|                       400.0|
760.0|
|   472|                                   127.5|                        393.0|                        50.0|
343.0|
|   465|                                   116.0|                        915.0|                       255.0|
660.0|
+------+----------------------------------------+-----------------------------+----------------------------+
--------------------------------+
```

12 Weekday wise films released,Display week name and no of films released

In [90]:
```
# exracting the week days from date
p_boxf["WeekDay"] = p_boxf["Release Date"].dt.strftime("%A")
```

In [91]:
```
# With Pandas
p_boxf.groupby("WeekDay").agg(Total_Film=("FilmID","count"))\
    .sort_values("Total_Film",ascending = False).reset_index()
```

Out[91]:

	WeekDay	Total_Film
0	Friday	423
1	Thursday	124
2	Wednesday	35
3	Saturday	10
4	Sunday	7
5	Tuesday	5

In [92]:
```
# extracting the week days from date
s_boxf = s_boxf.withColumn("WeekDay", date_format("Release Date", "EEEE"))
```

In [93]:
```
# With PySpark
s_boxf.groupby("WeekDay").agg(count("FilmID")\
    .alias("Total_Films"))\
    .orderBy(desc("Total_Films")).show()
```

```
+---------+-----------+
|  WeekDay|Total_Films|
+---------+-----------+
|   Friday|        423|
| Thursday|        124|
|Wednesday|         35|
| Saturday|         10|
|   Sunday|          7|
|  Tuesday|          5|
+---------+-----------+
```

In [94]:
```
# With Spark SQL
spark.sql("""Select date_format(to_date(`Release Date`,'dd-MMM-yy'),\
'EEEE')as WeekDay, count(FilmID) as Total_Films from boxf_tb \
group by WeekDay \
order by 2 desc""").show()
```

```
+---------+-----------+
|  WeekDay|Total_Films|
+---------+-----------+
|   Friday|        423|
| Thursday|        124|
|Wednesday|         35|
| Saturday|         10|
|   Sunday|          7|
|  Tuesday|          5|
+---------+-----------+
```

13 Write a query to get OTT platofrm wise movies count

In [95]:
```
# With Pandas
p_boxf.groupby("OTT Platform")[['FilmID']].count()\
    .sort_values("FilmID",ascending = False).reset_index()
```

Out[95]:

	OTT Platform	FilmID
0	Amazon Prime Video	384
1	Netflix	99
2	Disney+ Hotstar	54
3	ZEE5	39
4	Sun NXT	11
5	Aha	8
6	SonyLIV	4
7	JioCinema	3
8	Google Play Movies	1
9	Voot	1

In [96]:
```
# With PySpark
s_boxf.groupby("OTT Platform")\
    .agg(count("FilmID")\
        .alias("Film_count"))\
    .sort(desc("Film_count")).show()
```

```
+------------------+----------+
|      OTT Platform|Film_count|
+------------------+----------+
|Amazon Prime Video|       384|
|           Netflix|        99|
|   Disney+ Hotstar|        54|
|              Zee5|        39|
|           Sun Nxt|        11|
|               Aha|         8|
|           Sonyliv|         4|
|         Jiocinema|         3|
|              Voot|         1|
|Google Play Movies|         1|
+------------------+----------+
```

In [97]:
```
# With Spark SQL
spark.sql("select `OTT platform`,\
count(FilmID) as Count_film from boxf_tb \
group by `OTT Platform`\
order by Count_film desc").show()
```

```
+------------------+----------+
|      OTT platform|Count_film|
+------------------+----------+
|Amazon Prime Video|       384|
|           Netflix|        99|
|   Disney+ Hotstar|        54|
|              Zee5|        39|
|           Sun Nxt|        11|
|               Aha|         8|
|           Sonyliv|         4|
|         Jiocinema|         3|
|              Voot|         1|
|Google Play Movies|         1|
+------------------+----------+
```

14 Top 10 Directors by films released

In [98]:
```
# With Pandas
p_boxf.merge(p_dir,left_on ="DirectorID", right_on = "Director ID", how = "right")\
    .groupby("Director")[["FilmID"]].count().sort_values("FilmID",ascending = False)\
    .reset_index().head(10)
```

Out[98]:

	Director	FilmID
0	Trivikram Srinivas	6
1	Rohit Shetty	6
2	A.R. Murugadoss	5
3	Siva	5
4	Anil Ravipudi	5
5	Boyapati Srinu	5
6	Koratala Siva	5
7	Remo D'Souza	4
8	Atlee	4
9	Siddharth Anand	4

In [99]:
```
# With PySpark
s_boxf.join(s_dir, s_boxf["DirectorID"]==s_dir["Director ID"],"right")\
    .groupby("Director").agg(count("FilmID").alias("Film_Count"))\
    .sort(desc("Film_Count")).show(10)
```

```
+------------------+----------+
|          Director|Film_Count|
+------------------+----------+
|Trivikram Srinivas|         6|
|      Rohit Shetty|         6|
|     Anil Ravipudi|         5|
|     Koratala Siva|         5|
|   A.R. Murugadoss|         5|
|    Boyapati Srinu|         5|
|              Siva|         5|
|    Maruthi Dasari|         4|
|         Sundar C.|         4|
|   Siddharth Anand|         4|
+------------------+----------+
only showing top 10 rows
```

In [100…
```
# With Spark SQL
spark.sql("select Director, count(FilmID) as Film_Count \
from boxf_tb boxf \
right join dir_tb dir on dir.`Director ID`=boxf.DirectorID \
group by Director \
order by Film_Count desc limit 10").show()
```

```
+------------------+----------+
|          Director|Film_Count|
+------------------+----------+
|Trivikram Srinivas|         6|
|      Rohit Shetty|         6|
|     Anil Ravipudi|         5|
|     Koratala Siva|         5|
|   A.R. Murugadoss|         5|
|    Boyapati Srinu|         5|
|              Siva|         5|
|    Maruthi Dasari|         4|
|  Lokesh Kanagaraj|         4|
|   Siddharth Anand|         4|
+------------------+----------+
```

15 Top 10 directors by world wide collection

In [101…
```
# With Pandas
p_boxf.merge(p_dir, left_on = "DirectorID", right_on = "Director ID",how= "right")\
    .groupby("Director").agg(Total_WWC=("Worldwide Collection  in Crores","sum"))\
    .sort_values("Total_WWC",ascending = False).reset_index().head(10)
```

Out[101…

	Director	Total_WWC
0	S.S. Rajamouli	3668.00
1	Siddharth Anand	2229.45
2	Nitesh Tiwari	2122.30
3	Prashanth Neel	2101.75
4	Atlee	1865.35
5	Rajkumar Hirani	1834.50
6	Rohit Shetty	1656.60
7	Lokesh Kanagaraj	1350.13
8	Sandeep Reddy Vanga	1342.00
9	Kabir Khan	1220.55

In [102…

```
# With PySpark
s_boxf.join(s_dir, s_boxf["DirectorID"]==s_dir["Director ID"],"right")\
    .groupby("Director").agg(sum("Worldwide Collection  in Crores")\
        .alias("Total_WWC")).sort(desc("Total_WWC")).show(10)
```

```
+-------------------+------------------+
|           Director|         Total_WWC|
+-------------------+------------------+
|     S.S. Rajamouli|            3668.0|
|    Siddharth Anand|           2229.45|
|      Nitesh Tiwari|            2122.3|
|     Prashanth Neel|           2101.75|
|              Atlee|           1865.35|
|    Rajkumar Hirani|            1834.5|
|       Rohit Shetty|            1656.6|
|   Lokesh Kanagaraj|1350.1299999999999|
|Sandeep Reddy Vanga|            1342.0|
|         Kabir Khan|           1220.55|
+-------------------+------------------+
only showing top 10 rows
```

In [103…

```
# With Spark SQL
spark.sql("select Director,sum(`Worldwide Collection  in Crores`) as Total_WWC \
from boxf_tb boxf \
right join dir_tb dir on dir.`Director ID` = boxf.DirectorID \
group by Director \
order by Total_WWC desc limit 10 ").show()
```

```
+-------------------+------------------+
|           Director|         Total_WWC|
+-------------------+------------------+
|     S.S. Rajamouli|            3668.0|
|    Siddharth Anand|           2229.45|
|      Nitesh Tiwari|            2122.3|
|     Prashanth Neel|           2101.75|
|              Atlee|           1865.35|
|    Rajkumar Hirani|            1834.5|
|       Rohit Shetty|            1656.6|
|   Lokesh Kanagaraj|1350.1299999999999|
|Sandeep Reddy Vanga|            1342.0|
|         Kabir Khan|           1220.55|
+-------------------+------------------+
```

16 Top10 lead actors by world wide collection

In [106…

```
# With Pandas
p_boxf.groupby("Lead Actor/Actress")\
    .agg(Total_WWC=('Worldwide Collection  in Crores','sum'))\
    .sort_values("Total_WWC", ascending = False)\
    .reset_index().head(10)
```

Out[106…

	Lead Actor/Actress	Total_WWC
0	Prabhas	5091.50
1	Salman Khan	4515.20
2	Shah Rukh Khan	3800.10
3	Akshay Kumar	3168.22
4	Joseph Vijay	3076.11
5	Aamir Khan	2914.30
6	Rajinikanth	2695.55
7	Ajay Devgn	2571.21
8	N.T. Rama Rao Jr.	2242.60
9	Ranbir Kapoor	2111.25

In [107…

```
# With PySpark
s_boxf.groupby("Lead Actor/Actress").agg(sum("Worldwide Collection  in Crores")\
    .alias("Total_WWC")).sort(desc("Total_WWC")).show(10)
```

```
+------------------+------------------+
|Lead Actor/Actress|         Total_WWC|
+------------------+------------------+
|           Prabhas|            5091.5|
|       Salman Khan|            4515.2|
|    Shah Rukh Khan|3800.1000000000004|
|      Akshay Kumar|           3168.22|
|      Joseph Vijay|3076.1099999999997|
|        Aamir Khan|            2914.3|
|       Rajinikanth|           2695.55|
|        Ajay Devgn|2571.2100000000005|
| N.T. Rama Rao Jr.|            2242.6|
|     Ranbir Kapoor|           2111.25|
+------------------+------------------+
only showing top 10 rows
```

In [108…

```
# With Spark SQL
spark.sql("select `Lead Actor/Actress`,\
sum(`Worldwide Collection  in Crores`) as Total_WWC \
from boxf_tb \
group by `Lead Actor/Actress` \
order by Total_WWC desc limit 10").show()
```

```
+------------------+------------------+
|Lead Actor/Actress|         Total_WWC|
+------------------+------------------+
|           Prabhas|            5091.5|
|       Salman Khan|            4515.2|
|    Shah Rukh Khan|3800.1000000000004|
|      Akshay Kumar|           3168.22|
|      Joseph Vijay|3076.1099999999997|
|        Aamir Khan|            2914.3|
|       Rajinikanth|           2695.55|
|        Ajay Devgn|2571.2100000000005|
| N.T. Rama Rao Jr.|            2242.6|
|     Ranbir Kapoor|           2111.25|
+------------------+------------------+
```

17 Top 10 movies by IMDb rating

In [109…

```
# With Pandas
p_boxf[["Title","IMDb Rating"]]\
    .sort_values("IMDb Rating",ascending=False)\
    .reset_index().head(10)
```

Out[109…

	index	Title	IMDb Rating
0	121	12th Fail	8.8
1	316	Rocketry: The Nambi Effect	8.7
2	83	777 Charlie	8.7
3	140	Kishkindha Kaandam	8.6
4	79	The Kashmir Files	8.6
5	437	Jersey	8.5
6	25	96	8.5
7	512	Sachin	8.5
8	350	Meiyazhagan	8.5
9	204	Kumbalangi Nights	8.5

In [110…

```
# With PySpark
s_boxf.select("Title","IMDb Rating")\
    .sort(desc("IMDb Rating")).show(10)
```

```
+--------------------+-----------+
|               Title|IMDb Rating|
+--------------------+-----------+
|           12th Fail|        8.8|
|         777 Charlie|        8.7|
|Rocketry: The Nam...|        8.7|
|   The Kashmir Files|        8.6|
|  Kishkindha Kaandam|        8.6|
|          Sita Ramam|        8.5|
|                  96|        8.5|
|            Maharaja|        8.5|
|   Kumbalangi Nights|        8.5|
|         Meiyazhagan|        8.5|
+--------------------+-----------+
only showing top 10 rows
```

In [111…

```
# With Spark SQL
spark.sql("select Title, `IMDb Rating` from boxf_tb \
order by `IMDb Rating` desc limit 10").show()
```

```
+--------------------+-----------+
|               Title|IMDb Rating|
+--------------------+-----------+
|           12th Fail|        8.8|
|         777 Charlie|        8.7|
|Rocketry: The Nam...|        8.7|
|   The Kashmir Files|        8.6|
|  Kishkindha Kaandam|        8.6|
|                  96|        8.5|
|          Sita Ramam|        8.5|
|   Kumbalangi Nights|        8.5|
|            Maharaja|        8.5|
|         Meiyazhagan|        8.5|
+--------------------+-----------+
```

18 Bottom 10 movies by IMDb rating

In [112…

```
# With Pandas
p_boxf[["Title","IMDb Rating"]]\
    .sort_values("IMDb Rating",ascending = True)\
    .reset_index().head(10)
```

Out[112…

	index	Title	IMDb Rating
0	468	Race 3	1.9
1	491	Baaghi 3	2.2
2	571	Student of the Year 2	2.2
3	496	Heropanti 2	2.3
4	338	Chandramukhi 2	2.6
5	417	Liger	2.6
6	478	Gunday	2.7
7	213	Adipurush	2.7
8	565	Dabangg 3	3.0
9	495	A Flying Jatt	3.1

In [113…

```
# With PySpark
s_boxf.select("Title","IMDb Rating")\
    .sort(asc("IMDb Rating")).show(10)
```

```
+--------------------+-----------+
|               Title|IMDb Rating|
+--------------------+-----------+
|              Race 3|        1.9|
|            Baaghi 3|        2.2|
|Student of the Ye...|        2.2|
|         Heropanti 2|        2.3|
|      Chandramukhi 2|        2.6|
|               Liger|        2.6|
|           Adipurush|        2.7|
|              Gunday|        2.7|
|           Dabangg 3|        3.0|
|       A Flying Jatt|        3.1|
+--------------------+-----------+
only showing top 10 rows
```

In [114…

```
# With Spark SQl
spark.sql("Select Title, `IMDb Rating` from boxf_tb \
order by `IMDb Rating` asc limit 10").show()
```

```
+--------------------+-----------+
|               Title|IMDb Rating|
+--------------------+-----------+
|              Race 3|        1.9|
|            Baaghi 3|        2.2|
|Student of the Ye...|        2.2|
|         Heropanti 2|        2.3|
|      Chandramukhi 2|        2.6|
|               Liger|        2.6|
|           Adipurush|        2.7|
|              Gunday|        2.7|
|           Dabangg 3|        3.0|
|       A Flying Jatt|        3.1|
+--------------------+-----------+
```

19 Write a query to get 5 longest run time movies

In [115…

```
# With Pandas
p_boxf[["Title","Runtime (mins)"]]\
    .sort_values("Runtime (mins)",ascending = False)\
    .reset_index().head(5)
```

Out[115…

	index	Title	Runtime (mins)
0	114	Animal	204
1	39	I	188
2	45	RRR (Rise Roar Revolt)	187
3	103	Avane Srimannarayana	186
4	261	Jilla	185

In [116…

```
# With PySpark
s_boxf.select("Title","Runtime (mins)")\
    .sort(desc("Runtime (mins)")).show(5)
```

```
+--------------------+--------------+
|               Title|Runtime (mins)|
+--------------------+--------------+
|              Animal|           204|
|                   I|           188|
|RRR (Rise Roar Re...|           187|
|Avane Srimannarayana|           186|
|         Kurukshetra|           185|
+--------------------+--------------+
only showing top 5 rows
```

In [117…

```python
# With Spark SQL
spark.sql("select Title, `Runtime (mins)` from boxf_tb \
order by `Runtime (mins)` desc limit 5").show()
```

```
+--------------------+--------------+
|               Title|Runtime (mins)|
+--------------------+--------------+
|              Animal|           204|
|                   I|           188|
|RRR (Rise Roar Re...|           187|
|Avane Srimannarayana|           186|
|               Jilla|           185|
+--------------------+--------------+
```

20 Write a query to get 5 shortest run time movies

In [118…

```python
# With Pandas
p_boxf[["Title","Runtime (mins)"]]\
    .sort_values("Runtime (mins)",ascending = True)\
    .reset_index().head(5)
```

Out[118…

	index	Title	Runtime (mins)
0	599	Kill	105
1	497	Raksha Bandhan	108
2	560	Bhoot: Part One - The Haunted Ship	114
3	477	Hichki	116
4	407	The Ghazi Attack	116

In [119…

```python
# With PySpark
s_boxf.select("Title","Runtime (mins)")\
    .sort(asc("Runtime (mins)")).show(5)
```

```
+--------------------+--------------+
|               Title|Runtime (mins)|
+--------------------+--------------+
|                Kill|           105|
|      Raksha Bandhan|           108|
|Bhoot: Part One -...|           114|
|    The Ghazi Attack|           116|
|                Guru|           116|
+--------------------+--------------+
only showing top 5 rows
```

In [120…

```python
# With Spark SQL
spark.sql("select Title,`Runtime (mins)` from boxf_tb \
order by `Runtime (mins)` asc limit 5").show()
```

```
+--------------------+--------------+
|               Title|Runtime (mins)|
+--------------------+--------------+
|                Kill|           105|
|      Raksha Bandhan|           108|
|Bhoot: Part One -...|           114|
|                Guru|           116|
|              Hichki|           116|
+--------------------+--------------+
```

21 Top7 movies by world wide collection in Bollywood

In [121…

```python
# With Pandas
p_boxf.query("Industry=='Bollywood'").groupby(["Industry","Title"])\
    .agg(Total_WWC = ("Worldwide Collection  in Crores","sum"))\
    .sort_values("Total_WWC",ascending = False).reset_index().head(7)
```

Out[121…

	Industry	Title	Total_WWC
0	Bollywood	Dangal	2122.30
1	Bollywood	Jawan	1160.00
2	Bollywood	Pathaan	1055.00
3	Bollywood	Bajrangi Bhaijaan	922.10
4	Bollywood	Animal	915.00
5	Bollywood	Secret Superstar	912.60
6	Bollywood	Stree 2: Sarkate Ka Aatank	857.07

In [122…

```
# With PySpark
s_boxf.filter(col("Industry")=="Bollywood").groupby("Industry","Title")\
    .agg(sum("Worldwide Collection  in Crores")\
    .alias("Total_WWC")).sort(desc("Total_WWC")).show(7)
```

```
+---------+--------------------+---------+
| Industry|               Title|Total_WWC|
+---------+--------------------+---------+
|Bollywood|              Dangal|   2122.3|
|Bollywood|               Jawan|   1160.0|
|Bollywood|             Pathaan|   1055.0|
|Bollywood|   Bajrangi Bhaijaan|    922.1|
|Bollywood|              Animal|    915.0|
|Bollywood|    Secret Superstar|    912.6|
|Bollywood|Stree 2: Sarkate ...|   857.07|
+---------+--------------------+---------+
only showing top 7 rows
```

In [123…

```
# With Spark SQL
spark.sql("select Industry,Title, sum(`Worldwide Collection  in Crores`) \
as Total_WWC from boxf_tb \
Where Industry = 'Bollywood' \
group by Industry,Title \
order by Total_WWC desc limit 7").show()
```

```
+---------+--------------------+---------+
| Industry|               Title|Total_WWC|
+---------+--------------------+---------+
|Bollywood|              Dangal|   2122.3|
|Bollywood|               Jawan|   1160.0|
|Bollywood|             Pathaan|   1055.0|
|Bollywood|   Bajrangi Bhaijaan|    922.1|
|Bollywood|              Animal|    915.0|
|Bollywood|    Secret Superstar|    912.6|
|Bollywood|Stree 2: Sarkate ...|   857.07|
+---------+--------------------+---------+
```

22 Top7 movies by world wide collection in Tollywood

In [124…

```
# With Pandas
p_boxf.query("Industry=='Tollywood'").groupby(["Industry","Title"])\
    .agg(Total_WWC = ("Worldwide Collection  in Crores","sum"))\
    .sort_values("Total_WWC",ascending = False).reset_index().head(7)
```

Out[124…

	Industry	Title	Total_WWC
0	Tollywood	Bāhubali 2: The Conclusion	1788.00
1	Tollywood	RRR (Rise Roar Revolt)	1230.00
2	Tollywood	Kalki 2898-AD	1042.25
3	Tollywood	Bāhubali: The Beginning	650.00
4	Tollywood	Salaar: Part 1 - Ceasefire	617.75
5	Tollywood	Saaho	451.00
6	Tollywood	Devara: Part 1	421.60

In [125…

```
# With PySpark
s_boxf.filter(col("Industry")=="Tollywood").groupby("Industry","Title")\
    .agg(sum("Worldwide Collection  in Crores")\
    .alias("Total_WWC")).sort(desc("Total_WWC")).show(7)
```

```
+---------+--------------------+---------+
| Industry|               Title|Total_WWC|
+---------+--------------------+---------+
|Tollywood|Bãhubali 2: The C...|   1788.0|
|Tollywood|RRR (Rise Roar Re...|   1230.0|
|Tollywood|       Kalki 2898-AD|  1042.25|
|Tollywood|Bãhubali: The Beg...|    650.0|
|Tollywood|Salaar: Part 1 - ...|   617.75|
|Tollywood|               Saaho|    451.0|
|Tollywood|      Devara: Part 1|    421.6|
+---------+--------------------+---------+
only showing top 7 rows
```

In [126…

```python
# With Spark SQL
spark.sql("select Industry,Title, sum(`Worldwide Collection  in Crores`) \
as Total_WWC from boxf_tb \
Where Industry = 'Tollywood' \
group by Industry,Title \
order by Total_WWC desc limit 7").show()
```

```
+---------+--------------------+---------+
| Industry|               Title|Total_WWC|
+---------+--------------------+---------+
|Tollywood|Bãhubali 2: The C...|   1788.0|
|Tollywood|RRR (Rise Roar Re...|   1230.0|
|Tollywood|       Kalki 2898-AD|  1042.25|
|Tollywood|Bãhubali: The Beg...|    650.0|
|Tollywood|Salaar: Part 1 - ...|   617.75|
|Tollywood|               Saaho|    451.0|
|Tollywood|      Devara: Part 1|    421.6|
+---------+--------------------+---------+
```

23 Top7 movies by world wide collection in Kollywood

In [127…

```python
# With Pandas
p_boxf.query("Industry=='Kollywood'").groupby(["Industry","Title"])\
    .agg(Total_WWC = ("Worldwide Collection  in Crores","sum"))\
    .sort_values("Total_WWC",ascending = False).reset_index().head(7)
```

Out[127…

	Industry	Title	Total_WWC
0	Kollywood	2	701.00
1	Kollywood	Leo	605.90
2	Kollywood	Jailer	604.50
3	Kollywood	Ponniyin Selvan: Part One	488.36
4	Kollywood	The GOAT	451.23
5	Kollywood	Vikram	414.43
6	Kollywood	Ponniyin Selvan: Part Two	344.63

In [128…

```python
# With PySpark
s_boxf.filter(col("Industry")=="Kollywood").groupby("Industry","Title")\
    .agg(sum("Worldwide Collection  in Crores")\
    .alias("Total_WWC")).sort(desc("Total_WWC")).show(7)
```

```
+---------+--------------------+---------+
| Industry|               Title|Total_WWC|
+---------+--------------------+---------+
|Kollywood|                   2|    701.0|
|Kollywood|                 Leo|    605.9|
|Kollywood|              Jailer|    604.5|
|Kollywood|Ponniyin Selvan: ...|   488.36|
|Kollywood|            The GOAT|   451.23|
|Kollywood|              Vikram|   414.43|
|Kollywood|Ponniyin Selvan: ...|   344.63|
+---------+--------------------+---------+
only showing top 7 rows
```

In [129…

```python
# With Spark SQL
spark.sql("select Industry,Title, sum(`Worldwide Collection  in Crores`) \
as Total_WWC from boxf_tb \
Where Industry = 'Kollywood' \
group by Industry,Title \
order by Total_WWC desc limit 7").show()
```

```
+---------+--------------------+---------+
| Industry|               Title|Total_WWC|
+---------+--------------------+---------+
|Kollywood|                   2|    701.0|
|Kollywood|                 Leo|    605.9|
|Kollywood|              Jailer|    604.5|
|Kollywood|Ponniyin Selvan: ...|   488.36|
|Kollywood|            The GOAT|   451.23|
|Kollywood|              Vikram|   414.43|
|Kollywood|Ponniyin Selvan: ...|   344.63|
+---------+--------------------+---------+
```

24 Top7 movies by world wide collection in Sandalwood

In [130…

```python
# With Pandas
p_boxf.query("Industry=='Sandalwood'").groupby(["Industry","Title"])\
    .agg(Total_WWC = ("Worldwide Collection  in Crores","sum"))\
    .sort_values("Total_WWC",ascending = False).reset_index().head(7)
```

Out[130…

	Industry	Title	Total_WWC
0	Sandalwood	K.G.F: Chapter 2	1215.00
1	Sandalwood	Kantara	407.82
2	Sandalwood	K.G.F: Chapter 1	238.00
3	Sandalwood	777 Charlie	102.75
4	Sandalwood	VR (Vikrant Rona)	100.35
5	Sandalwood	James	94.20
6	Sandalwood	Kurukshetra	81.00

In [131…

```python
# With PySpark
s_boxf.filter(col("Industry")=="Sandalwood").groupby("Industry","Title")\
    .agg(sum("Worldwide Collection  in Crores")\
    .alias("Total_WWC")).sort(desc("Total_WWC")).show(7)
```

```
+----------+-----------------+---------+
|  Industry|            Title|Total_WWC|
+----------+-----------------+---------+
|Sandalwood| K.G.F: Chapter 2|   1215.0|
|Sandalwood|          Kantara|   407.82|
|Sandalwood| K.G.F: Chapter 1|    238.0|
|Sandalwood|      777 Charlie|   102.75|
|Sandalwood|VR (Vikrant Rona)|   100.35|
|Sandalwood|            James|     94.2|
|Sandalwood|      Kurukshetra|     81.0|
+----------+-----------------+---------+
only showing top 7 rows
```

In [132…

```python
# With Spark SQL
spark.sql("select Industry,Title, sum(`Worldwide Collection  in Crores`) \
as Total_WWC from boxf_tb \
Where Industry = 'Sandalwood' \
group by Industry,Title \
order by Total_WWC desc limit 7").show()
```

```
+----------+-----------------+---------+
|  Industry|            Title|Total_WWC|
+----------+-----------------+---------+
|Sandalwood| K.G.F: Chapter 2|   1215.0|
|Sandalwood|          Kantara|   407.82|
|Sandalwood| K.G.F: Chapter 1|    238.0|
|Sandalwood|      777 Charlie|   102.75|
|Sandalwood|VR (Vikrant Rona)|   100.35|
|Sandalwood|            James|     94.2|
|Sandalwood|      Kurukshetra|     81.0|
+----------+-----------------+---------+
```

25 Top7 movies by world wide collection in Mollywood

In [133…

```python
# With Pandas
p_boxf.query("Industry=='Mollywood'").groupby(["Industry","Title"])\
    .agg(Total_WWC = ("Worldwide Collection  in Crores","sum"))\
    .sort_values("Total_WWC",ascending = False).reset_index().head(7)
```

Out[133…

	Industry	Title	Total_WWC
0	Mollywood	Manjummel Boys	241.10
1	Mollywood	2018	180.03
2	Mollywood	The Goat Life	157.35
3	Mollywood	Aavesham	154.79
4	Mollywood	Pulimurugan	150.00
5	Mollywood	Premalu	131.18
6	Mollywood	Lucifer	129.20

In [134…

```
# With PySpark
s_boxf.filter(col("Industry")=="Mollywood").groupby("Industry","Title")\
    .agg(sum("Worldwide Collection  in Crores")\
    .alias("Total_WWC")).sort(desc("Total_WWC")).show(7)
```

```
+---------+--------------+---------+
| Industry|         Title|Total_WWC|
+---------+--------------+---------+
|Mollywood|Manjummel Boys|    241.1|
|Mollywood|          2018|   180.03|
|Mollywood| The Goat Life|   157.35|
|Mollywood|      Aavesham|   154.79|
|Mollywood|   Pulimurugan|    150.0|
|Mollywood|       Premalu|   131.18|
|Mollywood|       Lucifer|    129.2|
+---------+--------------+---------+
only showing top 7 rows
```

In [135…

```
# With Spark SQL
spark.sql("select Industry,Title, sum(`Worldwide Collection  in Crores`) \
as Total_WWC from boxf_tb \
Where Industry = 'Mollywood' \
group by Industry,Title \
order by Total_WWC desc limit 7").show()
```

```
+---------+--------------+---------+
| Industry|         Title|Total_WWC|
+---------+--------------+---------+
|Mollywood|Manjummel Boys|    241.1|
|Mollywood|          2018|   180.03|
|Mollywood| The Goat Life|   157.35|
|Mollywood|      Aavesham|   154.79|
|Mollywood|   Pulimurugan|    150.0|
|Mollywood|       Premalu|   131.18|
|Mollywood|       Lucifer|    129.2|
+---------+--------------+---------+
```

26 Write query to display industry and verdict wise films count

In [136…

```
# With Pandas
p_boxf.groupby(["Industry","Verdict"])\
    .agg(Total_Films=("FilmID","count"))\
    .sort_values("Total_Films",ascending=False)\
    .reset_index().head(10)
```

Out[136…

	Industry	Verdict	Total_Films
0	Mollywood	Blockbuster	46
1	Kollywood	Blockbuster	44
2	Bollywood	Hit	39
3	Tollywood	Blockbuster	39
4	Bollywood	Blockbuster	30
5	Tollywood	Hit	25
6	Bollywood	Disaster	25
7	Tollywood	Disaster	24
8	Kollywood	SuperHit	23
9	Tollywood	SuperHit	23

In [137…

```
# With PySpark
s_boxf.groupby(["Industry","Verdict"])\
    .agg(count("FilmID").alias("Total_Films"))\
```

```
    .sort(desc("Total_Films")).show(10)
```

```
+---------+-----------+-----------+
| Industry|    Verdict|Total_Films|
+---------+-----------+-----------+
|Mollywood|Blockbuster|         46|
|Kollywood|Blockbuster|         44|
|Tollywood|Blockbuster|         39|
|Bollywood|        Hit|         39|
|Bollywood|Blockbuster|         30|
|Bollywood|   Disaster|         25|
|Tollywood|        Hit|         25|
|Tollywood|   Disaster|         24|
|Kollywood|  Super Hit|         23|
|Tollywood|  Super Hit|         23|
+---------+-----------+-----------+
only showing top 10 rows
```

In [138…

```
# With Spark SQL
spark.sql("select Industry, Verdict, count(FilmID)as \
Total_Film from boxf_tb \
group by Industry, Verdict \
order by Total_Film desc limit 10").show()
```

```
+---------+-----------+----------+
| Industry|    Verdict|Total_Film|
+---------+-----------+----------+
|Mollywood|Blockbuster|        46|
|Kollywood|Blockbuster|        44|
|Bollywood|        Hit|        39|
|Tollywood|Blockbuster|        39|
|Bollywood|Blockbuster|        30|
|Bollywood|   Disaster|        25|
|Tollywood|        Hit|        25|
|Tollywood|   Disaster|        24|
|Kollywood|  Super Hit|        23|
|Tollywood|  Super Hit|        23|
+---------+-----------+----------+
```

27 Write query to get films based on budget in Bollowood

In [139…

```
# With Pandas
p_boxf.query("Industry =='Bollywood'")\
    .groupby(["Industry","Title"])\
    .agg(Budget = ( "Budget in Crores","sum"))\
    .sort_values("Budget", ascending = False)\
    .reset_index().head(10)
```

Out[139…

	Industry	Title	Budget
0	Bollywood	Brahmastra Part One: Shiva	400.0
1	Bollywood	Jawan	300.0
2	Bollywood	Thugs of Hindostan	275.0
3	Bollywood	Pathaan	250.0
4	Bollywood	Tiger 3	250.0
5	Bollywood	Bade Miyan Chote Miyan	250.0
6	Bollywood	Fighter	250.0
7	Bollywood	Samrat Prithviraj	220.0
8	Bollywood	Tiger Zinda Hai	200.0
9	Bollywood	83	200.0

In [140…

```
# With PySpark
s_boxf.filter(col("Industry")=="Bollywood")\
    .groupby("Industry","Title")\
    .agg(sum("Budget in Crores").alias("Budget"))\
    .sort(desc("Budget")).show(10)
```

```
+---------+--------------------+------+
| Industry|               Title|Budget|
+---------+--------------------+------+
|Bollywood|Brahmastra Part O...| 400.0|
|Bollywood|               Jawan| 300.0|
|Bollywood|  Thugs of Hindostan| 275.0|
|Bollywood|             Pathaan| 250.0|
|Bollywood|Bade Miyan Chote ...| 250.0|
|Bollywood|             Fighter| 250.0|
|Bollywood|             Tiger 3| 250.0|
|Bollywood|   Samrat Prithviraj| 220.0|
|Bollywood|                Zero| 200.0|
|Bollywood|     Tiger Zinda Hai| 200.0|
+---------+--------------------+------+
only showing top 10 rows
```

In [141…

```python
# With Spark SQL
spark.sql("select Industry, Title,`Budget in Crores` as Budget \
from boxf_tb \
where Industry = 'Bollywood' \
order by Budget desc limit 10").show()
```

```
+---------+--------------------+------+
| Industry|               Title|Budget|
+---------+--------------------+------+
|Bollywood|Brahmastra Part O...| 400.0|
|Bollywood|               Jawan| 300.0|
|Bollywood|  Thugs of Hindostan| 275.0|
|Bollywood|             Pathaan| 250.0|
|Bollywood|             Tiger 3| 250.0|
|Bollywood|             Fighter| 250.0|
|Bollywood|Bade Miyan Chote ...| 250.0|
|Bollywood|   Samrat Prithviraj| 220.0|
|Bollywood|                Zero| 200.0|
|Bollywood|                  83| 200.0|
+---------+--------------------+------+
```

28 Write query to get films based on budget in Tollywood

In [142…

```python
# With Pandas
p_boxf.query("Industry =='Tollywood'")\
    .groupby(["Industry","Title"])\
    .agg(Budget = ( "Budget in Crores","sum"))\
    .sort_values("Budget", ascending = False)\
    .reset_index().head(10)
```

Out[142…

	Industry	Title	Budget
0	Tollywood	RRR (Rise Roar Revolt)	550.0
1	Tollywood	Adipurush	500.0
2	Tollywood	Kalki 2898-AD	500.0
3	Tollywood	Radhe Shyam	300.0
4	Tollywood	Salaar: Part 1 - Ceasefire	280.0
5	Tollywood	Bāhubali 2: The Conclusion	250.0
6	Tollywood	Devara: Part 1	250.0
7	Tollywood	Bāhubali: The Beginning	180.0
8	Tollywood	Guntur Kaaram	150.0
9	Tollywood	Pushpa: The Rise - Part 1	150.0

In [143…

```python
# With PySpark
s_boxf.filter(col("Industry")=="Tollywood")\
    .groupby("Industry","Title")\
    .agg(sum("Budget in Crores").alias("Budget"))\
    .sort(desc("Budget")).show(10)
```

```
+---------+--------------------+------+
| Industry|               Title|Budget|
+---------+--------------------+------+
|Tollywood|RRR (Rise Roar Re...| 550.0|
|Tollywood|           Adipurush| 500.0|
|Tollywood|       Kalki 2898-AD| 500.0|
|Tollywood|         Radhe Shyam| 300.0|
|Tollywood|Salaar: Part 1 - ...| 280.0|
|Tollywood|      Devara: Part 1| 250.0|
|Tollywood|Bāhubali 2: The C...| 250.0|
|Tollywood|Bāhubali: The Beg...| 180.0|
|Tollywood|Pushpa: The Rise ...| 150.0|
|Tollywood|       Guntur Kaaram| 150.0|
+---------+--------------------+------+
only showing top 10 rows
```

In [144…

```python
# With Spark SQL
spark.sql("select Industry, Title,`Budget in Crores` as Budget \
from boxf_tb \
where Industry = 'Tollywood' \
order by Budget desc limit 10").show()
```

```
+---------+--------------------+------+
| Industry|               Title|Budget|
+---------+--------------------+------+
|Tollywood|RRR (Rise Roar Re...| 550.0|
|Tollywood|       Kalki 2898-AD| 500.0|
|Tollywood|           Adipurush| 500.0|
|Tollywood|         Radhe Shyam| 300.0|
|Tollywood|Salaar: Part 1 - ...| 280.0|
|Tollywood|Bāhubali 2: The C...| 250.0|
|Tollywood|      Devara: Part 1| 250.0|
|Tollywood|Bāhubali: The Beg...| 180.0|
|Tollywood|Pushpa: The Rise ...| 150.0|
|Tollywood|       Guntur Kaaram| 150.0|
+---------+--------------------+------+
```

29 Write query to get films based on budget in Kollywood

In [145…

```python
# With Pandas
p_boxf.query("Industry =='Kollywood'")\
    .groupby(["Industry","Title"])\
    .agg(Budget = ( "Budget in Crores","sum"))\
    .sort_values("Budget", ascending = False)\
    .reset_index().head(10)
```

Out[145…

	Industry	Title	Budget
0	Kollywood	2	540.0
1	Kollywood	The GOAT	320.0
2	Kollywood	Leo	250.0
3	Kollywood	Ponniyin Selvan: Part One	250.0
4	Kollywood	Indian 2	250.0
5	Kollywood	Ponniyin Selvan: Part Two	250.0
6	Kollywood	Jailer	200.0
7	Kollywood	Vettaiyan	200.0
8	Kollywood	Varisu	200.0
9	Kollywood	Darbar	200.0

In [146…

```python
# With PySpark
s_boxf.filter(col("Industry")=="Kollywood")\
    .groupby("Industry","Title")\
    .agg(sum("Budget in Crores").alias("Budget"))\
    .sort(desc("Budget")).show(10)
```

```
+---------+--------------------+------+
| Industry|               Title|Budget|
+---------+--------------------+------+
|Kollywood|                   2| 540.0|
|Kollywood|            The GOAT| 320.0|
|Kollywood|Ponniyin Selvan: ...| 250.0|
|Kollywood|                 Leo| 250.0|
|Kollywood|Ponniyin Selvan: ...| 250.0|
|Kollywood|            Indian 2| 250.0|
|Kollywood|              Varisu| 200.0|
|Kollywood|              Darbar| 200.0|
|Kollywood|           Vettaiyan| 200.0|
|Kollywood|              Jailer| 200.0|
+---------+--------------------+------+
only showing top 10 rows
```

In [147…

```python
# With Spark SQL
spark.sql("select Industry, Title,`Budget in Crores` as Budget \
from boxf_tb \
where Industry = 'Kollywood' \
order by Budget desc limit 10").show()
```

```
+---------+--------------------+------+
| Industry|               Title|Budget|
+---------+--------------------+------+
|Kollywood|                   2| 540.0|
|Kollywood|            The GOAT| 320.0|
|Kollywood|Ponniyin Selvan: ...| 250.0|
|Kollywood|                 Leo| 250.0|
|Kollywood|Ponniyin Selvan: ...| 250.0|
|Kollywood|            Indian 2| 250.0|
|Kollywood|              Jailer| 200.0|
|Kollywood|              Varisu| 200.0|
|Kollywood|              Darbar| 200.0|
|Kollywood|           Vettaiyan| 200.0|
+---------+--------------------+------+
```

30 Write query to get films based on budget in Sandalwood

In [148…

```python
# With Pandas
p_boxf.query("Industry =='Sandalwood'")\
    .groupby(["Industry","Title"])\
    .agg(Budget = ( "Budget in Crores","sum"))\
    .sort_values("Budget", ascending = False)\
    .reset_index().head(10)
```

Out[148…

	Industry	Title	Budget
0	Sandalwood	K.G.F: Chapter 2	100.0
1	Sandalwood	VR (Vikrant Rona)	80.0
2	Sandalwood	Kabzaa	70.0
3	Sandalwood	Kurukshetra	60.0
4	Sandalwood	K.G.F: Chapter 1	50.0
5	Sandalwood	The Villain	45.0
6	Sandalwood	Kaatera	45.0
7	Sandalwood	James	40.0
8	Sandalwood	Pailwaan	35.0
9	Sandalwood	Kranti	35.0

In [149…

```python
# With PySpark
s_boxf.filter(col("Industry")=="Sandalwood")\
    .groupby("Industry","Title")\
    .agg(sum("Budget in Crores").alias("Budget"))\
    .sort(desc("Budget")).show(10)
```

```
+----------+-----------------+------+
|  Industry|            Title|Budget|
+----------+-----------------+------+
|Sandalwood| K.G.F: Chapter 2| 100.0|
|Sandalwood|VR (Vikrant Rona)|  80.0|
|Sandalwood|           Kabzaa|  70.0|
|Sandalwood|      Kurukshetra|  60.0|
|Sandalwood| K.G.F: Chapter 1|  50.0|
|Sandalwood|      The Villain|  45.0|
|Sandalwood|          Kaatera|  45.0|
|Sandalwood|            James|  40.0|
|Sandalwood|         Pailwaan|  35.0|
|Sandalwood|           Kranti|  35.0|
+----------+-----------------+------+
only showing top 10 rows
```

In [150…

```python
# With Spark SQL
spark.sql("select Industry, Title,`Budget in Crores` as Budget \
from boxf_tb \
where Industry = 'Sandalwood' \
order by Budget desc limit 10").show()
```

```
+----------+-----------------+------+
|  Industry|            Title|Budget|
+----------+-----------------+------+
|Sandalwood| K.G.F: Chapter 2| 100.0|
|Sandalwood|VR (Vikrant Rona)|  80.0|
|Sandalwood|           Kabzaa|  70.0|
|Sandalwood|      Kurukshetra|  60.0|
|Sandalwood| K.G.F: Chapter 1|  50.0|
|Sandalwood|      The Villain|  45.0|
|Sandalwood|          Kaatera|  45.0|
|Sandalwood|            James|  40.0|
|Sandalwood|         Pailwaan|  35.0|
|Sandalwood|           Kranti|  35.0|
+----------+-----------------+------+
```

31 Write query to get films based on budget in Mollywood

In [151…

```python
# With Pandas
p_boxf.query("Industry =='Mollywood'")\
    .groupby(["Industry","Title"])\
    .agg(Budget = ( "Budget in Crores","sum"))\
    .sort_values("Budget", ascending = False)\
    .reset_index().head(10)
```

Out[151…

	Industry	Title	Budget
0	Mollywood	Marakkar: Lion of the Arabian Sea	100.0
1	Mollywood	Malaikottai Vaaliban	60.0
2	Mollywood	The Goat Life	50.0
3	Mollywood	Odiyan	50.0
4	Mollywood	Mamangam: History of the Brave	45.0
5	Mollywood	King of Kotha	45.0
6	Mollywood	Kayamkulam Kochunni	40.0
7	Mollywood	Lucifer	35.0
8	Mollywood	Kurup	35.0
9	Mollywood	Turbo	35.0

In [152…

```python
# With PySpark
s_boxf.filter(col("Industry")=="Mollywood")\
    .groupby("Industry","Title")\
    .agg(sum("Budget in Crores").alias("Budget"))\
    .sort(desc("Budget")).show(10)
```

```
+---------+--------------------+------+
| Industry|               Title|Budget|
+---------+--------------------+------+
|Mollywood|Marakkar: Lion of...| 100.0|
|Mollywood|Malaikottai Vaaliban|  60.0|
|Mollywood|              Odiyan|  50.0|
|Mollywood|       The Goat Life|  50.0|
|Mollywood|       King of Kotha|  45.0|
|Mollywood|Mamangam: History...|  45.0|
|Mollywood| Kayamkulam Kochunni|  40.0|
|Mollywood|               Turbo|  35.0|
|Mollywood|             Lucifer|  35.0|
|Mollywood|              Trance|  35.0|
+---------+--------------------+------+
only showing top 10 rows
```

In [153…

```python
# With Spark SQL
spark.sql("select Industry, Title,`Budget in Crores` as Budget \
from boxf_tb \
where Industry = 'Mollywood' \
order by Budget desc limit 10").show()
```

```
+---------+--------------------+------+
| Industry|               Title|Budget|
+---------+--------------------+------+
|Mollywood|Marakkar: Lion of...| 100.0|
|Mollywood|Malaikottai Vaaliban|  60.0|
|Mollywood|       The Goat Life|  50.0|
|Mollywood|              Odiyan|  50.0|
|Mollywood|Mamangam: History...|  45.0|
|Mollywood|       King of Kotha|  45.0|
|Mollywood| Kayamkulam Kochunni|  40.0|
|Mollywood|             Lucifer|  35.0|
|Mollywood|               Kurup|  35.0|
|Mollywood|              Trance|  35.0|
+---------+--------------------+------+
```

32 Top 5 movies by IMDb rating from Bollowood

In [154…

```python
# With Pandas
p_boxf.query("Industry == 'Bollywood'")[["Title","IMDb Rating"]]\
    .sort_values("IMDb Rating",ascending = False).head(7).head(5)
```

Out[154…

	Title	IMDb Rating
121	12th Fail	8.8
79	The Kashmir Files	8.6
512	Sachin	8.5
602	Laapataa Ladies	8.4
524	Dangal	8.3

In [155…

```python
# With PySpark
s_boxf.filter("Industry == 'Bollywood'")\
    .select("Title","IMDb Rating")\
    .sort(desc("IMDb Rating")).show(5)
```

```
+-----------------+-----------+
|            Title|IMDb Rating|
+-----------------+-----------+
|        12th Fail|        8.8|
|The Kashmir Files|        8.6|
|           Sachin|        8.5|
|  Laapataa Ladies|        8.4|
|           Dangal|        8.3|
+-----------------+-----------+
only showing top 5 rows
```

In [156…

```python
# With Spark SQL
spark.sql("Select Title,`IMDb Rating` from boxf_tb \
where Industry = 'Bollywood' \
order by `IMDb Rating` desc limit 5").show()
```

```
+-----------------+-----------+
|            Title|IMDb Rating|
+-----------------+-----------+
|        12th Fail|        8.8|
|The Kashmir Files|        8.6|
|           Sachin|        8.5|
|  Laapataa Ladies|        8.4|
|           Dangal|        8.3|
+-----------------+-----------+
```

33 Top 5 movies by IMDb rating from Kollywood

In [157…

```python
# With Pandas
p_boxf.query("Industry == 'Kollywood'")[["Title","IMDb Rating"]]\
    .sort_values("IMDb Rating",ascending = False).head(7).head(5)
```

Out[157…

	Title	IMDb Rating
316	Rocketry: The Nambi Effect	8.7
350	Meiyazhagan	8.5
135	Maharaja	8.5
25	96	8.5
274	Vada Chennai	8.4

In [158…

```python
# With PySpark
s_boxf.filter("Industry == 'Kollywood'")\
    .select("Title","IMDb Rating")\
    .sort(desc("IMDb Rating")).show(5)
```

```
+--------------------+-----------+
|               Title|IMDb Rating|
+--------------------+-----------+
|Rocketry: The Nam...|        8.7|
|                  96|        8.5|
|            Maharaja|        8.5|
|         Meiyazhagan|        8.5|
|              Asuran|        8.4|
+--------------------+-----------+
only showing top 5 rows
```

In [159…

```python
# With Spark SQL
spark.sql("Select Title,`IMDb Rating` from boxf_tb \
where Industry = 'Kollywood' \
order by `IMDb Rating` desc limit 5").show()
```

```
+--------------------+-----------+
|               Title|IMDb Rating|
+--------------------+-----------+
|Rocketry: The Nam...|        8.7|
|                  96|        8.5|
|            Maharaja|        8.5|
|         Meiyazhagan|        8.5|
|              Asuran|        8.4|
+--------------------+-----------+
```

34 Top 5 movies by IMDb rating from Tollywood

In [160…

```python
# With Pandas
p_boxf.query("Industry == 'Tollywood'")[["Title","IMDb Rating"]]\
    .sort_values("IMDb Rating",ascending = False).head(7).head(5)
```

Out[160…

	Title	IMDb Rating
437	Jersey	8.5
84	Sita Ramam	8.5
35	Mahanati	8.4
360	Rangasthalam	8.2
226	Hi Nanna	8.2

In [161…

```python
# With PySpark
s_boxf.filter("Industry == 'Tollywood'")\
    .select("Title","IMDb Rating")\
    .sort(desc("IMDb Rating")).show(5)
```

```
+------------+-----------+
|       Title|IMDb Rating|
+------------+-----------+
|  Sita Ramam|        8.5|
|      Jersey|        8.5|
|    Mahanati|        8.4|
|Rangasthalam|        8.2|
|    Hi Nanna|        8.2|
+------------+-----------+
only showing top 5 rows
```

In [162…

```
# With Spark SQL
spark.sql("Select Title,`IMDb Rating` from boxf_tb \
where Industry = 'Tollywood' \
order by `IMDb Rating` desc limit 5").show()
```

```
+------------+-----------+
|       Title|IMDb Rating|
+------------+-----------+
|  Sita Ramam|        8.5|
|      Jersey|        8.5|
|    Mahanati|        8.4|
|Rangasthalam|        8.2|
|    Hi Nanna|        8.2|
+------------+-----------+
```

35 Top 5 movies by IMDb rating from Sandalwood

In [163…

```
# With Pandas
p_boxf.query("Industry == 'Sandalwood'")[["Title","IMDb Rating"]]\
    .sort_values("IMDb Rating",ascending = False).head(7).head(5)
```

Out[163…

	Title	IMDb Rating
83	777 Charlie	8.7
148	K.G.F: Chapter 1	8.2
149	K.G.F: Chapter 2	8.2
237	Sapta Sagaradaache Ello: Side A	8.2
186	Kantara	8.2

In [164…

```
# With PySpark
s_boxf.filter("Industry == 'Sandalwood'")\
    .select("Title","IMDb Rating")\
    .sort(desc("IMDb Rating")).show(5)
```

```
+--------------------+-----------+
|               Title|IMDb Rating|
+--------------------+-----------+
|         777 Charlie|        8.7|
|             Kantara|        8.2|
|    K.G.F: Chapter 1|        8.2|
|    K.G.F: Chapter 2|        8.2|
|Sapta Sagaradaach...|        8.2|
+--------------------+-----------+
only showing top 5 rows
```

In [165…

```
# With Spark SQL
spark.sql("Select Title,`IMDb Rating` from boxf_tb \
where Industry = 'Sandalwood' \
order by `IMDb Rating` desc limit 5").show()
```

```
+--------------------+-----------+
|               Title|IMDb Rating|
+--------------------+-----------+
|         777 Charlie|        8.7|
|    K.G.F: Chapter 2|        8.2|
|    K.G.F: Chapter 1|        8.2|
|             Kantara|        8.2|
|Sapta Sagaradaach...|        8.2|
+--------------------+-----------+
```

36 Top 5 movies by IMDb rating from Mollywood

In [166…

```
# With Pandas
p_boxf.query("Industry == 'Mollywood'")[["Title","IMDb Rating"]]\
    .sort_values("IMDb Rating",ascending = False).head(7).head(5)
```

Out[166…

	Title	IMDb Rating
140	Kishkindha Kaandam	8.6
204	Kumbalangi Nights	8.5
189	Jana Gana Mana	8.3
217	2018	8.3
147	Bangalore Days	8.3

In [167…

```
# With PySpark
s_boxf.filter("Industry == 'Mollywood'")\
    .select("Title","IMDb Rating")\
    .sort(desc("IMDb Rating")).show(5)
```

```
+------------------+-----------+
|             Title|IMDb Rating|
+------------------+-----------+
|Kishkindha Kaandam|        8.6|
| Kumbalangi Nights|        8.5|
|    Bangalore Days|        8.3|
|              2018|        8.3|
|            Premam|        8.3|
+------------------+-----------+
only showing top 5 rows
```

In [168…

```
# With Spark SQL
spark.sql("Select Title,`IMDb Rating` from boxf_tb \
where Industry = 'Mollywood' \
order by `IMDb Rating` desc limit 5").show()
```

```
+------------------+-----------+
|             Title|IMDb Rating|
+------------------+-----------+
|Kishkindha Kaandam|        8.6|
| Kumbalangi Nights|        8.5|
|    Bangalore Days|        8.3|
|            Premam|        8.3|
|    Jana Gana Mana|        8.3|
+------------------+-----------+
```

37 Write a query to get language wise budget?

In [169…

```
# With Pandas
p_boxf.merge(p_lan,on = "LanguageID",how="right")\
    .groupby("Language").agg(Total_Budget=("Budget in Crores","sum"))\
    .sort_values("Total_Budget",ascending = False).reset_index()
```

Out[169…

	Language	Total_Budget
0	Hindi	14535.0
1	Telugu	9144.5
2	Tamil	8682.0
3	Malayalam	1342.5
4	Kannada	949.0

In [170…

```
# With PySpark
s_boxf.join(s_lan,s_boxf["LanguageID"]==s_lan["LanguageID"],"right")\
    .groupby("Language").agg(sum("Budget in Crores").alias("Total_Budget"))\
    .sort(desc("Total_Budget")).show()
```

```
+---------+------------+
| Language|Total_Budget|
+---------+------------+
|    Hindi|     14535.0|
|   Telugu|      9144.5|
|    Tamil|      8682.0|
|Malayalam|      1342.5|
|  Kannada|       949.0|
+---------+------------+
```

In [171…

```
# With Spark SQL
spark.sql("Select Language, sum(`Budget in Crores`) \
as Total_Budget from boxf_tb boxf \
right join lan_tb lang on lang.LanguageID = boxf.LanguageID \
group by Language \
order by Total_Budget desc").show()
```

```
+---------+------------+
| Language|Total_Budget|
+---------+------------+
|    Hindi|     14535.0|
|   Telugu|      9144.5|
|    Tamil|      8682.0|
|Malayalam|      1342.5|
|  Kannada|       949.0|
+---------+------------+
```

38 Write a query to get language wise how many directors are there?

In [172…

```
# With Pandas
p_boxf.merge(p_lan,on="LanguageID",how="inner")\
    .merge(p_dir,left_on = "DirectorID",\
           right_on = "Director ID",how="inner")\
    .groupby("Language").agg(Total_Directors=("DirectorID","count"))\
    .sort_values("Total_Directors",ascending = False)\
    .reset_index().head(10)
```

Out[172…

	Language	Total_Directors
0	Hindi	182
1	Telugu	166
2	Tamil	140
3	Malayalam	84
4	Kannada	32

In [173…

```
# With PySpark
s_boxf.join(s_lan,s_boxf["LanguageID"]==s_lan["LanguageID"],"inner")\
    .join(s_dir, s_boxf["DirectorID"]==s_dir["Director ID"],"inner")\
    .groupby("Language").agg(count("DirectorID").alias("Total_Directors"))\
    .sort(desc("Total_Directors")).show()
```

```
+---------+---------------+
| Language|Total_Directors|
+---------+---------------+
|    Hindi|            182|
|   Telugu|            166|
|    Tamil|            140|
|Malayalam|             84|
|  Kannada|             32|
+---------+---------------+
```

In [174…

```
# With Spark SQL
spark.sql("select Language, count(`Director ID`) \
as Total_Directors from boxf_tb boxf \
inner join lan_tb lan on lan.LanguageID = boxf.LanguageID \
inner join dir_tb dir on dir.`Director ID` = boxf.DirectorID \
group by Language \
order by Total_Directors desc").show()
```

```
+---------+---------------+
| Language|Total_Directors|
+---------+---------------+
|    Hindi|            182|
|   Telugu|            166|
|    Tamil|            140|
|Malayalam|             84|
|  Kannada|             32|
+---------+---------------+
```

39 Write a query to get language wise worldwide total collection ?

In [175…

```
# With Pandas
p_boxf.merge(p_lan,on="LanguageID",how="inner")\
    .groupby("Language")\
    .agg(Total_WWC=("Worldwide Collection  in Crores","sum"))\
    .sort_values("Total_WWC",ascending = False).reset_index()
```

Out[175…

	Language	Total_WWC
0	Hindi	38610.79
1	Telugu	18065.27
2	Tamil	15235.99
3	Malayalam	4597.32
4	Kannada	3369.42

In [176…

```
# With PySpark
s_boxf.join(s_lan,s_boxf["LanguageID"]==s_lan["LanguageID"],"right")\
    .groupby("Language").agg(sum("Worldwide Collection  in Crores")\
        .alias("Total_WWC")).sort(desc("Total_WWC")).show()
```

```
+---------+------------------+
| Language|         Total_WWC|
+---------+------------------+
|    Hindi|          38610.79|
|   Telugu|18065.270000000004|
|    Tamil|          15235.99|
|Malayalam|           4597.32|
|  Kannada|           3369.42|
+---------+------------------+
```

In [177…

```
# With Spark SQL
spark.sql("select Language, sum(`Worldwide Collection  in Crores`) \
as Total_WWC from boxf_tb boxf \
right join lan_tb lan on lan.LanguageID =boxf.LanguageID \
group by Language \
order by 2 desc").show()
```

```
+---------+------------------+
| Language|         Total_WWC|
+---------+------------------+
|    Hindi|          38610.79|
|   Telugu|18065.270000000004|
|    Tamil|          15235.99|
|Malayalam|           4597.32|
|  Kannada|           3369.42|
+---------+------------------+
```

40 Write a query to get language, lead actor/actress wise films they acted?

In [178…

```
# With Pandas
p_boxf.merge(p_lan,on="LanguageID",how="right")\
    .groupby(["Language","Lead Actor/Actress"])\
    .agg(Total_Films=("FilmID","count"))\
    .sort_values("Total_Films",ascending = False)\
    .reset_index().head(15)
```

Out[178…

	Language	Lead Actor/Actress	Total_Films
0	Hindi	Akshay Kumar	20
1	Hindi	Ajay Devgn	14
2	Tamil	Dhanush	13
3	Malayalam	Mammootty	13
4	Telugu	Nani	12
5	Tamil	Joseph Vijay	12
6	Hindi	Salman Khan	12
7	Tamil	Sivakarthikeyan	11
8	Malayalam	Mohanlal	11
9	Telugu	Ravi Teja	10
10	Telugu	Mahesh Babu	10
11	Tamil	Karthi	10
12	Tamil	Rajinikanth	10
13	Malayalam	Prithviraj Sukumaran	9
14	Telugu	Vijay Deverakonda	9

In [179…

```
# With PySpark
```

```
s_boxf.join(s_lan,s_boxf["LanguageID"]==s_lan["LanguageID"],"right")\
    .groupby("Language","Lead Actor/Actress").agg(count("FilmID")\
        .alias("Total_Films")).sort(desc("Total_Films")).show(15)
```

```
+---------+--------------------+-----------+
| Language|  Lead Actor/Actress|Total_Films|
+---------+--------------------+-----------+
|    Hindi|        Akshay Kumar|         20|
|    Hindi|          Ajay Devgn|         14|
|Malayalam|           Mammootty|         13|
|    Tamil|             Dhanush|         13|
|    Tamil|        Joseph Vijay|         12|
|    Hindi|         Salman Khan|         12|
|   Telugu|                Nani|         12|
|    Tamil|     Sivakarthikeyan|         11|
|Malayalam|            Mohanlal|         11|
|   Telugu|         Mahesh Babu|         10|
|   Telugu|           Ravi Teja|         10|
|    Tamil|         Rajinikanth|         10|
|    Tamil|              Karthi|         10|
|Malayalam|Prithviraj Sukumaran|          9|
|    Tamil|              Vikram|          9|
+---------+--------------------+-----------+
only showing top 15 rows
```

In [180…

```
# With Spark SQL
spark.sql("select Language, `Lead Actor/Actress`,\
count(FilmID) as Total_Films from boxf_tb boxf \
right join lan_tb lan on lan.LanguageID = boxf.LanguageID \
group by Language, `Lead Actor/Actress` \
order by 3 desc limit 15").show()
```

```
+---------+--------------------+-----------+
| Language|  Lead Actor/Actress|Total_Films|
+---------+--------------------+-----------+
|    Hindi|        Akshay Kumar|         20|
|    Hindi|          Ajay Devgn|         14|
|Malayalam|           Mammootty|         13|
|    Tamil|             Dhanush|         13|
|    Tamil|        Joseph Vijay|         12|
|    Hindi|         Salman Khan|         12|
|   Telugu|                Nani|         12|
|    Tamil|     Sivakarthikeyan|         11|
|Malayalam|            Mohanlal|         11|
|   Telugu|         Mahesh Babu|         10|
|   Telugu|           Ravi Teja|         10|
|    Tamil|         Rajinikanth|         10|
|    Tamil|              Karthi|         10|
|Malayalam|Prithviraj Sukumaran|          9|
|    Tamil|              Vikram|          9|
+---------+--------------------+-----------+
```

41 Write a query to get language, year wise films released?

In [181…

```
# With Pandas
p_boxf.merge(p_lan, on = "LanguageID", how="inner")\
    .groupby(["Language","Year"]).agg(Total_Films=("FilmID","count"))\
    .sort_values("Total_Films",ascending = False).reset_index().head(10)
```

Out[181…

	Language	Year	Total_Films
0	Telugu	2023	25
1	Telugu	2017	24
2	Hindi	2017	24
3	Hindi	2022	24
4	Hindi	2023	23
5	Telugu	2018	23
6	Telugu	2022	22
7	Tamil	2018	22
8	Hindi	2024	20
9	Telugu	2019	19

In [182…

```
s_boxf = s_boxf.withColumn("Year", year(col("Release Date")))
```

In [183…

```
# With PySpark
s_boxf.join(s_lan,s_boxf["LanguageID"]==s_lan["LanguageID"],"inner")\
```

```
    .groupby("Language","Year").agg(count("FilmID").alias("Total_Films"))\
    .sort(desc("Total_Films")).show(10)
```

```
+--------+----+-----------+
|Language|Year|Total_Films|
+--------+----+-----------+
|  Telugu|2023|         25|
|   Hindi|2022|         24|
|   Hindi|2017|         24|
|  Telugu|2017|         24|
|   Hindi|2023|         23|
|  Telugu|2018|         23|
|  Telugu|2022|         22|
|   Tamil|2018|         22|
|   Hindi|2024|         20|
|   Tamil|2019|         19|
+--------+----+-----------+
only showing top 10 rows
```

In [184…

```
# With Spark SQL
spark.sql("select Language,YEAR(try_to_timestamp(`Release Date`, 'dd-MMM-yy'))\
as Yr , count(FilmID) as Total_Films from boxf_tb boxf \
inner join lan_tb lan on lan.LanguageID = Boxf.LanguageID \
group by 1,2 \
order by 3 desc limit 10").show()
```

```
+--------+----+-----------+
|Language|  Yr|Total_Films|
+--------+----+-----------+
|  Telugu|2023|         25|
|   Hindi|2022|         24|
|   Hindi|2017|         24|
|  Telugu|2017|         24|
|   Hindi|2023|         23|
|  Telugu|2018|         23|
|   Tamil|2018|         22|
|  Telugu|2022|         22|
|   Hindi|2024|         20|
|   Tamil|2019|         19|
+--------+----+-----------+
```

42 Write a query to get films which was not released on overseas?

In [185…

```
# With PySpark
p_boxf.loc[p_boxf["Overseas Collection in Crores"]==0,\
    ["Title","Overseas Collection in Crores"]].reset_index()
```

Out[185…

	index	Title	Overseas Collection in Crores
0	4	Maanikya	0.0
1	13	Ugramm	0.0
2	36	Tagaru	0.0
3	57	Hebbuli	0.0
4	169	Anjaniputra	0.0
5	171	Chakravarthy	0.0
6	446	118	0.0

In [186…

```
# With PySpark
s_boxf.select("Title","Overseas Collection in Crores")\
    .filter(col("Overseas Collection in Crores")==0).show()
```

```
+------------+----------------------------+
|       Title|Overseas Collection in Crores|
+------------+----------------------------+
|    Maanikya|                         0.0|
|      Ugramm|                         0.0|
|      Tagaru|                         0.0|
|     Hebbuli|                         0.0|
| Anjaniputra|                         0.0|
|Chakravarthy|                         0.0|
|         118|                         0.0|
+------------+----------------------------+
```

In [187…

```
# With Spark SQL
spark.sql("select Title,`Overseas Collection in Crores` \
from boxf_tb where `Overseas Collection in Crores` = 0 ").show()
```

```
+------------+----------------------------+
|       Title|Overseas Collection in Crores|
+------------+----------------------------+
|    Maanikya|                         0.0|
|      Ugramm|                         0.0|
|      Tagaru|                         0.0|
|     Hebbuli|                         0.0|
| Anjaniputra|                         0.0|
|Chakravarthy|                         0.0|
|         118|                         0.0|
+------------+----------------------------+
```

43 Write a query to get language wise top 3 longest runtime moves?

In [188…

```
p_boxf.merge(p_lan,on = "LanguageID",how="inner")[["Title","Language","Runtime (mins)"]]\
    .sort_values(["Language","Runtime (mins)"],ascending=[True,False])\
    .assign(row_number=lambda x:x.groupby("Language").cumcount()+1)\
    .query("row_number <=3")
```

Out[188…

	Title	Language	Runtime (mins)	row_number
114	Animal	Hindi	204	1
176	M.S. Dhoni: The Untold Story	Hindi	184	2
594	Maidaan	Hindi	181	3
103	Avane Srimannarayana	Kannada	186	1
201	Kurukshetra	Kannada	185	2
225	Kaatera	Kannada	183	3
178	Marakkar: Lion of the Arabian Sea	Malayalam	181	1
92	Ayyappanum Koshiyum	Malayalam	177	2
98	Lucifer	Malayalam	175	3
39	I	Tamil	188	1
261	Jilla	Tamil	185	2
270	Cobra	Tamil	183	3
45	RRR (Rise Roar Revolt)	Telugu	187	1
54	Arjun Reddy	Telugu	182	2
126	Kalki 2898-AD	Telugu	180	3

In [189…

```
# With PySpark
window_spec = Window.partitionBy("Language").orderBy(desc("Runtime (mins)"))
s_boxf.join(s_lan,s_boxf["LanguageID"]==s_lan["LanguageID"],"inner")\
    .withColumn("Rnk",row_number().over(window_spec))\
    .filter(col("Rnk")<=3).select("Title","Language","Runtime (mins)","Rnk")\
    .sort("Language",desc("Runtime (mins)")).show()
```

```
+--------------------+---------+--------------+---+
|               Title| Language|Runtime (mins)|Rnk|
+--------------------+---------+--------------+---+
|              Animal|    Hindi|           204|  1|
|M.S. Dhoni: The U...|    Hindi|           184|  2|
|             Maidaan|    Hindi|           181|  3|
|Avane Srimannarayana|  Kannada|           186|  1|
|         Kurukshetra|  Kannada|           185|  2|
|             Kaatera|  Kannada|           183|  3|
|Marakkar: Lion of...|Malayalam|           181|  1|
| Ayyappanum Koshiyum|Malayalam|           177|  2|
|             Lucifer|Malayalam|           175|  3|
|                   I|    Tamil|           188|  1|
|               Jilla|    Tamil|           185|  2|
|               Cobra|    Tamil|           183|  3|
|RRR (Rise Roar Re...|   Telugu|           187|  1|
|         Arjun Reddy|   Telugu|           182|  2|
|       Kalki 2898-AD|   Telugu|           180|  3|
+--------------------+---------+--------------+---+
```

In [190…

```
# With Spark SQL
spark.sql("select * from (select Language,Title,`Runtime (mins)`,\
row_number() over(partition by Language order by `Runtime (mins)` desc)as rnk \
from boxf_tb boxf \
inner join lan_tb lan on lan.LanguageID = boxf.LanguageID) a \
where rnk <= 3 \
order by Language, `Runtime (mins)` desc").show()
```

```
+---------+--------------------+--------------+---+
| Language|               Title|Runtime (mins)|rnk|
+---------+--------------------+--------------+---+
|    Hindi|              Animal|           204|  1|
|    Hindi|M.S. Dhoni: The U...|           184|  2|
|    Hindi|             Maidaan|           181|  3|
|  Kannada|Avane Srimannarayana|           186|  1|
|  Kannada|         Kurukshetra|           185|  2|
|  Kannada|             Kaatera|           183|  3|
|Malayalam|Marakkar: Lion of...|           181|  1|
|Malayalam| Ayyappanum Koshiyum|           177|  2|
|Malayalam|             Lucifer|           175|  3|
|    Tamil|                   I|           188|  1|
|    Tamil|               Jilla|           185|  2|
|    Tamil|               Cobra|           183|  3|
|   Telugu|RRR (Rise Roar Re...|           187|  1|
|   Telugu|         Arjun Reddy|           182|  2|
|   Telugu|       Kalki 2898-AD|           180|  3|
+---------+--------------------+--------------+---+
```

44 Write a query to get language wise bottom 5 shortest runtime moves?

In [191…

```python
# With Pandas
p_boxf.merge(p_lan,on = "LanguageID", how= "inner")[["Title","Language","Runtime (mins)"]]\
    .sort_values(["Language","Runtime (mins)"],ascending = [True,True])\
    .assign(rnk =lambda x : x.groupby("Language").cumcount()+1)\
    .query('rnk <=3')
```

Out[191…

	Title	Language	Runtime (mins)	rnk
599	Kill	Hindi	105	1
497	Raksha Bandhan	Hindi	108	2
560	Bhoot: Part One - The Haunted Ship	Hindi	114	3
185	Salaga	Kannada	124	1
36	Tagaru	Kannada	129	2
13	Ugramm	Kannada	132	3
87	Malikappuram	Malayalam	121	1
42	Sudani from Nigeria	Malayalam	123	2
143	Vaazha: Biopic of a Billion Boys	Malayalam	125	3
279	Kochadaiiyaan	Tamil	118	1
332	Dhilluku Dhuddu 2	Tamil	119	2
326	Kadaram Kondan	Tamil	121	3
407	The Ghazi Attack	Telugu	116	1
411	Guru	Telugu	116	2
418	Raju Gari Gadhi 2	Telugu	117	3

In [192…

```python
# With PySpark
window_spec = Window.partitionBy("Language").orderBy("Runtime (mins)")
s_boxf.join(s_lan,s_boxf["LanguageID"]==s_lan["LanguageID"],"inner")\
    .withColumn("rnk",row_number().over(window_spec))\
    .filter(col("rnk")<=3).select("Title","Language","Runtime (mins)","rnk")\
    .sort("Language",asc("Runtime (mins)")).show()
```

```
+--------------------+---------+--------------+---+
|               Title| Language|Runtime (mins)|rnk|
+--------------------+---------+--------------+---+
|                Kill|    Hindi|           105|  1|
|      Raksha Bandhan|    Hindi|           108|  2|
|Bhoot: Part One -...|    Hindi|           114|  3|
|              Salaga|  Kannada|           124|  1|
|              Tagaru|  Kannada|           129|  2|
|              Ugramm|  Kannada|           132|  3|
|        Malikappuram|Malayalam|           121|  1|
| Sudani from Nigeria|Malayalam|           123|  2|
|Vaazha: Biopic of...|Malayalam|           125|  3|
|       Kochadaiiyaan|    Tamil|           118|  1|
|   Dhilluku Dhuddu 2|    Tamil|           119|  2|
|      Kadaram Kondan|    Tamil|           121|  3|
|    The Ghazi Attack|   Telugu|           116|  1|
|                Guru|   Telugu|           116|  2|
|   Raju Gari Gadhi 2|   Telugu|           117|  3|
+--------------------+---------+--------------+---+
```

In [193…

```python
# With Spark SQL
spark.sql("select * from (select Title, Language,`Runtime (mins)`,\
row_number() over(partition by Language order by `Runtime (mins)`)as rnk \
from boxf_tb boxf \
inner join lan_tb lan on lan.LanguageID = boxf.LanguageID) a \
where rnk <= 3 \
order by Language, `Runtime (mins)`").show()
```

```
+--------------------+---------+--------------+---+
|               Title| Language|Runtime (mins)|rnk|
+--------------------+---------+--------------+---+
|                Kill|    Hindi|           105|  1|
|      Raksha Bandhan|    Hindi|           108|  2|
|Bhoot: Part One -...|    Hindi|           114|  3|
|              Salaga|  Kannada|           124|  1|
|              Tagaru|  Kannada|           129|  2|
|              Ugramm|  Kannada|           132|  3|
|        Malikappuram|Malayalam|           121|  1|
| Sudani from Nigeria|Malayalam|           123|  2|
|Vaazha: Biopic of...|Malayalam|           125|  3|
|       Kochadaiiyaan|    Tamil|           118|  1|
|   Dhilluku Dhuddu 2|    Tamil|           119|  2|
|      Kadaram Kondan|    Tamil|           121|  3|
|    The Ghazi Attack|   Telugu|           116|  1|
|                Guru|   Telugu|           116|  2|
|   Raju Gari Gadhi 2|   Telugu|           117|  3|
+--------------------+---------+--------------+---+
```

45 Write a query to get language wise top 5 films based first day collections?

In [194…

```python
p_boxf.merge(p_lan, on = "LanguageID",how = "inner")\
    [["Title","Language","First Day Collection Worldwide  in Crores"]]\
    .sort_values(["Language","First Day Collection Worldwide  in Crores"],\
                 ascending = [True,False])\
    .assign(rnk=lambda x: x.groupby("Language").cumcount()+1)\
    .query("rnk <=3")
```

Out[194…

	Title	Language	First Day Collection Worldwide in Crores	rnk
572	Jawan	Hindi	129.10	1
114	Animal	Hindi	116.00	2
573	Pathaan	Hindi	104.80	3
149	K.G.F: Chapter 2	Kannada	159.00	1
187	VR (Vikrant Rona)	Kannada	25.50	2
148	K.G.F: Chapter 1	Kannada	25.00	3
178	Marakkar: Lion of the Arabian Sea	Malayalam	20.30	1
159	Odiyan	Malayalam	18.30	2
180	Kurup	Malayalam	18.30	3
116	Leo	Tamil	142.70	1
285	2	Tamil	105.00	2
135	Maharaja	Tamil	104.84	3
45	RRR (Rise Roar Revolt)	Telugu	223.00	1
396	Bãhubali 2: The Conclusion	Telugu	217.00	2
126	Kalki 2898-AD	Telugu	177.70	3

In [195…

```
# With PySpark
window_spec = Window.partitionBy("Language")\
    .orderBy(desc("First Day Collection Worldwide  in Crores"))
s_boxf.join(s_lan,s_boxf["LanguageID"]==s_lan["LanguageID"],"inner")\
    .withColumn("rnk",row_number().over(window_spec))\
    .filter(col("rnk")<=3)\
    .select("Title","Language",\
            "First Day Collection Worldwide  in Crores","rnk")\
    .show()
```

```
+--------------------+---------+----------------------------------------+---+
|               Title| Language|First Day Collection Worldwide  in Crores|rnk|
+--------------------+---------+----------------------------------------+---+
|               Jawan|    Hindi|                                   129.1|  1|
|              Animal|    Hindi|                                   116.0|  2|
|             Pathaan|    Hindi|                                   104.8|  3|
|    K.G.F: Chapter 2|  Kannada|                                   159.0|  1|
|   VR (Vikrant Rona)|  Kannada|                                    25.5|  2|
|    K.G.F: Chapter 1|  Kannada|                                    25.0|  3|
|Marakkar: Lion of...|Malayalam|                                    20.3|  1|
|              Odiyan|Malayalam|                                    18.3|  2|
|               Kurup|Malayalam|                                    18.3|  3|
|                 Leo|    Tamil|                                   142.7|  1|
|                   2|    Tamil|                                   105.0|  2|
|            Maharaja|    Tamil|                                  104.84|  3|
|RRR (Rise Roar Re...|   Telugu|                                   223.0|  1|
|Bãhubali 2: The C...|   Telugu|                                   217.0|  2|
|       Kalki 2898-AD|   Telugu|                                   177.7|  3|
+--------------------+---------+----------------------------------------+---+
```

In [196…

```
# With Spark SQL
spark.sql("select * from (select Title,Language,\
`First Day Collection Worldwide  in Crores`,row_number() \
over(partition by Language order by\
`First Day Collection Worldwide  in Crores` desc) as rnk \
from boxf_tb boxf \
inner join lan_tb lan on lan.LanguageID = boxf.LanguageID) a \
where rnk <= 3 \
order by Language, \
`First Day Collection Worldwide  in Crores` desc").show()
```

```
+--------------------+---------+----------------------------------------+---+
|               Title| Language|First Day Collection Worldwide  in Crores|rnk|
+--------------------+---------+----------------------------------------+---+
|               Jawan|    Hindi|                                   129.1|  1|
|              Animal|    Hindi|                                   116.0|  2|
|             Pathaan|    Hindi|                                   104.8|  3|
|    K.G.F: Chapter 2|  Kannada|                                   159.0|  1|
|   VR (Vikrant Rona)|  Kannada|                                    25.5|  2|
|    K.G.F: Chapter 1|  Kannada|                                    25.0|  3|
|Marakkar: Lion of...|Malayalam|                                    20.3|  1|
|              Odiyan|Malayalam|                                    18.3|  2|
|               Kurup|Malayalam|                                    18.3|  3|
|                 Leo|    Tamil|                                   142.7|  1|
|                   2|    Tamil|                                   105.0|  2|
|            Maharaja|    Tamil|                                  104.84|  3|
|RRR (Rise Roar Re...|   Telugu|                                   223.0|  1|
|Bāhubali 2: The C...|   Telugu|                                   217.0|  2|
|       Kalki 2898-AD|   Telugu|                                   177.7|  3|
+--------------------+---------+----------------------------------------+---+
```

46 Write a query to get language wise top 5 films based India gross collections?

In [197…

```
# with Pandas
p_boxf.merge(p_lan,on = "LanguageID", how="inner")\
    [["Title","Language","First Day Collection Worldwide  in Crores"]]\
    .sort_values(["Language","First Day Collection Worldwide  in Crores"]\
                 ,ascending = [True,True])\
    .assign(rnk = lambda x: x.groupby("Language").cumcount()+1)\
    .query("rnk <= 3")
```

Out[197…

	Title	Language	First Day Collection Worldwide in Crores	rnk
602	Laapataa Ladies	Hindi	1.40	1
121	12th Fail	Hindi	1.50	2
599	Kill	Hindi	1.50	3
13	Ugramm	Kannada	2.00	1
237	Sapta Sagaradaache Ello: Side A	Kannada	2.30	2
207	Odeya	Kannada	2.50	3
140	Kishkindha Kaandam	Malayalam	0.50	1
227	Romancham	Malayalam	0.50	2
87	Malikappuram	Malayalam	0.52	3
29	Ratsasan	Tamil	0.50	1
125	Dada	Tamil	0.85	2
331	Thadam	Tamil	1.20	3
236	Balagam	Telugu	0.50	1
43	Chalo	Telugu	1.20	2
63	Krack	Telugu	1.40	3

In [198…

```
# With PySpark
window_spec = Window.partitionBy("Language")\
    .orderBy(asc("First Day Collection Worldwide  in Crores"))
s_boxf.join(s_lan,s_boxf["LanguageID"]==s_lan["LanguageID"],"inner")\
    .withColumn("rnk",row_number().over(window_spec))\
    .filter(col("rnk")<=3)\
    .select("Title","Language",\
            "First Day Collection Worldwide  in Crores","rnk").show()
```

```
+--------------------+---------+----------------------------------------+---+
|               Title| Language|First Day Collection Worldwide  in Crores|rnk|
+--------------------+---------+----------------------------------------+---+
|     Laapataa Ladies|    Hindi|                                     1.4|  1|
|           12th Fail|    Hindi|                                     1.5|  2|
|                Kill|    Hindi|                                     1.5|  3|
|              Ugramm|  Kannada|                                     2.0|  1|
|Sapta Sagaradaach...|  Kannada|                                     2.3|  2|
|               Odeya|  Kannada|                                     2.5|  3|
|  Kishkindha Kaandam|Malayalam|                                     0.5|  1|
|           Romancham|Malayalam|                                     0.5|  2|
|        Malikappuram|Malayalam|                                    0.52|  3|
|            Ratsasan|    Tamil|                                     0.5|  1|
|                Dada|    Tamil|                                    0.85|  2|
|              Thadam|    Tamil|                                     1.2|  3|
|             Balagam|   Telugu|                                     0.5|  1|
|               Chalo|   Telugu|                                     1.2|  2|
|               Krack|   Telugu|                                     1.4|  3|
+--------------------+---------+----------------------------------------+---+
```

In [199…

```python
# With Spark SQL
spark.sql("select * from (select Title,Language,\
`First Day Collection Worldwide  in Crores`,\
row_number() over(partition by Language \
order by `First Day Collection Worldwide  in Crores`) as rnk \
from boxf_tb boxf \
inner join lan_tb lan on lan.LanguageID = boxf.LanguageID) a \
where rnk <= 3 \
order by Language,\
`First Day Collection Worldwide  in Crores`").show()
```

```
+--------------------+---------+----------------------------------------+---+
|               Title| Language|First Day Collection Worldwide  in Crores|rnk|
+--------------------+---------+----------------------------------------+---+
|     Laapataa Ladies|    Hindi|                                     1.4|  1|
|           12th Fail|    Hindi|                                     1.5|  2|
|                Kill|    Hindi|                                     1.5|  3|
|              Ugramm|  Kannada|                                     2.0|  1|
|Sapta Sagaradaach...|  Kannada|                                     2.3|  2|
|               Odeya|  Kannada|                                     2.5|  3|
|  Kishkindha Kaandam|Malayalam|                                     0.5|  1|
|           Romancham|Malayalam|                                     0.5|  2|
|        Malikappuram|Malayalam|                                    0.52|  3|
|            Ratsasan|    Tamil|                                     0.5|  1|
|                Dada|    Tamil|                                    0.85|  2|
|              Thadam|    Tamil|                                     1.2|  3|
|             Balagam|   Telugu|                                     0.5|  1|
|               Chalo|   Telugu|                                     1.2|  2|
|               Krack|   Telugu|                                     1.4|  3|
+--------------------+---------+----------------------------------------+---+
```

47 Write a query to get language, Director wise films count?

In [200…

```python
# With Pandas
p_boxf.merge(p_lan,on="LanguageID",how="inner")\
    .merge(p_dir,left_on = "DirectorID",\
           right_on = "Director ID",how="inner")\
    .groupby(["Language","Director"]).agg(Total_Films=("FilmID","count"))\
    .sort_values("Total_Films",ascending = False)\
    .reset_index().head(10)
```

Out[200…

	Language	Director	Total_Films
0	Hindi	Rohit Shetty	6
1	Telugu	Trivikram Srinivas	6
2	Telugu	Anil Ravipudi	5
3	Tamil	Siva	5
4	Telugu	Boyapati Srinu	5
5	Telugu	Maruthi Dasari	4
6	Telugu	Koratala Siva	4
7	Telugu	K.S. Ravindra	4
8	Telugu	Sukumar	4
9	Telugu	Kalyan Krishna	4

In [201…

```python
# With PySpark
```

```
s_boxf.join(s_lan,s_boxf["LanguageID"]==s_lan["LanguageID"],"inner")\
    .join(s_dir, s_boxf["DirectorID"]==s_dir["Director ID"],"inner")\
    .groupby("Language","Director").agg(count("FilmID").alias("Total_Films"))\
    .sort(desc("Total_Films")).show(10)
```

```
+--------+-----------------+-----------+
|Language|          Director|Total_Films|
+--------+-----------------+-----------+
|  Telugu|Trivikram Srinivas|          6|
|   Hindi|      Rohit Shetty|          6|
|   Tamil|              Siva|          5|
|  Telugu|     Anil Ravipudi|          5|
|  Telugu|    Boyapati Srinu|          5|
|   Tamil|         Sundar C.|          4|
|   Hindi|   Siddharth Anand|          4|
|  Telugu|     K.S. Ravindra|          4|
|  Telugu|     Koratala Siva|          4|
|  Telugu|    Kalyan Krishna|          4|
+--------+-----------------+-----------+
only showing top 10 rows
```

In [202…

```
# With Spark SQL
spark.sql("select Language,Director, count(FilmID) \
as Total_Films from boxf_tb boxf \
inner join lan_tb lan on lan.LanguageID = boxf.LanguageID \
inner join dir_tb dir on dir.`Director ID` = boxf.DirectorID \
group by Language,Director \
order by Total_Films desc limit 10").show()
```

```
+--------+-----------------+-----------+
|Language|          Director|Total_Films|
+--------+-----------------+-----------+
|   Hindi|      Rohit Shetty|          6|
|  Telugu|Trivikram Srinivas|          6|
|   Tamil|              Siva|          5|
|  Telugu|     Anil Ravipudi|          5|
|  Telugu|    Boyapati Srinu|          5|
|   Hindi|   Ali Abbas Zafar|          4|
|   Hindi|        Mohit Suri|          4|
|  Telugu|    Maruthi Dasari|          4|
|   Tamil|         H. Vinoth|          4|
|   Hindi|      Remo D'Souza|          4|
+--------+-----------------+-----------+
```

48 Write a query to get language wise OTT platofrm wise fims count?

In [203…

```
# With Pandas
p_boxf.merge(p_lan,on="LanguageID",how="inner")\
    .groupby(["Language","OTT Platform"])\
    .agg(Total_Films=("FilmID","count"))\
    .sort_values("Total_Films",ascending = False)\
    .reset_index().head(10)
```

Out[203…

	Language	OTT Platform	Total_Films
0	Telugu	Amazon Prime Video	119
1	Tamil	Amazon Prime Video	100
2	Hindi	Amazon Prime Video	78
3	Malayalam	Amazon Prime Video	60
4	Hindi	Netflix	52
5	Hindi	Disney+ Hotstar	29
6	Kannada	Amazon Prime Video	27
7	Telugu	Netflix	23
8	Tamil	Netflix	19
9	Hindi	ZEE5	18

In [204…

```
# With PySpark
s_boxf.join(s_lan,s_boxf["LanguageID"]==s_lan["LanguageID"],"inner")\
    .groupby("Language","OTT Platform")\
    .agg(count("FilmID").alias("Total_Films"))\
    .sort(desc("Total_Films")).show(10)
```

```
+---------+------------------+-----------+
| Language|      OTT Platform|Total_Films|
+---------+------------------+-----------+
|   Telugu|Amazon Prime Video|        119|
|    Tamil|Amazon Prime Video|        100|
|    Hindi|Amazon Prime Video|         78|
|Malayalam|Amazon Prime Video|         60|
|    Hindi|           Netflix|         52|
|    Hindi|   Disney+ Hotstar|         29|
|  Kannada|Amazon Prime Video|         27|
|   Telugu|           Netflix|         23|
|    Tamil|           Netflix|         19|
|    Hindi|              Zee5|         18|
+---------+------------------+-----------+
only showing top 10 rows
```

In [205…

```python
# With Spark SQL
spark.sql("select Language,`OTT Platform`, count(FilmID) \
as Total_Films from boxf_tb boxf \
inner join lan_tb lan on lan.LanguageID = boxf.LanguageID \
group by 1,2 \
order by 3 desc limit 10").show()
```

```
+---------+------------------+-----------+
| Language|      OTT Platform|Total_Films|
+---------+------------------+-----------+
|   Telugu|Amazon Prime Video|        119|
|    Tamil|Amazon Prime Video|        100|
|    Hindi|Amazon Prime Video|         78|
|Malayalam|Amazon Prime Video|         60|
|    Hindi|           Netflix|         52|
|    Hindi|   Disney+ Hotstar|         29|
|  Kannada|Amazon Prime Video|         27|
|   Telugu|           Netflix|         23|
|    Tamil|           Netflix|         19|
|    Hindi|              Zee5|         18|
+---------+------------------+-----------+
```

49 What are top 10 films based language and first day collection?

In [206…

```python
# With Pandas
p_boxf.merge(p_lan,on = "LanguageID",how= 'inner')\
    [['Title','Language','First Day Collection Worldwide  in Crores']]\
    .sort_values(['Language','First Day Collection Worldwide  in Crores']\
                 ,ascending = [True,False])\
    .assign(rnk = lambda x:x.groupby("Language")\
        .cumcount()+1).query("rnk <=10")
```

Out[206…

	Title	Language	First Day Collection Worldwide in Crores	rnk
572	Jawan	Hindi	129.10	1
114	Animal	Hindi	116.00	2
573	Pathaan	Hindi	104.80	3
574	Tiger 3	Hindi	94.00	4
127	Stree 2: Sarkate Ka Aatank	Hindi	80.00	5
484	Thugs of Hindostan	Hindi	76.00	6
199	War	Hindi	74.30	7
27	Sultan	Hindi	71.30	8
544	Brahmastra Part One: Shiva	Hindi	70.00	9
519	Prem Ratan Dhan Payo	Hindi	66.20	10
149	K.G.F: Chapter 2	Kannada	159.00	1
187	VR (Vikrant Rona)	Kannada	25.50	2
148	K.G.F: Chapter 1	Kannada	25.00	3
188	James	Kannada	23.50	4
181	Roberrt	Kannada	16.90	5
225	Kaatera	Kannada	13.00	6
230	Kabzaa	Kannada	12.50	7
155	The Villain	Kannada	12.00	8
228	Kranti	Kannada	11.60	9
182	Pogaru	Kannada	10.25	10

178	Marakkar: Lion of the Arabian Sea	Malayalam	20.30	1
159	Odiyan	Malayalam	18.30	2
180	Kurup	Malayalam	18.30	3
129	The Goat Life	Malayalam	16.22	4
246	Turbo	Malayalam	15.70	5
231	King of Kotha	Malayalam	15.00	6
98	Lucifer	Malayalam	14.00	7
15	Bheeshmaparvam	Malayalam	12.20	8
252	Malaikottai Vaaliban	Malayalam	12.15	9
130	Aavesham	Malayalam	10.50	10
116	Leo	Tamil	142.70	1
285	2	Tamil	105.00	2
135	Maharaja	Tamil	104.84	3
343	The GOAT	Tamil	100.75	4
117	Jailer	Tamil	96.60	5
262	Kabali	Tamil	90.00	6
260	Beast	Tamil	82.40	7
77	Ponniyin Selvan: Part One	Tamil	80.70	8
263	Sarkar	Tamil	71.00	9
344	Vettaiyan	Tamil	67.00	10
45	RRR (Rise Roar Revolt)	Telugu	223.00	1
396	Bāhubali 2: The Conclusion	Telugu	217.00	2
126	Kalki 2898-AD	Telugu	177.70	3
212	Salaar: Part 1 - Ceasefire	Telugu	158.10	4
455	Devara: Part 1	Telugu	142.00	5
213	Adipurush	Telugu	127.50	6
433	Sye Raa Narasimha Reddy	Telugu	85.00	7
397	Bāhubali: The Beginning	Telugu	73.00	8
456	Guntur Kaaram	Telugu	68.70	9
364	Sarkaru Vaari Paata	Telugu	68.50	10

In [207…

```
# With PySpark
window_spec = Window.partitionBy("Language")\
    .orderBy(desc("First Day Collection Worldwide  in Crores"))
s_boxf.join(s_lan,s_boxf["LanguageID"]==s_lan["LanguageID"],"inner")\
    .withColumn("rnk",row_number().over(window_spec))\
    .filter(col("rnk")<=10)\
    .select('Title','Language',\
            'First Day Collection Worldwide  in Crores','rnk').show(50)
```

```
+--------------------+---------+---------------------------------------+---+
|               Title| Language|First Day Collection Worldwide  in Crores|rnk|
+--------------------+---------+---------------------------------------+---+
|               Jawan|    Hindi|                                  129.1|  1|
|              Animal|    Hindi|                                  116.0|  2|
|             Pathaan|    Hindi|                                  104.8|  3|
|             Tiger 3|    Hindi|                                   94.0|  4|
|Stree 2: Sarkate ...|    Hindi|                                   80.0|  5|
|  Thugs of Hindostan|    Hindi|                                   76.0|  6|
|                 War|    Hindi|                                   74.3|  7|
|              Sultan|    Hindi|                                   71.3|  8|
|Brahmastra Part O...|    Hindi|                                   70.0|  9|
|Prem Ratan Dhan Payo|    Hindi|                                   66.2| 10|
|    K.G.F: Chapter 2|  Kannada|                                  159.0|  1|
|   VR (Vikrant Rona)|  Kannada|                                   25.5|  2|
|    K.G.F: Chapter 1|  Kannada|                                   25.0|  3|
|               James|  Kannada|                                   23.5|  4|
|             Roberrt|  Kannada|                                   16.9|  5|
|             Kaatera|  Kannada|                                   13.0|  6|
|              Kabzaa|  Kannada|                                   12.5|  7|
|         The Villain|  Kannada|                                   12.0|  8|
|              Kranti|  Kannada|                                   11.6|  9|
|              Pogaru|  Kannada|                                  10.25| 10|
|Marakkar: Lion of...|Malayalam|                                   20.3|  1|
|              Odiyan|Malayalam|                                   18.3|  2|
|               Kurup|Malayalam|                                   18.3|  3|
|       The Goat Life|Malayalam|                                  16.22|  4|
|               Turbo|Malayalam|                                   15.7|  5|
|       King of Kotha|Malayalam|                                   15.0|  6|
|             Lucifer|Malayalam|                                   14.0|  7|
|      Bheeshmaparvam|Malayalam|                                   12.2|  8|
|Malaikottai Vaaliban|Malayalam|                                  12.15|  9|
|            Aavesham|Malayalam|                                   10.5| 10|
|                 Leo|    Tamil|                                  142.7|  1|
|                   2|    Tamil|                                  105.0|  2|
|            Maharaja|    Tamil|                                 104.84|  3|
|            The GOAT|    Tamil|                                 100.75|  4|
|              Jailer|    Tamil|                                   96.6|  5|
|              Kabali|    Tamil|                                   90.0|  6|
|               Beast|    Tamil|                                   82.4|  7|
|Ponniyin Selvan: ...|    Tamil|                                   80.7|  8|
|              Sarkar|    Tamil|                                   71.0|  9|
|           Vettaiyan|    Tamil|                                   67.0| 10|
|RRR (Rise Roar Re...|   Telugu|                                  223.0|  1|
|Bãhubali 2: The C...|   Telugu|                                  217.0|  2|
|       Kalki 2898-AD|   Telugu|                                  177.7|  3|
|Salaar: Part 1 - ...|   Telugu|                                  158.1|  4|
|      Devara: Part 1|   Telugu|                                  142.0|  5|
|           Adipurush|   Telugu|                                  127.5|  6|
|Sye Raa Narasimha...|   Telugu|                                   85.0|  7|
|Bãhubali: The Beg...|   Telugu|                                   73.0|  8|
|       Guntur Kaaram|   Telugu|                                   68.7|  9|
| Sarkaru Vaari Paata|   Telugu|                                   68.5| 10|
+--------------------+---------+---------------------------------------+---+
```

In [208…

```python
# With Spark SQL
spark.sql("select * from (select Title, Language,\
`First Day Collection Worldwide  in Crores`,\
row_number() over(Partition by Language \
order by `First Day Collection Worldwide  in Crores` desc) as rnk \
from boxf_tb boxf  \
inner join lan_tb lan on lan.LanguageID = boxf.LanguageID)a \
where rnk <= 10 \
order by Language,\
`First Day Collection Worldwide  in Crores` desc").show()
```

```
+--------------------+--------+----------------------------------------+---+
|               Title|Language|First Day Collection Worldwide  in Crores|rnk|
+--------------------+--------+----------------------------------------+---+
|               Jawan|   Hindi|                                   129.1|  1|
|              Animal|   Hindi|                                   116.0|  2|
|             Pathaan|   Hindi|                                   104.8|  3|
|             Tiger 3|   Hindi|                                    94.0|  4|
|Stree 2: Sarkate ...|   Hindi|                                    80.0|  5|
|  Thugs of Hindostan|   Hindi|                                    76.0|  6|
|                 War|   Hindi|                                    74.3|  7|
|              Sultan|   Hindi|                                    71.3|  8|
|Brahmastra Part O...|   Hindi|                                    70.0|  9|
|Prem Ratan Dhan Payo|   Hindi|                                    66.2| 10|
|    K.G.F: Chapter 2| Kannada|                                   159.0|  1|
|   VR (Vikrant Rona)| Kannada|                                    25.5|  2|
|    K.G.F: Chapter 1| Kannada|                                    25.0|  3|
|               James| Kannada|                                    23.5|  4|
|             Roberrt| Kannada|                                    16.9|  5|
|             Kaatera| Kannada|                                    13.0|  6|
|              Kabzaa| Kannada|                                    12.5|  7|
|         The Villain| Kannada|                                    12.0|  8|
|              Kranti| Kannada|                                    11.6|  9|
|              Pogaru| Kannada|                                   10.25| 10|
+--------------------+--------+----------------------------------------+---+
only showing top 20 rows
```

50 Write a query to get director wise number of fims released in from year 2017 to 2019

In [209…

```
# With Pandas
p_boxf.merge(p_dir,left_on ="DirectorID",right_on="Director ID",how="right")\
    .query("Year in [2017,2018,2019]").groupby(["Director","Year"])\
    .agg(Total_Films=("FilmID","count"))\
    .sort_values("Total_Films",ascending = False)\
    .reset_index().head(10)
```

Out[209…

	Director	Year	Total_Films
0	Trivikram Srinivas	2018	2
1	A.R. Murugadoss	2017	1
2	S Krishna	2019	1
3	Rajkumar Hirani	2018	1
4	Ram Kumar	2018	1
5	Rambhala	2019	1
6	Ramesh Varma	2019	1
7	Ravi Udyawar	2017	1
8	Reema Kagti	2018	1
9	Remo D'Souza	2018	1

In [210…

```
# With PySpark
s_boxf.join(s_dir,s_boxf["DirectorID"]==s_dir["Director ID"],"inner")\
    .filter(col("Year").isin(2017,2018,2019)).groupby("Director","Year")\
    .agg(count("FilmID").alias("Total_Films")).sort(desc("Total_Films"))\
    .show(10)
```

```
+--------------------+----+-----------+
|            Director|Year|Total_Films|
+--------------------+----+-----------+
|  Trivikram Srinivas|2018|          2|
|Puri Jagannadh, A...|2019|          1|
|     Abhishek Varman|2019|          1|
|       Milan Luthria|2017|          1|
|          K.V. Anand|2017|          1|
|          Milind Rau|2017|          1|
|     Rajkumar Hirani|2018|          1|
|    Shashank Khaitan|2017|          1|
|     Ali Abbas Zafar|2019|          1|
|      Suresh Triveni|2017|          1|
+--------------------+----+-----------+
only showing top 10 rows
```

In [211…

```
# With Spark SQL
spark.sql("""select Director, YEAR(try_to_timestamp(`Release Date`,\
'dd-MMM-yy')) as YR,count(FilmID) as Total_Films from boxf_tb boxf \
right join dir_tb dir on dir.`Director ID`=boxf.DirectorID \
group by Director, YR \
having YEAR(try_to_timestamp(`Release Date`,'dd-MMM-yy'))\
between 2017 and 2019 \
```

```
order by 3 desc limit 10""").show()
```

```
+--------------------+----+-----------+
|            Director|  YR|Total_Films|
+--------------------+----+-----------+
|  Trivikram Srinivas|2018|          2|
|Puri Jagannadh, A...|2019|          1|
|     Abhishek Varman|2019|          1|
|       Milan Luthria|2017|          1|
|          K.V. Anand|2017|          1|
|          Milind Rau|2017|          1|
|     Rajkumar Hirani|2018|          1|
|    Shashank Khaitan|2017|          1|
|     Ali Abbas Zafar|2019|          1|
|      Suresh Triveni|2017|          1|
+--------------------+----+-----------+
```

51 Write a query to get director wise world wide collections?

In [212…

```
# With Pandas
p_boxf.merge(p_dir,left_on = "DirectorID",right_on="Director ID",how="right")\
    .groupby("Director").agg(Total_WWC=("Worldwide Collection  in Crores","sum"))\
    .sort_values("Total_WWC",ascending = False).reset_index().head(10)
```

Out[212…

	Director	Total_WWC
0	S.S. Rajamouli	3668.00
1	Siddharth Anand	2229.45
2	Nitesh Tiwari	2122.30
3	Prashanth Neel	2101.75
4	Atlee	1865.35
5	Rajkumar Hirani	1834.50
6	Rohit Shetty	1656.60
7	Lokesh Kanagaraj	1350.13
8	Sandeep Reddy Vanga	1342.00
9	Kabir Khan	1220.55

In [213…

```
# With PySpark
s_boxf.join(s_dir,s_boxf["DirectorID"]==s_dir["Director ID"],"right")\
    .groupby("Director").agg(sum("Worldwide Collection  in Crores")\
        .alias("Total_WWC")).sort(desc("Total_WWC")).show(10)
```

```
+-------------------+------------------+
|           Director|         Total_WWC|
+-------------------+------------------+
|     S.S. Rajamouli|            3668.0|
|    Siddharth Anand|           2229.45|
|      Nitesh Tiwari|            2122.3|
|     Prashanth Neel|           2101.75|
|              Atlee|           1865.35|
|    Rajkumar Hirani|            1834.5|
|       Rohit Shetty|            1656.6|
|   Lokesh Kanagaraj|1350.1299999999999|
|Sandeep Reddy Vanga|            1342.0|
|         Kabir Khan|           1220.55|
+-------------------+------------------+
only showing top 10 rows
```

In [214…

```
# With Spark SQL
spark.sql("Select Director, sum(`Worldwide Collection  in Crores`) as Total_WWC \
from boxf_tb boxf right join dir_tb dir on dir.`Director ID`=boxf.DirectorID \
group by Director \
order by Total_WWC desc limit 10").show()
```

```
+-------------------+------------------+
|           Director|         Total_WWC|
+-------------------+------------------+
|     S.S. Rajamouli|            3668.0|
|    Siddharth Anand|           2229.45|
|      Nitesh Tiwari|            2122.3|
|     Prashanth Neel|           2101.75|
|              Atlee|           1865.35|
|    Rajkumar Hirani|            1834.5|
|       Rohit Shetty|            1656.6|
|   Lokesh Kanagaraj|1350.1299999999999|
|Sandeep Reddy Vanga|            1342.0|
|         Kabir Khan|           1220.55|
+-------------------+------------------+
```

52 Write a query to get director wise first day world wide collections?

In [215…

```python
# With Pandas
p_boxf.merge(p_dir,left_on = "DirectorID",right_on="Director ID",how="right")\
    .groupby("Director").agg(Total_WWC=("Worldwide Collection  in Crores","sum"))\
    .sort_values("Total_WWC",ascending = False).reset_index().head(10)
```

Out[215…

	Director	Total_WWC
0	S.S. Rajamouli	3668.00
1	Siddharth Anand	2229.45
2	Nitesh Tiwari	2122.30
3	Prashanth Neel	2101.75
4	Atlee	1865.35
5	Rajkumar Hirani	1834.50
6	Rohit Shetty	1656.60
7	Lokesh Kanagaraj	1350.13
8	Sandeep Reddy Vanga	1342.00
9	Kabir Khan	1220.55

In [216…

```python
# With PySpark
s_boxf.join(s_dir,s_boxf["DirectorID"]==s_dir["Director ID"],"right")\
    .groupby("Director").agg(sum("Worldwide Collection  in Crores")\
        .alias("Total_WWC")).sort(desc("Total_WWC")).show(10)
```

```
+-------------------+------------------+
|           Director|         Total_WWC|
+-------------------+------------------+
|     S.S. Rajamouli|            3668.0|
|    Siddharth Anand|           2229.45|
|      Nitesh Tiwari|            2122.3|
|     Prashanth Neel|           2101.75|
|              Atlee|           1865.35|
|    Rajkumar Hirani|            1834.5|
|       Rohit Shetty|            1656.6|
|   Lokesh Kanagaraj|1350.1299999999999|
|Sandeep Reddy Vanga|            1342.0|
|         Kabir Khan|           1220.55|
+-------------------+------------------+
only showing top 10 rows
```

In [217…

```python
# With Spark SQL
spark.sql("Select Director, sum(`Worldwide Collection  in Crores`) as Total_WWC \
from boxf_tb boxf right join dir_tb dir on dir.`Director ID`=boxf.DirectorID \
group by Director \
order by Total_WWC desc limit 10").show()
```

```
+-------------------+------------------+
|           Director|         Total_WWC|
+-------------------+------------------+
|     S.S. Rajamouli|            3668.0|
|    Siddharth Anand|           2229.45|
|      Nitesh Tiwari|            2122.3|
|     Prashanth Neel|           2101.75|
|              Atlee|           1865.35|
|    Rajkumar Hirani|            1834.5|
|       Rohit Shetty|            1656.6|
|   Lokesh Kanagaraj|1350.1299999999999|
|Sandeep Reddy Vanga|            1342.0|
|         Kabir Khan|           1220.55|
+-------------------+------------------+
```

53 Write a query to get director wise India gross collections?

In [218…

```python
# With Pandas
p_boxf.merge(p_dir,left_on = "DirectorID",right_on="Director ID",how="right")\
    .groupby("Director").agg(Total_IGC=("India Gross Collection in Crores","sum"))\
    .sort_values("Total_IGC",ascending = False).reset_index().head(10)
```

Out[218…

	Director	Total_IGC
0	S.S. Rajamouli	2848.85
1	Prashanth Neel	1747.60
2	Siddharth Anand	1560.91
3	Atlee	1256.05
4	Rohit Shetty	1246.20
5	Rajkumar Hirani	1199.50
6	Sandeep Reddy Vanga	1028.40
7	Amar Kaushik	961.48
8	Lokesh Kanagaraj	956.88
9	Ali Abbas Zafar	856.79

In [219…

```python
# With PySpark
s_boxf.join(s_dir,s_boxf["DirectorID"]==s_dir["Director ID"],"right")\
    .groupby("Director").agg(sum("India Gross Collection in Crores")\
        .alias("Total_IGC")).sort(desc("Total_IGC")).show(10)
```

```
+-------------------+------------------+
|           Director|         Total_IGC|
+-------------------+------------------+
|     S.S. Rajamouli|           2848.85|
|     Prashanth Neel|            1747.6|
|    Siddharth Anand|1560.9099999999999|
|              Atlee|           1256.05|
|       Rohit Shetty|            1246.2|
|    Rajkumar Hirani|            1199.5|
|Sandeep Reddy Vanga|            1028.4|
|       Amar Kaushik|            961.48|
|   Lokesh Kanagaraj| 956.8799999999999|
|    Ali Abbas Zafar|            856.79|
+-------------------+------------------+
only showing top 10 rows
```

In [220…

```python
# With Spark SQL
spark.sql("Select Director, sum(`India Gross Collection in Crores`) as Total_IGC \
from boxf_tb boxf right join dir_tb dir on dir.`Director ID`=boxf.DirectorID \
group by Director \
order by Total_IGC desc limit 10").show()
```

```
+-------------------+------------------+
|           Director|         Total_IGC|
+-------------------+------------------+
|     S.S. Rajamouli|           2848.85|
|     Prashanth Neel|            1747.6|
|    Siddharth Anand|1560.9099999999999|
|              Atlee|           1256.05|
|       Rohit Shetty|            1246.2|
|    Rajkumar Hirani|            1199.5|
|Sandeep Reddy Vanga|            1028.4|
|       Amar Kaushik|            961.48|
|   Lokesh Kanagaraj| 956.8799999999999|
|    Ali Abbas Zafar|            856.79|
+-------------------+------------------+
```

54 Write a query to get director wise overseas collections?

In [221…

```
# With Pandas
p_boxf.merge(p_dir,left_on = "DirectorID",right_on="Director ID",how="right")\
    .groupby("Director").agg(Total_OSC=("Overseas Collection in Crores","sum"))\
    .sort_values("Total_OSC",ascending = False).reset_index().head(10)
```

Out[221…

	Director	Total_OSC
0	Nitesh Tiwari	1535.30
1	Advait Chandan	892.40
2	Siddharth Anand	819.37
3	S.S. Rajamouli	819.15
4	Rajkumar Hirani	635.00
5	Atlee	609.30
6	Kabir Khan	549.05
7	Rohit Shetty	410.40
8	Lokesh Kanagaraj	393.25
9	Prashanth Neel	354.15

In [222…

```
# With PySpark
s_boxf.join(s_dir,s_boxf["DirectorID"]==s_dir["Director ID"],"right")\
    .groupby("Director").agg(sum("Overseas Collection in Crores")\
        .alias("Total_OSC")).sort(desc("Total_OSC")).show(10)
```

```
+----------------+---------+
|        Director|Total_OSC|
+----------------+---------+
|   Nitesh Tiwari|   1535.3|
|  Advait Chandan|    892.4|
| Siddharth Anand|   819.37|
|  S.S. Rajamouli|   819.15|
| Rajkumar Hirani|    635.0|
|           Atlee|    609.3|
|      Kabir Khan|   549.05|
|    Rohit Shetty|    410.4|
|Lokesh Kanagaraj|   393.25|
|  Prashanth Neel|   354.15|
+----------------+---------+
only showing top 10 rows
```

In [223…

```
# With Spark SQL
spark.sql("Select Director, sum(`Overseas Collection in Crores`) as Total_OSC \
from boxf_tb boxf right join dir_tb dir on dir.`Director ID`=boxf.DirectorID \
group by Director \
order by Total_OSC desc limit 10").show()
```

```
+----------------+---------+
|        Director|Total_OSC|
+----------------+---------+
|   Nitesh Tiwari|   1535.3|
|  Advait Chandan|    892.4|
| Siddharth Anand|   819.37|
|  S.S. Rajamouli|   819.15|
| Rajkumar Hirani|    635.0|
|           Atlee|    609.3|
|      Kabir Khan|   549.05|
|    Rohit Shetty|    410.4|
|Lokesh Kanagaraj|   393.25|
|  Prashanth Neel|   354.15|
+----------------+---------+
```

55 Write a query to get director, lead actor/actress and number of films?

In [224…

```
# With Pandas
p_boxf.merge(p_dir,left_on="DirectorID",right_on="Director ID",how="right")\
    .groupby(["Director","Lead Actor/Actress"]).agg(Total_Films=("FilmID","count"))\
    .sort_values("Total_Films",ascending=False).reset_index().head(10)
```

Out[224…

	Director	Lead Actor/Actress	Total_Films
0	Siva	Ajith Kumar	4
1	H. Vinoth	Ajith Kumar	3
2	Sabbir Khan	Tiger Shroff	3
3	Siddharth Anand	Hrithik Roshan	3
4	Atlee	Joseph Vijay	3
5	Ajay Devgn	Ajay Devgn	3
6	Prashanth Neel	Yash	2
7	Mani Ratnam, Sruti Harihara Subramanian	Vikram	2
8	Priyadarshan	Mohanlal	2
9	Koratala Siva	N.T. Rama Rao Jr.	2

In [225…

```
# With PySpark
s_boxf.join(s_dir,s_boxf["DirectorID"]==s_dir["Director ID"],"right")\
    .groupby("Director","Lead Actor/Actress").agg(count("FilmID")\
        .alias("Total_Films")).sort(desc("Total_Films")).show(10)
```

```
+-----------------+------------------+-----------+
|         Director|Lead Actor/Actress|Total_Films|
+-----------------+------------------+-----------+
|             Siva|       Ajith Kumar|          4|
|  Siddharth Anand|    Hrithik Roshan|          3|
|       Ajay Devgn|        Ajay Devgn|          3|
|      Sabbir Khan|      Tiger Shroff|          3|
|            Atlee|      Joseph Vijay|          3|
|        H. Vinoth|       Ajith Kumar|          3|
|Tinu Suresh Desai|      Akshay Kumar|          2|
|           Vysakh|         Mammootty|          2|
|  A.R. Murugadoss|      Joseph Vijay|          2|
|    Vivek Athreya|              Nani|          2|
+-----------------+------------------+-----------+
only showing top 10 rows
```

In [226…

```
# With Spark SQL
spark.sql("select Director,`Lead Actor/Actress`, \
count(FilmID) as Total_Films from boxf_tb boxf \
right join dir_tb dir on dir.`Director ID` = boxf.DirectorID \
group by Director, `Lead Actor/Actress` \
order by Total_Films desc limit 10").show()
```

```
+-----------------+------------------+-----------+
|         Director|Lead Actor/Actress|Total_Films|
+-----------------+------------------+-----------+
|             Siva|       Ajith Kumar|          4|
|       Ajay Devgn|        Ajay Devgn|          3|
|      Sabbir Khan|      Tiger Shroff|          3|
|  Siddharth Anand|    Hrithik Roshan|          3|
|            Atlee|      Joseph Vijay|          3|
|        H. Vinoth|       Ajith Kumar|          3|
|      Vetrimaaran|           Dhanush|          2|
|Tinu Suresh Desai|      Akshay Kumar|          2|
|    Koratala Siva| N.T. Rama Rao Jr.|          2|
|  A.R. Murugadoss|      Joseph Vijay|          2|
+-----------------+------------------+-----------+
```

56 Write a query to get films which is having budget on between 150 crores and 277 crores?

In [227…

```
# With Pandas
p_boxf[p_boxf["Budget in Crores"].between(150,277)][["Title","Budget in Crores"]]\
    .sort_values("Budget in Crores",ascending = False).reset_index().head(10)
```

Out[227…

	index	Title	Budget in Crores
0	484	Thugs of Hindostan	275.0
1	591	Bade Miyan Chote Miyan	250.0
2	77	Ponniyin Selvan: Part One	250.0
3	586	Fighter	250.0
4	574	Tiger 3	250.0
5	116	Leo	250.0
6	396	Bãhubali 2: The Conclusion	250.0
7	573	Pathaan	250.0
8	345	Indian 2	250.0
9	455	Devara: Part 1	250.0

In [228…

```
# With PySpark
s_boxf[s_boxf["Budget in Crores"].between(150,277)]\
    [["Title","Budget in Crores"]].sort(desc("Budget in Crores"))\
    .show(10)
```

```
+--------------------+----------------+
|               Title|Budget in Crores|
+--------------------+----------------+
|  Thugs of Hindostan|           275.0|
|Ponniyin Selvan: ...|           250.0|
|                 Leo|           250.0|
|Ponniyin Selvan: ...|           250.0|
|            Indian 2|           250.0|
|Bãhubali 2: The C...|           250.0|
|      Devara: Part 1|           250.0|
|             Pathaan|           250.0|
|             Tiger 3|           250.0|
|             Fighter|           250.0|
+--------------------+----------------+
only showing top 10 rows
```

In [229…

```
# With Spark SQL
spark.sql("select Title, `Budget in Crores` from boxf_tb \
where `Budget in Crores` between 150 and 277 \
order by `Budget in Crores` desc limit 10").show()
```

```
+--------------------+----------------+
|               Title|Budget in Crores|
+--------------------+----------------+
|  Thugs of Hindostan|           275.0|
|Ponniyin Selvan: ...|           250.0|
|                 Leo|           250.0|
|Ponniyin Selvan: ...|           250.0|
|            Indian 2|           250.0|
|Bãhubali 2: The C...|           250.0|
|      Devara: Part 1|           250.0|
|             Pathaan|           250.0|
|             Tiger 3|           250.0|
|             Fighter|           250.0|
+--------------------+----------------+
```

57 Write a query to get director, week name wise films released?

In [230…
```
# With Pandas
p_boxf.merge(p_dir, left_on ="DirectorID", right_on = "Director ID",how="right")\
    .groupby(["WeekDay","Director"]).agg(Total_Films=("FilmID","count"))\
    .sort_values("Total_Films",ascending = False).reset_index().head(10)
```

Out[230…

	WeekDay	Director	Total_Films
0	Friday	Rohit Shetty	6
1	Friday	Siva	4
2	Friday	Mohit Suri	4
3	Friday	Farhad Samji	3
4	Friday	Anil Ravipudi	3
5	Friday	Anees Bazmee	3
6	Friday	Meghna Gulzar	3
7	Friday	Remo D'Souza	3
8	Friday	Boyapati Srinu	3
9	Wednesday	Ali Abbas Zafar	3

In [231…
```
# With PySpark
s_boxf.join(s_dir,s_boxf["DirectorID"]==s_dir["Director ID"],"right")\
    .groupby("WeekDay","Director").agg(count("FilmID").alias("Total_Films"))\
    .sort(desc("Total_Films")).show(10)
```

```
+-------+--------------+-----------+
|WeekDay|      Director|Total_Films|
+-------+--------------+-----------+
| Friday|  Rohit Shetty|          6|
| Friday|          Siva|          4|
| Friday|    Mohit Suri|          4|
| Friday| Anil Ravipudi|          3|
| Friday|   Sabbir Khan|          3|
| Friday|     H. Vinoth|          3|
| Friday|  P.S. Mithran|          3|
| Friday|    K.V. Anand|          3|
| Friday|Surender Reddy|          3|
| Friday|Boyapati Srinu|          3|
+-------+--------------+-----------+
only showing top 10 rows
```

In [232…
```
s_boxf.createOrReplaceTempView("boxf_tb")
```

In [233…
```
# With Spark SQL
spark.sql("select WeekDay, Director,count(FilmID)as Total_Films from \
boxf_tb boxf right join dir_tb dir on dir.`Director ID`=boxf.DirectorID \
group by WeekDay, Director \
order by Total_Films desc limit 10").show()
```

```
+-------+--------------+-----------+
|WeekDay|      Director|Total_Films|
+-------+--------------+-----------+
| Friday|  Rohit Shetty|          6|
| Friday|          Siva|          4|
| Friday|    Mohit Suri|          4|
| Friday|Surender Reddy|          3|
| Friday|   Sabbir Khan|          3|
| Friday|     H. Vinoth|          3|
| Friday|  P.S. Mithran|          3|
| Friday|Prashanth Neel|          3|
| Friday|    K.V. Anand|          3|
| Friday|  Anees Bazmee|          3|
+-------+--------------+-----------+
```

58 Write a query to get OTT platofrm and director wise films released?

In [234…
```
# With Pandas
p_boxf.merge(p_dir,left_on="DirectorID",right_on="Director ID",how="right")\
    .groupby(["OTT Platform","Director"]).agg(Total_Films=("FilmID","count"))\
    .sort_values("Total_Films",ascending = False).reset_index().head(10)
```

Out[234…

	OTT Platform	Director	Total_Films
0	Amazon Prime Video	Anil Ravipudi	5
1	Amazon Prime Video	Trivikram Srinivas	5
2	Amazon Prime Video	A.R. Murugadoss	4
3	Amazon Prime Video	K.S. Ravindra	4
4	Netflix	Rohit Shetty	4
5	Amazon Prime Video	Prashanth Neel	4
6	Amazon Prime Video	Siva	4
7	Amazon Prime Video	Sukumar	4
8	Amazon Prime Video	Vikram K. Kumar	4
9	Amazon Prime Video	Sundar C.	4

In [235…

```
# With PySpark
s_boxf.join(s_dir,s_boxf["DirectorID"]==s_dir["Director ID"],"right")\
    .groupby("OTT Platform","Director").agg(count("FilmID").alias("Total_Films"))\
    .sort(desc("Total_Films")).show(10)
```

```
+------------------+------------------+-----------+
|      OTT Platform|          Director|Total_Films|
+------------------+------------------+-----------+
|Amazon Prime Video|     Anil Ravipudi|          5|
|Amazon Prime Video|Trivikram Srinivas|          5|
|           Netflix|      Rohit Shetty|          4|
|Amazon Prime Video|     K.S. Ravindra|          4|
|Amazon Prime Video|   Vikram K. Kumar|          4|
|Amazon Prime Video| Vamshi Paidipally|          4|
|Amazon Prime Video|    Prashanth Neel|          4|
|Amazon Prime Video|           Sukumar|          4|
|Amazon Prime Video|   A.R. Murugadoss|          4|
|Amazon Prime Video|              Siva|          4|
+------------------+------------------+-----------+
only showing top 10 rows
```

In [236…

```
# with Spark SQL
spark.sql("select `OTT Platform`, Director , count(FilmID) as \
Total_Films from boxf_tb boxf right join dir_tb dir on \
dir.`Director ID` = boxf.DirectorID \
group by 1,2 \
order by 3 desc limit 10").show()
```

```
+------------------+------------------+-----------+
|      OTT Platform|          Director|Total_Films|
+------------------+------------------+-----------+
|Amazon Prime Video|     Anil Ravipudi|          5|
|Amazon Prime Video|Trivikram Srinivas|          5|
|Amazon Prime Video|     K.S. Ravindra|          4|
|Amazon Prime Video|    Prashanth Neel|          4|
|Amazon Prime Video| Vamshi Paidipally|          4|
|           Netflix|      Rohit Shetty|          4|
|Amazon Prime Video|   Vikram K. Kumar|          4|
|Amazon Prime Video|           Sukumar|          4|
|Amazon Prime Video|   A.R. Murugadoss|          4|
|Amazon Prime Video|              Siva|          4|
+------------------+------------------+-----------+
```

59 Write a query to get director wise films released on Friday only?

In [237…

```
# With Pandas
p_boxf.merge(p_dir, left_on ="DirectorID", right_on = "Director ID",how="right")\
    .query("WeekDay == 'Friday'").groupby(["WeekDay","Director"])\
    .agg(Total_Films=("FilmID","count"))\
    .sort_values("Total_Films",ascending = False)\
    .reset_index().head(10)
```

Out[237…

	WeekDay	Director	Total_Films
0	Friday	Rohit Shetty	6
1	Friday	Siva	4
2	Friday	Mohit Suri	4
3	Friday	Farhad Samji	3
4	Friday	Surender Reddy	3
5	Friday	Meghna Gulzar	3
6	Friday	Anil Ravipudi	3
7	Friday	S.S. Rajamouli	3
8	Friday	Boyapati Srinu	3
9	Friday	Remo D'Souza	3

In [238…

```
# With PySpark
s_boxf.join(s_dir,s_boxf["DirectorID"]==s_dir["Director ID"],"right")\
    .filter(col("Weekday")=="Friday")\
    .groupby("WeekDay","Director").agg(count("FilmID").alias("Total_Films"))\
    .sort(desc("Total_Films")).show(10)
```

```
+-------+--------------+-----------+
|WeekDay|      Director|Total_Films|
+-------+--------------+-----------+
| Friday|  Rohit Shetty|          6|
| Friday|          Siva|          4|
| Friday|    Mohit Suri|          4|
| Friday| Anil Ravipudi|          3|
| Friday|   Sabbir Khan|          3|
| Friday|     H. Vinoth|          3|
| Friday|  P.S. Mithran|          3|
| Friday|    K.V. Anand|          3|
| Friday|  Anees Bazmee|          3|
| Friday|Boyapati Srinu|          3|
+-------+--------------+-----------+
only showing top 10 rows
```

In [239…

```
# With Spark SQL
spark.sql("select WeekDay, Director,count(FilmID)as Total_Films from \
boxf_tb boxf right join dir_tb dir on dir.`Director ID`=boxf.DirectorID \
where Weekday ='Friday' \
group by WeekDay, Director \
order by Total_Films desc limit 10").show()
```

```
+-------+--------------+-----------+
|WeekDay|      Director|Total_Films|
+-------+--------------+-----------+
| Friday|  Rohit Shetty|          6|
| Friday|          Siva|          4|
| Friday|    Mohit Suri|          4|
| Friday|Surender Reddy|          3|
| Friday|   Sabbir Khan|          3|
| Friday|     H. Vinoth|          3|
| Friday|  P.S. Mithran|          3|
| Friday|Prashanth Neel|          3|
| Friday|    K.V. Anand|          3|
| Friday|  Anees Bazmee|          3|
+-------+--------------+-----------+
```

60 Write a query to get films based on IMDb reating between 6.5 and 7.7?

In [240…

```
# With Pandas
p_boxf[p_boxf["IMDb Rating"].between(6.5,7.7)][["Title","IMDb Rating"]]\
    .sort_values("Title",ascending = False).head(10)
```

Out[240…

	Title	IMDb Rating
307	Yennai Arindhaal	7.3
356	Yashoda	6.5
199	War	6.5
321	Viswasam	6.5
211	Virus	7.7
221	Virupaksha	7.2
546	Vikram Vedha	7.1
308	Vendhu Thanindhathu Kaadu	7.3
299	Velaikkaran	7.2
9	Veeram	6.5

In [241…

```
# With PySpark
s_boxf.filter(col("IMDb Rating").isin(6.5,7.7))\
    .select("Title","IMDb Rating")\
    .sort(desc("Title")).show(10)
```

```
+--------------------+-----------+
|               Title|IMDb Rating|
+--------------------+-----------+
|             Yashoda|        6.5|
|                 War|        6.5|
|            Viswasam|        6.5|
|               Virus|        7.7|
|              Veeram|        6.5|
|              Uppena|        6.5|
|                Unda|        7.7|
|     Thamizh Padam 2|        6.5|
|Swatantrya Veer S...|        7.7|
|                Star|        6.5|
+--------------------+-----------+
only showing top 10 rows
```

In [242…

```
# With Spark SQL
spark.sql("select Title, `IMDb Rating` from boxf_tb \
where `IMDb Rating` between 6.5 and 7.7 \
order by Title desc").show()
```

```
+--------------------+-----------+
|               Title|IMDb Rating|
+--------------------+-----------+
|    Yennai Arindhaal|        7.3|
|             Yashoda|        6.5|
|                 War|        6.5|
|            Viswasam|        6.5|
|               Virus|        7.7|
|          Virupaksha|        7.2|
|        Vikram Vedha|        7.1|
|Vendhu Thanindhat...|        7.3|
|         Velaikkaran|        7.2|
|              Veeram|        6.5|
|Varshangalkku She...|        6.6|
|            Varathan|        7.5|
|  Varane Avashyamund|        6.9|
|         Vakeel Saab|        6.9|
|Vaazha: Biopic of...|        7.3|
|              Vaathi|        7.3|
|   VR (Vikrant Rona)|        6.9|
|             Uunchai|        7.0|
|              Uppena|        6.5|
|                Unda|        7.7|
+--------------------+-----------+
only showing top 20 rows
```

61 Write a query to get director,films and IMDb ratings?

In [243…

```
# With Pandas
p_boxf.merge(p_dir,left_on="DirectorID",right_on="Director ID",how="right")\
    [["Title","Director","IMDb Rating"]].sort_values("IMDb Rating",ascending = False)\
    .head(10)
```

Out[243…

	Title	Director	IMDb Rating
539	12th Fail	Vidhu Vinod Chopra	8.8
442	Rocketry: The Nambi Effect	Madhavan	8.7
423	777 Charlie	Kiranraj K	8.7
414	The Kashmir Files	Vivek Agnihotri	8.6
583	Kishkindha Kaandam	Dinjith Ayyathan	8.6
579	Maharaja	Nithilan Saminathan	8.5
425	Sita Ramam	Hanu Raghavapudi	8.5
458	Jersey	Gowtam Tinnanuri	8.5
265	Sachin	James Erskine	8.5
505	Kumbalangi Nights	Madhu C. Narayanan	8.5

In [244…

```
# With PySpark
s_boxf.join(s_dir, s_boxf["DirectorID"]==s_dir["Director ID"],"right")\
    .select("Title","Director","IMDb Rating").sort(desc("IMDb Rating"))\
    .show(10)
```

```
+--------------------+------------------+-----------+
|               Title|          Director|IMDb Rating|
+--------------------+------------------+-----------+
|           12th Fail|Vidhu Vinod Chopra|        8.8|
|Rocketry: The Nam...|          Madhavan|        8.7|
|         777 Charlie|        Kiranraj K|        8.7|
|   The Kashmir Files|   Vivek Agnihotri|        8.6|
|  Kishkindha Kaandam|  Dinjith Ayyathan|        8.6|
|         Meiyazhagan|     C. Prem Kumar|        8.5|
|                  96|     C. Prem Kumar|        8.5|
|              Sachin|     James Erskine|        8.5|
|          Sita Ramam|  Hanu Raghavapudi|        8.5|
|              Jersey|  Gowtam Tinnanuri|        8.5|
+--------------------+------------------+-----------+
only showing top 10 rows
```

In [245…

```
# With Spark SQL
spark.sql("select Title, Director, `IMDb Rating` from boxf_tb boxf \
right join dir_tb dir on dir.`Director ID`=boxf.DirectorID \
order by `IMDb Rating` desc limit 10").show()
```

```
+--------------------+------------------+-----------+
|               Title|          Director|IMDb Rating|
+--------------------+------------------+-----------+
|           12th Fail|Vidhu Vinod Chopra|        8.8|
|         777 Charlie|        Kiranraj K|        8.7|
|Rocketry: The Nam...|          Madhavan|        8.7|
|  Kishkindha Kaandam|  Dinjith Ayyathan|        8.6|
|   The Kashmir Files|   Vivek Agnihotri|        8.6|
|              Jersey|  Gowtam Tinnanuri|        8.5|
|   Kumbalangi Nights|Madhu C. Narayanan|        8.5|
|          Sita Ramam|  Hanu Raghavapudi|        8.5|
|         Meiyazhagan|     C. Prem Kumar|        8.5|
|              Sachin|     James Erskine|        8.5|
+--------------------+------------------+-----------+
```

62 Write a query to get films with highest budget based flop verdict?

In [246…

```
# With Pandas
p_boxf.query("Verdict =='Flop'")[["Title","Verdict","Budget in Crores"]]\
    .sort_values("Budget in Crores",ascending = False).head(10)
```

Out[246…

	Title	Verdict	Budget in Crores
484	Thugs of Hindostan	Flop	275.0
541	83	Flop	200.0
277	Annaatthe	Flop	180.0
301	Valimai	Flop	150.0
546	Vikram Vedha	Flop	150.0
468	Race 3	Flop	150.0
366	Spyder	Flop	120.0
280	Vivegam	Flop	120.0
491	Baaghi 3	Flop	100.0
403	Godfather	Flop	100.0

In [247…

```
# With PySpark
s_boxf.filter(col("Verdict")=="Flop")\
    .select("Title","Verdict","Budget in Crores")\
    .sort(desc("Budget in Crores")).show(10)
```

```
+------------------+-------+----------------+
|             Title|Verdict|Budget in Crores|
+------------------+-------+----------------+
|Thugs of Hindostan|   Flop|           275.0|
|                83|   Flop|           200.0|
|         Annaatthe|   Flop|           180.0|
|           Valimai|   Flop|           150.0|
|      Vikram Vedha|   Flop|           150.0|
|            Race 3|   Flop|           150.0|
|           Vivegam|   Flop|           120.0|
|            Spyder|   Flop|           120.0|
|               Bro|   Flop|           100.0|
|          Baaghi 3|   Flop|           100.0|
+------------------+-------+----------------+
only showing top 10 rows
```

In [248…

```
# With Spark SQL
spark.sql("select Title,Verdict,`Budget in Crores` from boxf_tb \
where Verdict ='Flop' \
order by `Budget in Crores` desc limit 10").show()
```

```
+------------------+-------+----------------+
|             Title|Verdict|Budget in Crores|
+------------------+-------+----------------+
|Thugs of Hindostan|   Flop|           275.0|
|                83|   Flop|           200.0|
|         Annaatthe|   Flop|           180.0|
|           Valimai|   Flop|           150.0|
|            Race 3|   Flop|           150.0|
|      Vikram Vedha|   Flop|           150.0|
|            Spyder|   Flop|           120.0|
|           Vivegam|   Flop|           120.0|
|               Bro|   Flop|           100.0|
|         Godfather|   Flop|           100.0|
+------------------+-------+----------------+
```

63 Write a query to get total number of directors?

In [249…

```
# With Pandas
Total_directors = p_dir["Director"].nunique()
print("Total Directors: ",Total_directors)
```

```
Total Directors:  384
```

In [250…

```
# With PySpark
s_dir.select("Director").distinct().count()
```

Out[250… 384

In [251…

```
# With Spark SQL
spark.sql("select distinct(count(Director)) \
as Total_Directors from dir_tb").show()
```

```
+---------------+
|Total_Directors|
+---------------+
|            384|
+---------------+
```

64 Write a query to get vedridct wise total films released?

In [252…
```python
# With Pandas
p_boxf.groupby("Verdict").agg(Total_Films=("FilmID","count"))\
    .sort_values("Total_Films",ascending = False).reset_index()
```

Out[252…

	Verdict	Total_Films
0	Blockbuster	166
1	Hit	96
2	SuperHit	89
3	Disaster	68
4	Flop	57
5	BelowAverage	47
6	Average	42
7	AboveAverage	23
8	AllTimeBlockbuster	16

In [253…
```python
# With PySpark
s_boxf.groupby("Verdict").agg(count("FilmID")\
    .alias("Total_Films")).sort(desc("Total_Films")).show()
```

```
+--------------------+-----------+
|             Verdict|Total_Films|
+--------------------+-----------+
|         Blockbuster|        166|
|                 Hit|         96|
|           Super Hit|         89|
|            Disaster|         68|
|                Flop|         57|
|       Below Average|         47|
|             Average|         42|
|       Above Average|         23|
|All Time Blockbuster|         16|
+--------------------+-----------+
```

In [254…
```python
# With Spark SQL
spark.sql("select Verdict, count(FilmID) as Total_Films \
from boxf_tb \
group by Verdict \
order by Total_Films desc").show()
```

```
+--------------------+-----------+
|             Verdict|Total_Films|
+--------------------+-----------+
|         Blockbuster|        166|
|                 Hit|         96|
|           Super Hit|         89|
|            Disaster|         68|
|                Flop|         57|
|       Below Average|         47|
|             Average|         42|
|       Above Average|         23|
|All Time Blockbuster|         16|
+--------------------+-----------+
```

65 Write a query to get top 10 directors based number of films?

In [255…
```python
# With Pandas
p_boxf.merge(p_dir,left_on = "DirectorID",\
             right_on = "Director ID", how = "right")\
    .groupby("Director").agg(Total_Films=("FilmID","count"))\
    .sort_values("Total_Films",ascending = False)\
    .reset_index().head(10)
```

Out[255…

	Director	Total_Films
0	Trivikram Srinivas	6
1	Rohit Shetty	6
2	A.R. Murugadoss	5
3	Siva	5
4	Anil Ravipudi	5
5	Boyapati Srinu	5
6	Koratala Siva	5
7	Remo D'Souza	4
8	Atlee	4
9	Siddharth Anand	4

In [256…

```
# With PySpark
s_boxf.join(s_dir,s_boxf["DirectorID"]==s_dir["Director ID"],"right")\
    .groupby("Director").agg(count("FilmID").alias("Total_Films"))\
    .sort(desc("Total_Films")).show(10)
```

```
+------------------+-----------+
|          Director|Total_Films|
+------------------+-----------+
|Trivikram Srinivas|          6|
|      Rohit Shetty|          6|
|     Anil Ravipudi|          5|
|     Koratala Siva|          5|
|   A.R. Murugadoss|          5|
|    Boyapati Srinu|          5|
|              Siva|          5|
|    Maruthi Dasari|          4|
|         Sundar C.|          4|
|   Siddharth Anand|          4|
+------------------+-----------+
only showing top 10 rows
```

In [257…

```
# With PySpark
spark.sql("select Director, count(FilmID) as Total_Films \
from boxf_tb boxf \
right join dir_tb dir on dir.`Director ID`=boxf.DirectorID \
group by Director \
order by Total_Films desc limit 10").show()
```

```
+------------------+-----------+
|          Director|Total_Films|
+------------------+-----------+
|Trivikram Srinivas|          6|
|      Rohit Shetty|          6|
|     Anil Ravipudi|          5|
|     Koratala Siva|          5|
|   A.R. Murugadoss|          5|
|    Boyapati Srinu|          5|
|              Siva|          5|
|    Maruthi Dasari|          4|
|  Lokesh Kanagaraj|          4|
|   Siddharth Anand|          4|
+------------------+-----------+
```

66 Write a query to get top 5 directors based on world wide collections and also industry wise?

In [258…

```
# With Pandas
p_boxf.merge(p_dir,left_on= "DirectorID",right_on = "Director ID",how="inner")\
    .groupby(["Director","Industry"])\
    .agg(Total_WWC=("Worldwide Collection  in Crores","sum"))\
    .sort_values('Total_WWC',ascending = False)\
    .assign(rnk=lambda x:x\
        .groupby("Industry").cumcount()+1)\
    .query("rnk == 1").reset_index().head(5)
```

Out[258…

	Director	Industry	Total_WWC	rnk
0	S.S. Rajamouli	Tollywood	3668.00	1
1	Siddharth Anand	Bollywood	2229.45	1
2	Prashanth Neel	Sandalwood	1484.00	1
3	Lokesh Kanagaraj	Kollywood	1350.13	1
4	Vysakh	Mollywood	300.65	1

In [259…

```
# With PySpark
window_spec = Window.partitionBy("Industry")\
    .orderBy(desc("Total_WWC"))
s_boxf.join(s_dir,s_boxf["DirectorID"]==s_dir["Director ID"],"right")\
    .groupby("Director","Industry")\
    .agg(sum("Worldwide Collection  in Crores")\
        .alias("Total_WWC"))\
    .withColumn("rnk",row_number().over(window_spec))\
    .sort(desc("Total_WWC"))\
    .filter(col("rnk")==1).show()
```

```
+----------------+----------+------------------+---+
|        Director|  Industry|         Total_WWC|rnk|
+----------------+----------+------------------+---+
|  S.S. Rajamouli| Tollywood|            3668.0|  1|
| Siddharth Anand| Bollywood|           2229.45|  1|
|  Prashanth Neel|Sandalwood|            1484.0|  1|
|Lokesh Kanagaraj| Kollywood|1350.1299999999999|  1|
|          Vysakh| Mollywood|            300.65|  1|
+----------------+----------+------------------+---+
```

In [260…

```
# With Spark SQL
spark.sql("select * from (select Director,Industry,Total_WWC \
,row_number() over(partition by Industry order by Total_WWC desc)\
as rnk from (select Director, Industry, \
sum(`Worldwide Collection  in Crores`) as Total_WWC \
from boxf_tb boxf \
right join dir_tb dir on dir.`Director Id`=boxf.DirectorID \
group by Director, Industry)a )b \
where rnk = 1 ").show()
```

```
+----------------+----------+------------------+---+
|        Director|  Industry|         Total_WWC|rnk|
+----------------+----------+------------------+---+
| Siddharth Anand| Bollywood|           2229.45|  1|
|Lokesh Kanagaraj| Kollywood|1350.1299999999999|  1|
|          Vysakh| Mollywood|            300.65|  1|
|  Prashanth Neel|Sandalwood|            1484.0|  1|
|  S.S. Rajamouli| Tollywood|            3668.0|  1|
+----------------+----------+------------------+---+
```

67 Write a query to get top 3 directors based on India gross collections in Bollowood industry?

In [261…

```
# With Pandas
p_boxf.merge(p_dir, left_on = "DirectorID",right_on = "Director ID", how="right")\
    .query("Industry=='Bollywood'").groupby(["Director","Industry"])\
    .agg(Total_IGC= ("India Gross Collection in Crores","sum"))\
    .sort_values("Total_IGC",ascending = False)\
    .assign(rnk=lambda x:x.groupby("Industry").cumcount()+1)\
    .query("rnk <=3").reset_index()
```

Out[261…

	Director	Industry	Total_IGC	rnk
0	Siddharth Anand	Bollywood	1560.91	1
1	Rohit Shetty	Bollywood	1246.20	2
2	Rajkumar Hirani	Bollywood	1199.50	3

In [262…

```
# PySpark
window_spec = Window.partitionBy("Industry")\
    .orderBy(desc("Total_IGC"))
s_boxf.join(s_dir,s_boxf["DirectorID"]==s_dir["Director ID"],"right")\
    .filter(col("Industry")=="Bollywood")\
    .groupby("Director","Industry")\
    .agg(sum("India Gross Collection in Crores")\
        .alias("Total_IGC"))\
    .withColumn("rnk",row_number().over(window_spec))\
    .sort(desc("Total_IGC"))\
    .filter(col("rnk")<=3).show()
```

```
+---------------+---------+------------------+---+
|       Director| Industry|         Total_IGC|rnk|
+---------------+---------+------------------+---+
|Siddharth Anand|Bollywood|1560.9099999999999|  1|
|   Rohit Shetty|Bollywood|            1246.2|  2|
|Rajkumar Hirani|Bollywood|            1199.5|  3|
+---------------+---------+------------------+---+
```

In [263…

```
# With Spark SQL
spark.sql("select * from (select Director, Industry, Total_IGC, \
row_number() over(partition by Industry order by Total_IGC desc)\
as rnk from (select Director,Industry,\
sum(`India Gross Collection in Crores`) as Total_IGC from \
boxf_tb boxf \
right join dir_tb dir on dir.`Director ID`=boxf.DirectorID \
where Industry = 'Bollywood' \
group by Director, Industry)a)b where rnk <=3").show()
```

```
+---------------+---------+------------------+---+
|       Director| Industry|         Total_IGC|rnk|
+---------------+---------+------------------+---+
|Siddharth Anand|Bollywood|1560.9099999999999|  1|
|   Rohit Shetty|Bollywood|            1246.2|  2|
|Rajkumar Hirani|Bollywood|            1199.5|  3|
+---------------+---------+------------------+---+
```

68 Write a query to get top 3 directors based on India gross collections in Tollywood industry?

In [264…

```
# Pandas
p_boxf.merge(p_dir,left_on="DirectorID",right_on="Director ID",how="right")\
    .query("Industry =='Tollywood'").groupby(["Director","Industry"])\
    .agg(Total_IGC=("India Gross Collection in Crores","sum"))\
    .sort_values("Total_IGC",ascending = False)\
    .assign(rnk = lambda x:x.groupby("Industry").cumcount()+1)\
    .query("rnk <=3").reset_index()
```

Out[264…

	Director	Industry	Total_IGC	rnk
0	S.S. Rajamouli	Tollywood	2848.85	1
1	Nag Ashwin	Tollywood	840.25	2
2	Trivikram Srinivas	Tollywood	711.65	3

In [265…

```
# With PySpark
window_spec = Window.partitionBy("Industry")\
    .orderBy(desc("Total_IGC"))
s_boxf.join(s_dir,s_boxf["DirectorID"]==s_dir["Director ID"],"right")\
    .filter(col("Industry")=="Tollywood")\
    .groupby("Director","Industry")\
    .agg(sum("India Gross Collection in Crores")\
        .alias("Total_IGC"))\
    .sort(desc("Total_IGC"))\
    .withColumn("rnk",row_number().over(window_spec))\
    .filter(col("rnk")<=3).show()
```

```
+------------------+---------+-----------------+---+
|          Director| Industry|        Total_IGC|rnk|
+------------------+---------+-----------------+---+
|    S.S. Rajamouli|Tollywood|          2848.85|  1|
|        Nag Ashwin|Tollywood|           840.25|  2|
|Trivikram Srinivas|Tollywood|711.6500000000001|  3|
+------------------+---------+-----------------+---+
```

In [266…

```
# With Spark SQL
spark.sql("Select * from (select Director, Industry, Total_IGC,\
row_number() over(partition by Industry order by Total_IGC desc) \
as rnk from (select Director, Industry, \
sum(`India Gross Collection in Crores`) as Total_IGC \
from boxf_tb boxf \
right join dir_tb dir on dir.`Director ID`==boxf.DirectorID \
where Industry = 'Tollywood' \
group by Industry, Director)a) b \
where rnk <= 3").show()
```

```
+------------------+---------+-----------------+---+
|          Director| Industry|        Total_IGC|rnk|
+------------------+---------+-----------------+---+
|    S.S. Rajamouli|Tollywood|          2848.85|  1|
|        Nag Ashwin|Tollywood|           840.25|  2|
|Trivikram Srinivas|Tollywood|711.6500000000001|  3|
+------------------+---------+-----------------+---+
```

69 Write a query to get top 3 directors based on India gross collections in Kollowood industry?

In [267…

```
# With Pandas
p_boxf.merge(p_dir,left_on="DirectorID",right_on="Director ID",how="right")\
    .query("Industry == 'Kollywood'")\
    .groupby(["Director","Industry"])\
    .agg(Total_IGC=("India Gross Collection in Crores","sum"))\
    .sort_values("Total_IGC",ascending = False)\
    .assign(rnk= lambda x:x.groupby("Industry")\
        .cumcount()+1)\
    .query("rnk <=3").reset_index()
```

Out[267…

	Director	Industry	Total_IGC	rnk
0	Lokesh Kanagaraj	Kollywood	956.88	1
1	S. Shankar	Kollywood	826.83	2
2	Nelson Dilipkumar	Kollywood	669.88	3

In [317…

```
# With PySpark
window_spec = Window.partitionBy("Industry")\
    .orderBy(desc("Total_IGC"))
s_boxf.join(s_dir,s_boxf["DirectorID"]==s_dir["Director ID"],"right")\
    .filter(col("Industry")=="Kollywood")\
    .groupby("Director","Industry")\
    .agg(sum("India Gross Collection in Crores")\
        .alias("Total_IGC"))\
    .sort(desc("Total_IGC"))\
    .withColumn("rnk",row_number().over(window_spec))\
    .filter(col("rnk")<=3).show()
```

```
+-----------------+---------+---------+---+
|         Director| Industry|Total_IGC|rnk|
+-----------------+---------+---------+---+
| Lokesh Kanagaraj|Kollywood|   956.88|  1|
|       S. Shankar|Kollywood|   826.83|  2|
|Nelson Dilipkumar|Kollywood|   669.88|  3|
+-----------------+---------+---------+---+
```

In [ ]:

```
# With Spark SQL
spark.sql("select * from (select Director, Industry, Total_IGC,\
row_number() over(partition by Industry order by Total_IGC desc)\
as rnk from (select Director, Industry, \
sum(`India Gross Collection in Crores`) as Total_IGC \
from boxf_tb boxf \
right join dir_tb dir on dir.`Director ID`=boxf.DirectorID \
where Industry = 'Kollywood' \
group by Director, Industry)a) b \
where rnk <=3").show()
```

70 Write a query to get top 3 directors based on India gross collections in Mollowood industry?

In [ ]:

```
# With Pandas
p_boxf.merge(p_dir,left_on = "DirectorID", right_on = "Director ID", how="right")\
    .query("Industry == 'Mollywood'")\
    .groupby(["Director","Industry"])\
    .agg(Total_IGC=("India Gross Collection in Crores","sum"))\
    .sort_values("Total_IGC", ascending = False)\
    .assign(rnk=lambda x:x.groupby("Industry")\
        .cumcount()+1)\
    .query("rnk <=3").reset_index()
```

In [ ]:

```
# With PySpark
window_spec = Window.partitionBy("Industry")\
    .orderBy(desc("Total_IGC"))
s_boxf.join(s_dir,s_boxf["DirectorID"]==s_dir["Director ID"],"right")\
    .filter(col("Industry")=="Mollywood")\
    .groupby("Director","Industry")\
    .agg(sum("India Gross Collection in Crores")\
        .alias("Total_IGC"))\
    .sort(desc("Total_IGC"))\
```

```
    .withColumn("rnk",row_number().over(window_spec))\
    .filter(col("rnk")<=3).show()
```

In [ ]:
```
# With Spark SQL
spark.sql("select * from (select Director,Industry,Total_IGC, \
row_number()over(partition by Industry order by Total_IGC desc) \
as rnk from (select Director, Industry,\
sum(`India Gross Collection in Crores`)as Total_IGC \
from boxf_tb boxf \
right join dir_tb dir on dir.`Director Id`=boxf.DirectorID \
where Industry = 'Mollywood' \
group by Director, Industry)a)b \
where rnk <=3").show()
```

71 Write a query to get top 3 directors based on India gross collections in Sandalwood industry?

In [ ]:
```
# With Pandas
p_boxf.merge(p_dir,left_on = "DirectorID",\
             right_on = "Director ID",how="right")\
    .query("Industry == 'Sandalwood'")\
    .groupby(["Director","Industry"])\
    .agg(Total_IGC = ("India Gross Collection in Crores","sum"))\
    .sort_values("Total_IGC",ascending = False)\
    .assign(rnk=lambda x:x.groupby("Industry")\
        .cumcount()+1)\
    .query("rnk<=3").reset_index()
```

In [ ]:
```
# With PySpark
window_spec = Window.partitionBy("Industry")\
    .orderBy(desc("Total_IGC"))
s_boxf.join(s_dir,s_boxf["DirectorID"]==s_dir["Director ID"],"inner")\
    .filter(col("Industry")=="Sandalwood").groupby("Director","Industry")\
    .agg(sum("India Gross Collection in Crores").alias("Total_IGC"))\
    .withColumn("rnk",row_number().over(window_spec))\
    .sort(desc("Total_IGC")).filter(col("rnk")<=3).show()
```

In [ ]:
```
# with Spark SQL
spark.sql("select * from (select Director,Industry,Total_IGC,\
row_number() over(partition by Industry order by Total_IGC desc)\
as rnk from (select Director, Industry,\
sum(`India Gross Collection in Crores`) as Total_IGC \
from boxf_tb boxf \
right join dir_tb dir on dir.`Director ID` = boxf.DirectorID \
where Industry ='Sandalwood' \
group by Industry, Director )a )b \
where rnk <=3").show()
```

In [ ]:
```
p_boxf.columns
```

72 Write to get total number of generes?

In [ ]:
```
# With Pandas
p_gen["Genre"].nunique()
```

In [ ]:
```
# With PySpark
s_gen.select("Genre").distinct().count()
```

In [ ]:
```
# With Spark SQL
spark.sql("select Distinct(count(Genre))as \
Total_Genres from gen_tb ").show()
```

73 Write query to get director,language,genere films count?

In [ ]:
```
# With Pandas
p_boxf.merge(p_dir, left_on="DirectorID",\
             right_on="Director ID",how="inner")\
    .merge(p_lan,on ="LanguageID",how="inner")\
    .merge(p_gen,on="GenreID",how="inner")\
    .groupby(["Director","Language","Genre"])\
    .agg(Total_Films = ("FilmID","count"))\
    .sort_values("Total_Films",ascending = False)\
    .reset_index().head(10)
```

In [ ]:
```
# With PySpark
s_boxf.join(s_dir,s_boxf["DirectorID"]==s_dir["Director ID"],"inner")\
    .join(s_lan,s_boxf["LanguageID"]==s_lan["LanguageID"],"inner")\
    .join(s_gen,s_boxf["GenreID"]==s_gen["GenreID"],"inner")\
    .groupby("Director","Language","Genre").agg(count("FilmID")\
        .alias("Total_Films")).sort(desc("Total_Films")).show(10)
```

In [ ]:
```
# With Spark SQL
spark.sql("select Director,Language,Genre, \
count(FilmID)as Total_Films from boxf_tb boxf \
inner join dir_tb dir on dir.`Director ID`=boxf.DirectorID \
inner join lan_tb lan on lan.LanguageID =boxf.LanguageID \
inner join gen_tb gen on gen.GenreID = boxf.GenreID \
group by Director, Language, Genre \
order by Total_Films desc limit 10").show()
```

74 Write a query to genere wise budget?

In [ ]:
```
# With Pandas
p_boxf.merge(p_gen,on = "GenreID",how="right")\
    .groupby("Genre")\
    .agg(Total_Budget=("Budget in Crores","sum"))\
    .sort_values("Total_Budget",ascending=False)\
    .reset_index().head(10)
```

In [ ]:
```
# With PySpark
s_boxf.join(s_gen,s_boxf["GenreID"]==s_gen["GenreID"],"right")\
    .groupby("Genre").agg(sum("Budget in Crores")\
        .alias("Total_Budget"))\
    .sort(desc("Total_Budget")).show(10)
```

In [ ]:
```
# With Spark SQL
spark.sql("Select Genre, sum(`Budget in Crores`)\
as Total_Budget from boxf_tb boxf \
right join gen_tb gen on gen.GenreID = boxf.GenreID \
group by Genre \
order by 2 desc limit 10").show()
```

75 Write a query to get genere wise first day worldwide collections?

In [ ]:
```
# With Pandas
p_boxf.merge(p_gen,on = "GenreID",how="right")\
    .groupby("Genre")\
    .agg(Total_WWC=("Worldwide Collection  in Crores","sum"))\
    .sort_values("Total_WWC",ascending=False)\
    .reset_index().head(10)
```

In [ ]:
```
# With PySpark
s_boxf.join(s_gen,s_boxf["GenreID"]==s_gen["GenreID"],"right")\
    .groupby("Genre")\
    .agg(sum("Worldwide Collection  in Crores")\
        .alias("Total_WWC"))\
    .sort(desc("Total_WWC")).show(10)
```

In [ ]:
```
# With Spark SQL
spark.sql("Select Genre, sum(`Worldwide Collection  in Crores`) \
as Total_WWC from boxf_tb boxf \
right join gen_tb gen on gen.GenreID = boxf.GenreID \
group by Genre \
order by 2 desc limit 10").show()
```

76 Write a query to get genere wise overseas collections?

In [ ]:
```
# With Pandas
p_boxf.merge(p_gen,on = "GenreID",how="right")\
    .groupby("Genre")\
    .agg(Total_OSC=("Overseas Collection in Crores","sum"))\
    .sort_values("Total_OSC",ascending=False)\
    .reset_index().head(10)
```

In [ ]:
```
# With PySpark
s_boxf.join(s_gen,s_boxf["GenreID"]==s_gen["GenreID"],"right")\
    .groupby("Genre").agg(sum("Overseas Collection in Crores")\
        .alias("Total_OSC")).sort(desc("Total_OSC")).show(10)
```

In [ ]:
```
# With Spark SQL
spark.sql("Select Genre,sum(`Overseas Collection in Crores`) \
as Total_OSC from boxf_tb boxf \
right join gen_tb gen on gen.GenreID = boxf.GenreID \
group by Genre \
order by 2 desc limit 10").show()
```

77 Write a query to get genere wise India gross collections?

In [ ]:
```
# With Pandas
p_boxf.merge(p_gen,on = "GenreID",how="right")\
    .groupby("Genre")\
    .agg(Total_IGC=("India Gross Collection in Crores","sum"))\
    .sort_values("Total_IGC",ascending=False)\
    .reset_index().head(10)
```

In [ ]:
```
# With PySpark
s_boxf.join(s_gen,s_boxf["GenreID"]==s_gen["GenreID"],"right")\
    .groupby("Genre").agg(sum("India Gross Collection in Crores")\
        .alias("Total_IGC")).sort(desc("Total_IGC")).show(10)
```

In [ ]:
```
# With Spark SQL
spark.sql("Select Genre,sum(`India Gross Collection in Crores`) \
as Total_IGC from boxf_tb boxf \
right join gen_tb gen on gen.GenreID = boxf.GenreID \
group by Genre \
order by 2 desc limit 10").show()
```

78 Write a query to get genere wise top 2 longest run time movies?

In [ ]:
```
# With Pandas
p_boxf.merge(p_gen,on = "GenreID",how="right")\
    [["Title","Genre","Runtime (mins)"]]\
    .sort_values("Runtime (mins)",ascending=False)\
    .reset_index().head(2)
```

In [ ]:
```
# With PySpark
s_boxf.join(s_gen,s_boxf["GenreID"]==s_gen["GenreID"],"right")\
    .select("Title","Genre","Runtime (mins)")\
    .sort(desc("Runtime (mins)")).show(2)
```

In [ ]:
```
# With Spark SQL
spark.sql("Select Title,Genre,`Runtime (mins)` \
from boxf_tb boxf right join gen_tb gen on gen.GenreID = boxf.GenreID \
order by 3 desc limit 2").show()
```

79 Write a query to get genere wise bottom shortest runtime movies?

In [ ]:
```
# With Pandas
p_boxf.merge(p_gen,on = "GenreID",how="right")\
    [["Title","Genre","Runtime (mins)"]]\
    .sort_values("Runtime (mins)",ascending=True)\
    .reset_index().head(2)
```

In [ ]:
```
# With PySpark
s_boxf.join(s_gen,s_boxf["GenreID"]==s_gen["GenreID"],"right")\
    .select("Title","Genre","Runtime (mins)")\
    .sort(asc("Runtime (mins)")).show(2)
```

In [ ]:
```
# With Spark SQL
spark.sql("Select Title,Genre,`Runtime (mins)` \
from boxf_tb boxf right join gen_tb gen on gen.GenreID = boxf.GenreID \
order by 3 asc limit 2").show()
```

80 Write a query to get verdict, genere wise films released?

In [ ]:
```
# With Pandas
p_boxf.merge(p_gen,on = "GenreID",how="right")\
    .groupby(["Verdict","Genre"]).agg(Total_Films=("FilmID","count"))\
    .sort_values("Total_Films",ascending=False).reset_index().head(10)
```

In [ ]:
```
# With PySpark
s_boxf.join(s_gen,s_boxf["GenreID"]==s_gen["GenreID"],"right")\
    .groupby("Verdict","Genre").agg(count("FilmID")\
        .alias("Total_Films")).sort("Total_Films",ascending = False).show(10)
```

In [ ]:
```
# With Spark SQL
spark.sql("Select Verdict,Genre,count(FilmID)as Total_Films \
from boxf_tb boxf right join gen_tb gen on gen.GenreID = boxf.GenreID \
group by Genre,Verdict \
order by 3 desc limit 10").show()
```

81 Write a query to get genere, OTT platform wise films count?

In [ ]:
```
# With Pandas
p_boxf.merge(p_gen,on = "GenreID",how="right")\
```

```
    .groupby(["OTT Platform","Genre"]).agg(Total_Films=("FilmID","count"))\
    .sort_values("Total_Films",ascending=False).reset_index().head(10)
```

In [ ]:
```
# With PySpark
s_boxf.join(s_gen,s_boxf["GenreID"]==s_gen["GenreID"],"right")\
    .groupby("OTT Platform","Genre").agg(count("FilmID")\
        .alias("Total_Films")).sort("Total_Films",ascending = False).show(10)
```

In [ ]:
```
# With Spark SQL
spark.sql("Select `OTT Platform`,Genre,count(FilmID)as Total_Films \
from boxf_tb boxf right join gen_tb gen on gen.GenreID = boxf.GenreID \
group by Genre,`OTT platform` \
order by 3 desc limit 10").show()
```

82 Write a query to get genere wise films count?

In [ ]:
```
# With Pandas
p_boxf.merge(p_gen,on = "GenreID",how="right")\
    .groupby("Genre").agg(Total_Films=("FilmID","count"))\
    .sort_values("Total_Films",ascending=False).reset_index().head(10)
```

In [ ]:
```
# With PySpark
s_boxf.join(s_gen,s_boxf["GenreID"]==s_gen["GenreID"],"right")\
    .groupby("Genre").agg(count("FilmID")\
        .alias("Total_Films")).sort("Total_Films",ascending = False).show(10)
```

In [ ]:
```
# With Spark SQL
spark.sql("Select Genre,count(FilmID)as Total_Films \
from boxf_tb boxf right join gen_tb gen on gen.GenreID = boxf.GenreID \
group by Genre \
order by 2 desc limit 10").show()
```

83 Write a query to get genere wise films count in Tollowood Industry?

In [269…
```
# With Pandas
p_boxf.merge(p_gen,on = "GenreID",how="right").query("Industry =='Tollywood'")\
    .groupby(["Industry","Genre"]).agg(Total_Films=("FilmID","count"))\
    .sort_values("Total_Films",ascending=False).reset_index().head(10)
```

Out[269…

	Industry	Genre	Total_Films
0	Tollywood	Action, Drama	19
1	Tollywood	Action, Crime, Drama, Thriller	9
2	Tollywood	Comedy, Drama, Romance	8
3	Tollywood	Action, Comedy	8
4	Tollywood	Action, Drama, Thriller	8
5	Tollywood	Action, Thriller	7
6	Tollywood	Drama, Romance	6
7	Tollywood	Action, Comedy, Drama	6
8	Tollywood	Action, Drama, Romance	5
9	Tollywood	Action, Crime, Thriller	4

In [270…
```
# With PySpark
s_boxf.join(s_gen,s_boxf["GenreID"]==s_gen["GenreID"],"right")\
    .filter(col("Industry")=="Tollywood").groupby("Industry","Genre")\
    .agg(count("FilmID").alias("Total_Films"))\
    .sort("Total_Films",ascending = False).show(10)
```

```
+---------+--------------------+-----------+
| Industry|               Genre|Total_Films|
+---------+--------------------+-----------+
|Tollywood|       Action, Drama|         19|
|Tollywood|Action, Crime, Dr...|          9|
|Tollywood|Action, Drama, Th...|          8|
|Tollywood|      Action, Comedy|          8|
|Tollywood|Comedy, Drama, Ro...|          8|
|Tollywood|    Action, Thriller|          7|
|Tollywood|Action, Comedy, D...|          6|
|Tollywood|      Drama, Romance|          6|
|Tollywood|Action, Drama, Ro...|          5|
|Tollywood|Action, Crime, Th...|          4|
+---------+--------------------+-----------+
only showing top 10 rows
```

In [271…

```
# With Spark SQL
spark.sql("Select Industry,Genre,count(FilmID)as Total_Films \
from boxf_tb boxf right join gen_tb gen on gen.GenreID = boxf.GenreID \
where Industry = 'Tollywood'\
group by Industry,Genre \
order by 3 desc limit 10").show()
```

```
+---------+--------------------+-----------+
| Industry|               Genre|Total_Films|
+---------+--------------------+-----------+
|Tollywood|       Action, Drama|         19|
|Tollywood|Action, Crime, Dr...|          9|
|Tollywood|Action, Drama, Th...|          8|
|Tollywood|      Action, Comedy|          8|
|Tollywood|Comedy, Drama, Ro...|          8|
|Tollywood|    Action, Thriller|          7|
|Tollywood|Action, Comedy, D...|          6|
|Tollywood|      Drama, Romance|          6|
|Tollywood|Action, Drama, Ro...|          5|
|Tollywood|               Drama|          4|
+---------+--------------------+-----------+
```

84 Write a query to get genere wise films count in Kollowood Industry?

In [272…

```
# With Pandas
p_boxf.merge(p_gen,on = "GenreID",how="right").query("Industry =='Kollywood'")\
    .groupby(["Industry","Genre"]).agg(Total_Films=("FilmID","count"))\
    .sort_values("Total_Films",ascending=False).reset_index().head(10)
```

Out[272…

	Industry	Genre	Total_Films
0	Kollywood	Action, Drama	17
1	Kollywood	Action, Crime, Drama, Thriller	14
2	Kollywood	Action, Crime, Thriller	13
3	Kollywood	Action, Drama, Thriller	8
4	Kollywood	Action, Thriller	7
5	Kollywood	Action, Crime, Drama	6
6	Kollywood	Action, Comedy, Drama	5
7	Kollywood	Comedy, Horror	4
8	Kollywood	Action	3
9	Kollywood	Comedy, Drama	3

In [273…

```
# With PySpark
s_boxf.join(s_gen,s_boxf["GenreID"]==s_gen["GenreID"],"right")\
    .filter(col("Industry")=="Kollywood").groupby("Industry","Genre")\
    .agg(count("FilmID").alias("Total_Films"))\
    .sort("Total_Films",ascending = False).show(10)
```

```
+---------+--------------------+-----------+
| Industry|               Genre|Total_Films|
+---------+--------------------+-----------+
|Kollywood|       Action, Drama|         17|
|Kollywood|Action, Crime, Dr...|         14|
|Kollywood|Action, Crime, Th...|         13|
|Kollywood|Action, Drama, Th...|          8|
|Kollywood|    Action, Thriller|          7|
|Kollywood|Action, Crime, Drama|          6|
|Kollywood|Action, Comedy, D...|          5|
|Kollywood|      Comedy, Horror|          4|
|Kollywood|              Action|          3|
|Kollywood|              Comedy|          3|
+---------+--------------------+-----------+
only showing top 10 rows
```

In [274…

```
# With Spark SQL
spark.sql("Select Industry,Genre,count(FilmID)as Total_Films \
from boxf_tb boxf right join gen_tb gen on gen.GenreID = boxf.GenreID \
where Industry = 'Kollywood'\
group by Industry,Genre \
order by 3 desc limit 10").show()
```

```
+---------+--------------------+-----------+
| Industry|               Genre|Total_Films|
+---------+--------------------+-----------+
|Kollywood|       Action, Drama|         17|
|Kollywood|Action, Crime, Dr...|         14|
|Kollywood|Action, Crime, Th...|         13|
|Kollywood|Action, Drama, Th...|          8|
|Kollywood|    Action, Thriller|          7|
|Kollywood|Action, Crime, Drama|          6|
|Kollywood|Action, Comedy, D...|          5|
|Kollywood|      Comedy, Horror|          4|
|Kollywood|              Action|          3|
|Kollywood|       Comedy, Drama|          3|
+---------+--------------------+-----------+
```

85 Write a query to get genere wise films count in Mollowood Industry?

In [275…

```python
# With Pandas
p_boxf.merge(p_gen,on = "GenreID",how="right").query("Industry =='Mollywood'")\
    .groupby(["Industry","Genre"]).agg(Total_Films=("FilmID","count"))\
    .sort_values("Total_Films",ascending=False).reset_index().head(10)
```

Out[275…

	Industry	Genre	Total_Films
0	Mollywood	Action, Drama, Thriller	6
1	Mollywood	Comedy, Drama, Romance	6
2	Mollywood	Comedy, Drama	5
3	Mollywood	Drama, Thriller	4
4	Mollywood	Comedy, Romance	4
5	Mollywood	Action, Crime, Drama, Thriller	4
6	Mollywood	Comedy	3
7	Mollywood	Drama	2
8	Mollywood	Crime, Drama, Mystery, Thriller	2
9	Mollywood	Action, Thriller	2

In [276…

```python
# With PySpark
s_boxf.join(s_gen,s_boxf["GenreID"]==s_gen["GenreID"],"right")\
    .filter(col("Industry")=="Mollywood").groupby("Industry","Genre")\
    .agg(count("FilmID").alias("Total_Films"))\
    .sort("Total_Films",ascending = False).show(10)
```

```
+---------+--------------------+-----------+
| Industry|               Genre|Total_Films|
+---------+--------------------+-----------+
|Mollywood|Action, Drama, Th...|          6|
|Mollywood|Comedy, Drama, Ro...|          6|
|Mollywood|       Comedy, Drama|          5|
|Mollywood|     Comedy, Romance|          4|
|Mollywood|     Drama, Thriller|          4|
|Mollywood|Action, Crime, Dr...|          4|
|Mollywood|              Comedy|          3|
|Mollywood|               Drama|          2|
|Mollywood|Crime, Drama, Mys...|          2|
|Mollywood|Action, Comedy, T...|          2|
+---------+--------------------+-----------+
only showing top 10 rows
```

In [277…

```python
# With Spark SQL
spark.sql("Select Industry,Genre,count(FilmID)as Total_Films \
from boxf_tb boxf right join gen_tb gen on gen.GenreID = boxf.GenreID \
where Industry = 'Mollywood'\
group by Industry,Genre \
order by 3 desc limit 10").show()
```

```
+---------+--------------------+-----------+
| Industry|               Genre|Total_Films|
+---------+--------------------+-----------+
|Mollywood|Action, Drama, Th...|          6|
|Mollywood|Comedy, Drama, Ro...|          6|
|Mollywood|       Comedy, Drama|          5|
|Mollywood|     Comedy, Romance|          4|
|Mollywood|     Drama, Thriller|          4|
|Mollywood|Action, Crime, Dr...|          4|
|Mollywood|              Comedy|          3|
|Mollywood|Action, Drama, Hi...|          2|
|Mollywood|Horror, Mystery, ...|          2|
|Mollywood|Action, Comedy, T...|          2|
+---------+--------------------+-----------+
```

86 Write a query to get genere wise films count in Bollowood Industry?

In [278…

```
# With Pandas
p_boxf.merge(p_gen,on = "GenreID",how="right").query("Industry =='Bollywood'")\
    .groupby(["Industry","Genre"]).agg(Total_Films=("FilmID","count"))\
    .sort_values("Total_Films",ascending=False).reset_index().head(10)
```

Out[278…

	Industry	Genre	Total_Films
0	Bollywood	Comedy, Drama	12
1	Bollywood	Comedy, Drama, Romance	12
2	Bollywood	Comedy, Romance	8
3	Bollywood	Action, Adventure, Thriller	7
4	Bollywood	Biography, Drama	6
5	Bollywood	Action, Crime, Drama, Thriller	6
6	Bollywood	Action, Drama, Thriller	5
7	Bollywood	Action, Crime, Thriller	4
8	Bollywood	Action, Crime, Drama	4
9	Bollywood	Drama, Romance	4

In [279…

```
# With PySpark
s_boxf.join(s_gen,s_boxf["GenreID"]==s_gen["GenreID"],"right")\
    .filter(col("Industry")=="Bollywood").groupby("Industry","Genre")\
    .agg(count("FilmID").alias("Total_Films"))\
    .sort("Total_Films",ascending = False).show(10)
```

```
+---------+--------------------+-----------+
| Industry|               Genre|Total_Films|
+---------+--------------------+-----------+
|Bollywood|Comedy, Drama, Ro...|         12|
|Bollywood|       Comedy, Drama|         12|
|Bollywood|     Comedy, Romance|          8|
|Bollywood|Action, Adventure...|          7|
|Bollywood|    Biography, Drama|          6|
|Bollywood|Action, Crime, Dr...|          6|
|Bollywood|Action, Drama, Th...|          5|
|Bollywood|      Drama, Romance|          4|
|Bollywood|Action, Crime, Th...|          4|
|Bollywood|      Comedy, Horror|          4|
+---------+--------------------+-----------+
only showing top 10 rows
```

In [280…

```
# With Spark SQL
spark.sql("Select Industry,Genre,count(FilmID)as Total_Films \
from boxf_tb boxf right join gen_tb gen on gen.GenreID = boxf.GenreID \
where Industry = 'Bollywood'\
group by Industry,Genre \
order by 3 desc limit 10").show()
```

```
+---------+--------------------+-----------+
| Industry|               Genre|Total_Films|
+---------+--------------------+-----------+
|Bollywood|Comedy, Drama, Ro...|         12|
|Bollywood|       Comedy, Drama|         12|
|Bollywood|     Comedy, Romance|          8|
|Bollywood|Action, Adventure...|          7|
|Bollywood|    Biography, Drama|          6|
|Bollywood|Action, Crime, Dr...|          6|
|Bollywood|Action, Drama, Th...|          5|
|Bollywood|Action, Crime, Th...|          4|
|Bollywood|      Drama, Romance|          4|
|Bollywood|      Comedy, Horror|          4|
+---------+--------------------+-----------+
```

87 Write a query to get genere wise films count in Sandalwood Industry?

In [281…

```python
# With Pandas
p_boxf.merge(p_gen,on = "GenreID",how="right").query("Industry =='Sandalwood'")\
    .groupby(["Industry","Genre"]).agg(Total_Films=("FilmID","count"))\
    .sort_values("Total_Films",ascending=False).reset_index().head(10)
```

Out[281…

	Industry	Genre	Total_Films
0	Sandalwood	Action, Drama	5
1	Sandalwood	Action	4
2	Sandalwood	Action, Crime	3
3	Sandalwood	Action, Crime, Drama, Thriller	3
4	Sandalwood	Action, Crime, Thriller	2
5	Sandalwood	Action, Drama, Romance	2
6	Sandalwood	Action, Thriller	2
7	Sandalwood	Action, Adventure, Drama, Thriller	1
8	Sandalwood	Action, Comedy, Drama	1
9	Sandalwood	Action, Comedy, Musical	1

In [282…

```python
# With PySpark
s_boxf.join(s_gen,s_boxf["GenreID"]==s_gen["GenreID"],"right")\
    .filter(col("Industry")=="Sandalwood").groupby("Industry","Genre")\
    .agg(count("FilmID").alias("Total_Films"))\
    .sort("Total_Films",ascending = False).show(10)
```

```
+----------+--------------------+-----------+
|  Industry|               Genre|Total_Films|
+----------+--------------------+-----------+
|Sandalwood|       Action, Drama|          5|
|Sandalwood|              Action|          4|
|Sandalwood|Action, Crime, Dr...|          3|
|Sandalwood|       Action, Crime|          3|
|Sandalwood|Action, Crime, Th...|          2|
|Sandalwood|    Action, Thriller|          2|
|Sandalwood|Action, Drama, Ro...|          2|
|Sandalwood|Adventure, Comedy...|          1|
|Sandalwood|Action, Adventure...|          1|
|Sandalwood|Action, Drama, Sport|          1|
+----------+--------------------+-----------+
only showing top 10 rows
```

In [283…

```python
# With Spark SQL
spark.sql("Select Industry,Genre,count(FilmID)as Total_Films \
from boxf_tb boxf right join gen_tb gen on gen.GenreID = boxf.GenreID \
where Industry = 'Sandalwood'\
group by Industry,Genre \
order by 3 desc limit 10").show()
```

```
+----------+--------------------+-----------+
|  Industry|               Genre|Total_Films|
+----------+--------------------+-----------+
|Sandalwood|       Action, Drama|          5|
|Sandalwood|              Action|          4|
|Sandalwood|Action, Crime, Dr...|          3|
|Sandalwood|       Action, Crime|          3|
|Sandalwood|Action, Crime, Th...|          2|
|Sandalwood|    Action, Thriller|          2|
|Sandalwood|Action, Drama, Ro...|          2|
|Sandalwood|Adventure, Comedy...|          1|
|Sandalwood|Action, Adventure...|          1|
|Sandalwood|Action, Drama, Sport|          1|
+----------+--------------------+-----------+
```

88 Write a query to get lead actors/actress wise,genere and films count?

In [284…

```
# With Pandas
p_boxf.merge(p_gen,on = "GenreID",how="right")\
    .groupby(["Lead Actor/Actress","Genre"])\
    .agg(Total_Films=("FilmID","count"))\
    .sort_values("Total_Films",ascending=False)\
    .reset_index().head(10)
```

Out[284…

	Lead Actor/Actress	Genre	Total_Films
0	Dhanush	Action, Drama	5
1	Mahesh Babu	Action, Drama	4
2	N.T. Rama Rao Jr.	Action, Drama, Thriller	3
3	Vijay Deverakonda	Comedy, Drama, Romance	3
4	Ravi Teja	Action, Crime, Drama, Thriller	3
5	Nani	Comedy, Romance	3
6	Ajith Kumar	Action, Crime, Drama, Thriller	3
7	Allu Arjun	Action, Drama	3
8	Ayushmann Khurrana	Comedy, Drama	2
9	Sivakarthikeyan	Action, Drama, Thriller	2

In [285…

```
# With PySpark
s_boxf.join(s_gen,s_boxf["GenreID"]==s_gen["GenreID"],"right")\
    .groupby("Lead Actor/Actress","Genre")\
    .agg(count("FilmID").alias("Total_Films"))\
    .sort("Total_Films",ascending = False).show(10)
```

```
+------------------+--------------------+-----------+
|Lead Actor/Actress|               Genre|Total_Films|
+------------------+--------------------+-----------+
|           Dhanush|       Action, Drama|          5|
|       Mahesh Babu|       Action, Drama|          4|
|        Allu Arjun|       Action, Drama|          3|
|              Nani|     Comedy, Romance|          3|
| N.T. Rama Rao Jr.|Action, Drama, Th...|          3|
|       Ajith Kumar|Action, Crime, Dr...|          3|
| Vijay Deverakonda|Comedy, Drama, Ro...|          3|
|         Ravi Teja|Action, Crime, Dr...|          3|
|           Prabhas|       Action, Drama|          2|
|         Mammootty|Action, Crime, Dr...|          2|
+------------------+--------------------+-----------+
only showing top 10 rows
```

In [286…

```
# With Spark SQL
spark.sql("Select `Lead Actor/Actress`,Genre,count(FilmID)as Total_Films \
from boxf_tb boxf right join gen_tb gen on gen.GenreID = boxf.GenreID \
group by `Lead Actor/Actress`,Genre \
order by 3 desc limit 10").show()
```

```
+------------------+--------------------+-----------+
|Lead Actor/Actress|               Genre|Total_Films|
+------------------+--------------------+-----------+
|           Dhanush|       Action, Drama|          5|
|       Mahesh Babu|       Action, Drama|          4|
|        Allu Arjun|       Action, Drama|          3|
|              Nani|     Comedy, Romance|          3|
| N.T. Rama Rao Jr.|Action, Drama, Th...|          3|
|       Ajith Kumar|Action, Crime, Dr...|          3|
| Vijay Deverakonda|Comedy, Drama, Ro...|          3|
|         Ravi Teja|Action, Crime, Dr...|          3|
|           Prabhas|       Action, Drama|          2|
|            Vikram|Action, Adventure...|          2|
+------------------+--------------------+-----------+
```

89 Write a query to get 5th rank movie based on Worldwide total collections?

In [287…
```
# With Pandas
p_boxf["rnk"]=p_boxf[["Worldwide Collection  in Crores"]]\
    .rank(method="dense",ascending=False)
p_boxf.query("rnk == 5")[["Title","Worldwide Collection  in Crores","rnk"]]
```

Out[287…

	Title	Worldwide Collection in Crores	rnk
572	Jawan	1160.0	5.0

In [288…
```
# With PySpark
s_boxf = s_boxf.withColumn("rnk",dense_rank()\
    .over(Window.orderBy(desc("Worldwide Collection  in Crores"))))
s_boxf.filter(col("rnk")==5)\
    .select("Title","Worldwide Collection  in Crores","rnk").show()
```

```
+-----+------------------------------+---+
|Title|Worldwide Collection  in Crores|rnk|
+-----+------------------------------+---+
|Jawan|                        1160.0|  5|
+-----+------------------------------+---+
```

In [289…
```
# With Spark SQL
spark.sql("select * from\
(select Title,`Worldwide Collection  in Crores`, \
dense_rank() over(Order by `Worldwide Collection  in Crores` desc) \
as rnk from boxf_tb) \
where rnk = 5").show()
```

```
+-----+------------------------------+---+
|Title|Worldwide Collection  in Crores|rnk|
+-----+------------------------------+---+
|Jawan|                        1160.0|  5|
+-----+------------------------------+---+
```

90 Write a query to get 5th rank movie by industry wise based on First day worldwide collections?

In [290…
```
p_boxf["rnk"]=p_boxf[["First Day Collection Worldwide  in Crores"]]\
    .rank(method="dense",ascending=False)
p_boxf.query("rnk==5")\
    [["Title","First Day Collection Worldwide  in Crores","rnk"]]
```

Out[290…

	Title	First Day Collection Worldwide in Crores	rnk
212	Salaar: Part 1 - Ceasefire	158.1	5.0

In [291…
```
# With PySpark
s_boxf=s_boxf.withColumn("rnk",dense_rank()\
    .over(Window.orderBy(desc("First Day Collection Worldwide  in Crores"))))
s_boxf.filter(col("rnk")==5)\
    .select("Title","First Day Collection Worldwide  in Crores","rnk").show()
```

```
+--------------------+----------------------------------------+---+
|               Title|First Day Collection Worldwide  in Crores|rnk|
+--------------------+----------------------------------------+---+
|Salaar: Part 1 - ...|                                   158.1|  5|
+--------------------+----------------------------------------+---+
```

In [292…
```
#With Spark SQL
spark.sql("select * from(select Title,\
`First Day Collection Worldwide  in Crores`,\
dense_rank() over(order by \
```

```
`First Day Collection Worldwide  in Crores` desc) as rnk \
from boxf_tb) \
where rnk = 5").show()
```

```
+--------------------+----------------------------------------+---+
|               Title|First Day Collection Worldwide  in Crores|rnk|
+--------------------+----------------------------------------+---+
|Salaar: Part 1 - ...|                                   158.1|  5|
+--------------------+----------------------------------------+---+
```

91 Write a query to get 3rd rank movie by industry wise based on IMDb Ratings?

In [293…

```
# With Pandas
p_boxf[["Industry","Title","IMDb Rating"]]\
    .sort_values(["Industry","IMDb Rating"],ascending=[True,False])\
    .assign(rnk= lambda x:x.groupby("Industry").cumcount()+1)\
    .query("rnk ==3")
```

Out[293…

	Industry	Title	IMDb Rating	rnk
512	Bollywood	Sachin	8.5	3
135	Kollywood	Maharaja	8.5	3
147	Mollywood	Bangalore Days	8.3	3
149	Sandalwood	K.G.F: Chapter 2	8.2	3
35	Tollywood	Mahanati	8.4	3

In [294…

```
# With PySpark
window_spec = Window.partitionBy("Industry").orderBy(desc("IMDb Rating"))
s_boxf.withColumn("rnk",row_number().over(window_spec))\
    .filter(col("rnk")==3).sort("Industry",desc("IMDb Rating"))\
    .select("Industry","Title","IMDb Rating","rnk").show()
```

```
+----------+----------------+-----------+---+
|  Industry|           Title|IMDb Rating|rnk|
+----------+----------------+-----------+---+
| Bollywood|          Sachin|        8.5|  3|
| Kollywood|        Maharaja|        8.5|  3|
| Mollywood|  Bangalore Days|        8.3|  3|
|Sandalwood|K.G.F: Chapter 2|        8.2|  3|
| Tollywood|        Mahanati|        8.4|  3|
+----------+----------------+-----------+---+
```

In [295…

```
#with Spark SQL
spark.sql("select * from(select Industry,Title,`IMDb Rating`,\
row_number() over(partition by Industry order by `IMDb Rating` desc)as rnk \
from boxf_tb ) where rnk =3 ").show()
```

```
+----------+----------------+-----------+---+
|  Industry|           Title|IMDb Rating|rnk|
+----------+----------------+-----------+---+
| Bollywood|          Sachin|        8.5|  3|
| Kollywood|        Maharaja|        8.5|  3|
| Mollywood|  Bangalore Days|        8.3|  3|
|Sandalwood|K.G.F: Chapter 2|        8.2|  3|
| Tollywood|        Mahanati|        8.4|  3|
+----------+----------------+-----------+---+
```

92 Calculate YoY% Budget growth?

With Pandas

In [296…

```
Yearly_Bud=p_boxf.groupby("Year")[["Budget in Crores"]].sum()\
    .sort_values("Year",ascending=True).reset_index()
```

In [297…

```
Yearly_Bud["YoY_Growth"]=Yearly_Bud["Budget in Crores"].pct_change()*100
```

In [298…

```
Yearly_Bud
```

Out[298…

	Year	Budget in Crores	YoY_Growth
0	2014	1820.0	NaN
1	2015	1906.0	4.725275
2	2016	2142.0	12.381952
3	2017	3291.0	53.641457
4	2018	3679.0	11.789730
5	2019	2544.0	-30.850775
6	2020	1387.0	-45.479560
7	2021	1855.0	33.741889
8	2022	5722.0	208.463612
9	2023	5877.0	2.708843
10	2024	4430.0	-24.621405

With PySpark

In [299…

```
Yearly_Bud = s_boxf.groupby("Year")\
    .agg(sum("Budget in Crores").alias("Total_Budget"))
```

In [300…

```
Yearly_Bud = Yearly_Bud.withColumn("PY_Budget",lag("Total_Budget")\
    .over(Window.orderBy("Year")))\
    .withColumn("YoY%",round(((\
        col("Total_Budget")-col("PY_Budget"))/col("PY_Budget")*100),2))
```

In [301…

```
Yearly_Bud.show()
```

```
+----+------------+---------+------+
|Year|Total_Budget|PY_Budget|  YoY%|
+----+------------+---------+------+
|2014|      1820.0|     NULL|  NULL|
|2015|      1906.0|   1820.0|  4.73|
|2016|      2142.0|   1906.0| 12.38|
|2017|      3291.0|   2142.0| 53.64|
|2018|      3679.0|   3291.0| 11.79|
|2019|      2544.0|   3679.0|-30.85|
|2020|      1387.0|   2544.0|-45.48|
|2021|      1855.0|   1387.0| 33.74|
|2022|      5722.0|   1855.0|208.46|
|2023|      5877.0|   5722.0|  2.71|
|2024|      4430.0|   5877.0|-24.62|
+----+------------+---------+------+
```

With Spark SQL

In [302…

```
spark.sql("select Year,Total_Bud,PY_Bud, \
round(((Total_Bud-PY_Bud)/PY_Bud)*100,2) YoY from \
(select Year, Total_Bud,lag(Total_Bud) \
over(order by Year) PY_Bud from \
(select Year, sum(`Budget in Crores`) \
as Total_Bud from boxf_tb \
group by Year)a)b").show()
```

```
+----+---------+------+------+
|Year|Total_Bud|PY_Bud|   YoY|
+----+---------+------+------+
|2014|   1820.0|  NULL|  NULL|
|2015|   1906.0|1820.0|  4.73|
|2016|   2142.0|1906.0| 12.38|
|2017|   3291.0|2142.0| 53.64|
|2018|   3679.0|3291.0| 11.79|
|2019|   2544.0|3679.0|-30.85|
|2020|   1387.0|2544.0|-45.48|
|2021|   1855.0|1387.0| 33.74|
|2022|   5722.0|1855.0|208.46|
|2023|   5877.0|5722.0|  2.71|
|2024|   4430.0|5877.0|-24.62|
+----+---------+------+------+
```

93 Calculate YoY% Worldwide total collelctions growth?

In [303…

```
Yearly_WWC = p_boxf.groupby("Year")\
    [["Worldwide Collection  in Crores"]].sum()\
```

```
    .sort_values("Year",ascending = True).reset_index()
```

In [304…

```
Yearly_WWC["YoY%_Growth"]=Yearly_WWC["Worldwide Collection  in Crores"]\
    .pct_change()*100
```

In [305…

```
Yearly_WWC
```

Out[305…

	Year	Worldwide Collection in Crores	YoY%_Growth
0	2014	4645.00	NaN
1	2015	5470.20	17.765339
2	2016	7088.80	29.589412
3	2017	9659.20	36.260016
4	2018	8275.30	-14.327273
5	2019	8242.72	-0.393702
6	2020	2348.75	-71.505158
7	2021	2711.65	15.450772
8	2022	9853.16	263.364003
9	2023	13461.72	36.623378
10	2024	8122.29	-39.663802

In [306…

```
Yearly_WWC = s_boxf.groupby("Year")\
    .agg(sum("Worldwide Collection  in Crores")\
    .alias("Total_WWC"))
```

In [307…

```
Yearly_WWC = Yearly_WWC.withColumn("PY_WWC",lag("Total_WWC")\
    .over(Window.orderBy("Year")))\
    .withColumn("YoY%",round(((\
        col("Total_WWC")-col("PY_WWC"))/col("PY_WWC")*100),2))
```

In [308…

```
Yearly_WWC.show()
```

```
+----+------------------+------------------+------+
|Year|         Total_WWC|            PY_WWC|  YoY%|
+----+------------------+------------------+------+
|2014| 4645.000000000001|              NULL|  NULL|
|2015| 5470.200000000001| 4645.000000000001| 17.77|
|2016| 7088.799999999999| 5470.200000000001| 29.59|
|2017|            9659.2| 7088.799999999999| 36.26|
|2018| 8275.300000000001|            9659.2|-14.33|
|2019|           8242.72| 8275.300000000001| -0.39|
|2020|           2348.75|           8242.72|-71.51|
|2021|2711.6499999999996|           2348.75| 15.45|
|2022| 9853.160000000002|2711.6499999999996|263.36|
|2023|13461.720000000001| 9853.160000000002| 36.62|
|2024|           8122.29|13461.720000000001|-39.66|
+----+------------------+------------------+------+
```

With Spark SQL

In [309…

```
spark.sql("select Year, Total_WWC,\
PY_WWC,round(((Total_WWC-PY_WWC)/PY_WWC)*100,2) YoY from \
(select Year, Total_WWC, lag(Total_WWC) over(order by Year)\
as PY_WWC from (select Year, \
sum(`Worldwide Collection  in Crores`) as Total_WWC from boxf_tb \
group by Year)a)b").show()
```

```
+----+------------------+------------------+------+
|Year|         Total_WWC|            PY_WWC|   YoY|
+----+------------------+------------------+------+
|2014| 4645.000000000001|              NULL|  NULL|
|2015| 5470.200000000001| 4645.000000000001| 17.77|
|2016| 7088.799999999999| 5470.200000000001| 29.59|
|2017|            9659.2| 7088.799999999999| 36.26|
|2018| 8275.300000000001|            9659.2|-14.33|
|2019|           8242.72| 8275.300000000001| -0.39|
|2020|           2348.75|           8242.72|-71.51|
|2021|2711.6499999999996|           2348.75| 15.45|
|2022| 9853.160000000002|2711.6499999999996|263.36|
|2023|13461.720000000001| 9853.160000000002| 36.62|
|2024|           8122.29|13461.720000000001|-39.66|
+----+------------------+------------------+------+
```

94 Calculate YoY% Indian Gross colelctions growth?

In [310…
```
Yearly_IGC = p_boxf.groupby("Year")\
    [["India Gross Collection in Crores"]].sum()\
    .sort_values("Year",ascending = True).reset_index()
```

In [311…
```
Yearly_IGC["YoY%_Growth"]=Yearly_IGC["India Gross Collection in Crores"]\
    .pct_change()*100
```

In [312…
```
Yearly_IGC
```

Out[312…

	Year	India Gross Collection in Crores	YoY%_Growth
0	2014	3711.10	NaN
1	2015	3935.30	6.041335
2	2016	4459.90	13.330623
3	2017	6852.90	53.655912
4	2018	6327.40	-7.668286
5	2019	6655.58	5.186649
6	2020	1966.35	-70.455618
7	2021	2238.40	13.835279
8	2022	7716.21	244.719889
9	2023	10080.17	30.636284
10	2024	6120.79	-39.278901

With PySpark

In [313…
```
Yearly_IGC = s_boxf.groupby("Year")\
    .agg(sum("India Gross Collection in Crores")\
        .alias("Total_IGC"))
```

In [314…
```
Yearly_IGC = Yearly_IGC.withColumn("PY_IGC",lag("Total_IGC")\
    .over(Window.orderBy("Year")))\
    .withColumn("YoY%",round(((\
        col("Total_IGC")-col("PY_IGC"))/col("PY_IGC")*100),2))
```

In [315…
```
Yearly_IGC.show()
```

```
+----+------------------+------------------+------+
|Year|         Total_IGC|            PY_IGC|  YoY%|
+----+------------------+------------------+------+
|2014| 3711.099999999999|              NULL|  NULL|
|2015|3935.2999999999997| 3711.099999999999|  6.04|
|2016| 4459.900000000001|3935.2999999999997| 13.33|
|2017|6852.9000000000015| 4459.900000000001| 53.66|
|2018| 6327.400000000001|6852.9000000000015| -7.67|
|2019| 6655.580000000001| 6327.400000000001|  5.19|
|2020|           1966.35| 6655.580000000001|-70.46|
|2021|            2238.4|           1966.35| 13.84|
|2022| 7716.210000000003|            2238.4|244.72|
|2023|          10080.17| 7716.210000000003| 30.64|
|2024|           6120.79|          10080.17|-39.28|
+----+------------------+------------------+------+
```

With Spark SQL

In [316…
```
spark.sql("select Year, Total_IGC, PY_IGC, \
round(((Total_IGC-PY_IGC)/PY_IGC)*100,2) YoY from \
(select Year, Total_IGC, lag(Total_IGC)\
over(order by Year) as PY_IGC from (select Year,\
sum(`India Gross Collection in Crores`)\
as Total_IGC from boxf_tb \
group by Year)a)b").show()
```

```
+----+------------------+------------------+------+
|Year|         Total_IGC|            PY_IGC|   YoY|
+----+------------------+------------------+------+
|2014| 3711.099999999999|              NULL|  NULL|
|2015|3935.2999999999997| 3711.099999999999|  6.04|
|2016| 4459.900000000001|3935.2999999999997| 13.33|
|2017|6852.9000000000015| 4459.900000000001| 53.66|
|2018| 6327.400000000001|6852.9000000000015| -7.67|
|2019| 6655.580000000001| 6327.400000000001|  5.19|
|2020|           1966.35| 6655.580000000001|-70.46|
|2021|            2238.4|           1966.35| 13.84|
|2022| 7716.210000000003|            2238.4|244.72|
|2023|          10080.17| 7716.210000000003| 30.64|
|2024|           6120.79|          10080.17|-39.28|
+----+------------------+------------------+------+
```

In [ ]: